The 40 Hour Cabin

2V Geodesic Dome Wedge Method

One small chainsaw, three trees, forty hours or less: a geodesic wedge cabin, winched up between the trees, skinned and floored in any order

A builder's manual: the geometry, fabrication, construction and shell of a hubless timber geodesic system, documented from one standing prototype. The frame was built on blocks in a clearing, winched up on three cables to saddles on two standing trees — because that is how this one was actually built — and left hanging there while the skin, the floor and the ground were decided in any order.

By Donovan Zeanah.

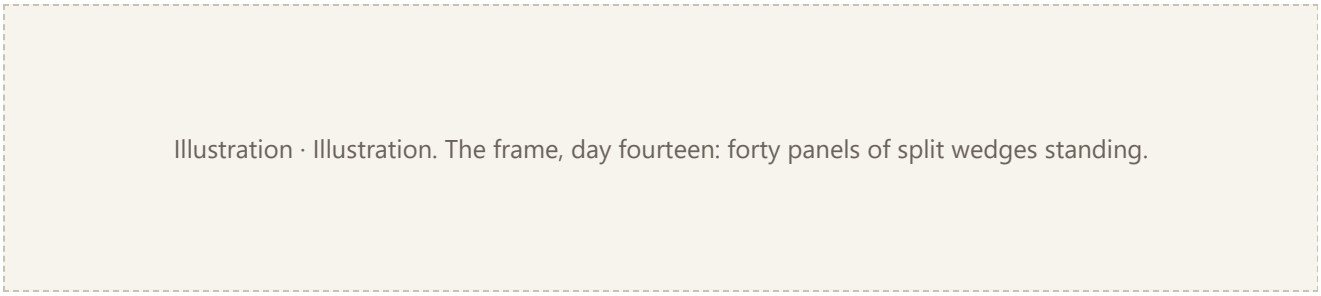


Illustration. The frame, day fourteen: forty panels of split wedges standing.

That is the whole claim in one picture: a frame of 120 members, cut from 3 trees, standing on day fourteen of the fortnight this book teaches — and the fortnight's hands-on total is the title's other claim, counted below.

The forty hours

The title is an arithmetic, not a slogan. The clock is the one the companion wedge book prints, and every rate in it is a measured session figure rather than a process estimate:

At the log. The frame's 120 members, split at the measured 30 wedges per 4-hour session: **16 hours** of splitting.

From wedge to standing. Each member, split through standing, measured at 15 minutes: **30 straight hours** in all, of which the splitting above is 16. The rest is the cut, the panels and the raise.

With the error allowance. A 25 per cent allowance on top of the straight time, because a first build always pays its tuition: **37.5 hours of hands-on work** — inside the 40-hour week the title is named after, with 2.5 hours to spare.

The honest reading, stated once and used everywhere after: the forty hours is the *building* — the frame, from split wedges to standing, in forty hours or less of hands-on work. Felling and bucking the logs is further work at the saw, and the skin, the floor and the ground are added later, in any order, which is the point of the next page and of the whole book.

The part you are allowed to get wrong

Your skull is not symmetrical either, and it protects the brain just fine. This book builds on that sentence.

A frame's job is to be a *frame*: forty triangles whose lengths set their angles, standing because the shape cannot fall without stretching a stick. It does not have to be pretty, milled, graded or symmetrical — the lumpy skull protects the brain, and the lumpy frame holds the cabin. The parts that must be right are the two that touch the weather and the earth: the **skin**, and the **ground**. Those are the chapters this book treats with the most care, and they are the parts that can be added, replaced or removed in any order — the frame is hot-swappable by design, because every panel, every layer, every floor and every pad is a separate object that bolts on and off.

The build order this book actually used is the order its title page promises: frame first, winched between the trees, then whatever the winter asks for — skin before floor, floor before skin, the ground last of all. The book is written so that any order works.

Which cabin this book is

Two books carry the name *The 40 Hour Cabin*, and this page states the difference so nobody is surprised later. **This one** — the paperback — is the 21.6-foot dome: 120 members, 365 square feet of floor, cut from 3 trees. The **Complete Edition**, sold on the site with the codebase, is built around a different, smaller dome (19.4 feet, 277 square feet) solved by the same software. Same method, same clock; two cabins.

Copyright, and the engineer’s note

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This book documents construction methods, geometric principles, prototype builds and design possibilities. It is not a substitute for site-specific structural engineering, and nothing in it is offered as code-compliant design. Codes, loads, materials, soil conditions and permitting requirements vary by place and by site. A structure intended for permanent occupancy — and anything carrying snow, wind or people you are not prepared to lose — should be reviewed by a qualified engineer and the local authority before it is built. The winch rig that raises the frame between the trees carries real loads — the trees, the saddles, the cables, the mast and the hoist — and none of those ratings is in this book.

Three kinds of claim run through this book, and each is labelled so they cannot be mistaken for one another:

Known geometry. A fact of the solved frame, true at every size: a 2V dome of this configuration has this many members, at these ratios. These are calculations, and the book shows the calculation behind each one.

Tested construction. Something that happened on the standing prototype this book is written around: the joint that was used, the fortnight that was clocked, the winch raise that put the cabin between the trees, the mistake that was made and corrected. These are reports, not promises.

Design possibilities. Things that are priced or drawn but not yet built — the pony wall, the brim, the later layers of the shell ladder. Each is marked as such where it appears, and the chapters state what each claim rests on.

A method whose three kinds of claim are kept apart is a method a reader can check. That separation is the point of this book, and the rest of it is the working out.

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What this book is

3 trees, one chainsaw off a hardware-store shelf, one hundred and twenty structural members, and a geodesic dome standing in fourteen days — 37.5 hours of hands-on work, inside the forty-hour week the title is named after.

That is the whole claim, and everything in this book exists to either explain it, prove it, or teach you to do it.

It is a real build

The dome in the plates is the simulator's solved frame — the same software computed every number printed in this book, and the build itself was cut for real: felled, split, panel by panel on a jig, and winched up between two standing trees. The process of cutting it produced a number of surprises, most of which are in here. Several chapters exist specifically to report a figure that does not flatter the method — Chapter 13 revises the recovery claim downward, Chapter 46 cuts an hourly rate this project publicised by more than half, once the honest section is used — and the corrections live in the chapters where the claims were made, named rather than silently edited. A book about a new building method that contains no corrections is a brochure.

It is three books at once

The reader I had in mind wanted three different things, and they read at three different speeds, so rather than blend them into a mush each chapter declares which one it is.

The story is what happened, in order, to somebody with a saw: the frankendome that came first, the afternoon the wedge was found, and the week lost to a problem that turned out not to exist. 6 chapters, and you can read them straight through as a narrative.

The manual is do this, then this. 19 chapters, and if you follow only those you get a dome. It is a complete set of instructions on its own.

The explanation is why any of it works — why a triangulated frame will accept a stick no mill would sell, why splitting a round log is free and squaring it is not, and why this method does not transfer to a rectangular house. 22 chapters, all skippable, all there because a method you do not understand is a method you cannot adapt.

The rest is reference: the tables you come back to at the saw rather than read in an armchair.

Every number here is computed

This is the unusual part, and it is worth a paragraph.

Not one figure in this book was typed into a sentence. Every dimension, count, angle, percentage and price is a token in the manuscript that gets filled in, at the moment the book is made, by the same software that draws the three-dimensional models. When it says the dome is 21.6 feet across, that number came out of a solver about a second before it reached this page.

The reason is not showing off. It is that a number typed into a paragraph is correct on the day it is typed and silently wrong from the first time anybody changes anything — and there is no way to find those. Nothing fails. The book just quietly becomes untrue in places nobody remembers.

So the arithmetic in this book is a test suite. It runs, it checks itself against the geometry, and where two ways of computing something disagree the book prints both and says which is which. Chapter 23 is nothing but a proof that the two methods in Part One are the same calculation, and it prints the residual — zero, to the last decimal the computer keeps — rather than asking you to take it on trust.

The tools that produced these numbers are published as open source at github.com/dkzeanah/DomeSimulator, and the Complete Edition — the same method carried further, with the codebase — is sold on the site. The last pages of this book point the way.

Before you start a saw

The felling chapter sends you back here, so this page says the things once, plainly, before any cutting is described.

A chainsaw is the most dangerous tool most people will ever hold, and felling is the operation that kills people who have done it a hundred times. If you have never felled a tree, learn on the ground with somebody who has — this book teaches the method, not the trade.

Never fell alone. Two people, minimum. The second person is the lookout, the caller, and the one who goes for help. Their one job during the fall is to be out of the fall.

Wear the kit. Chainsaw-protective chaps or trousers, a helmet with a mesh visor and ear defenders, gloves, and boots. Every one of them exists because a specific accident already happened to somebody without it.

The escape route is walked, not imagined. Before the saw starts, walk the route back and to the side, opposite the planned fall, and clear it. A tree falls in seconds and the escape takes seconds; the route being pre-walked is the entire margin.

Read the bend before you cut a downed trunk. A log lying between two supports is under tension on one side and compression on the other. Cut the side in tension first, or cut from the outside of the bend inward; cutting the compressed side pinches the bar.

Never stand under a hanging load. Not the frame on the winch, not a limb under a trunk, not for one second — including the moment you most want to, which is when it is working.

And this book is not an engineer's stamp. Nothing here has been checked by a licensed structural engineer. Where a figure depends on species, grade or moisture the book says so and tells you who to ask instead of guessing; where a load figure matters, it names the number as an engineer's job and leaves it there.

What it is not

It is not an engineer's stamp — the sentence above says why, and the title page says it at length. Nothing here has been checked by a licensed structural engineer, and where a figure depends on species, grade or moisture this book says so and tells you who to ask instead of guessing.

It is not a code compliance document. Your jurisdiction has opinions about round buildings and they are not in here.

And it is not a promise that you will do this in a fortnight. I did, on the second attempt, having already made every mistake the story chapters record. The forty hours is the honest claim — the frame, split wedges to standing — and the calendar around it is yours: a fortnight of part days, or a long weekend of full ones, with the skin, the floor and the ground added later, in any order.

PART 1

How to Build One

“Your skull is not symmetrical either, and it protects the brain just fine. The frame is the part you are allowed to get wrong. The skin and the ground are not.”

The method, in the order it is actually done: what a dome is, why the shape carries load, how three trees become forty panels, and the forty hours that raise those panels into a cabin winched between the trees -- with the skin, the floor and the ground added later, in any order.

CHAPTER 1 · STORY

The Tree That Was Already Down

A windfall pine, a small saw, and a question nobody had asked out loud

This book starts with one specific tree, because every number after this page descends from it, and because the story does not make sense if the tree stays abstract.

A pine came down in a storm on land I could walk to. Not a tree anybody planted for the purpose. Not a timber sale. A windfall: sixty or so feet of pine that had stood in a line of others, gone over in one night, root plate up, crown in the brush, and the trunk lying where it fell with the ground still pressed into its bark.

The useful part of a windfall like that is the straight part between the root flare and the first big limb. On this tree that ran 48 feet: 12 inches across at the butt, 8 at the top. The flare below it and the crown above it were firewood. In the middle was a cylinder of pine the length of a shipping container, and there was no way to use it.

That sentence is the whole book in miniature. The wood was not missing, rotten, or in some faraway forest. It was on the ground, within carrying distance of where the building now stands. What was missing was every machine, service, and middleman that stands between a fallen trunk and a usable stick of wood — and it is those, not the tree, that decide what a piece of wood is worth. Chapter 2 walks that chain one step at a time. This chapter is about the tree at the near end of it, before anybody got paid.

Sixty feet of pine, and no way to use it

Here is the problem stated plainly, because most of the ways it gets solved are ways of moving the problem somewhere more expensive.

A standing tree is worth almost nothing until somebody converts it. A windfall is worth even less: it cannot wait for a forester, it will not be part of any sale, and it will rot where it lies. Convert it the normal way and the trunk has to travel — dragged to a landing, loaded, trucked to a mill, and then hauled again as boards to wherever the building is. Every one of those journeys is paid by the mile

and the hour, and a mill will not touch a three-tree job without a setup charge that makes the exercise pointless. The honest summary of the conventional route is that the tree was worth less than the diesel to move it, which is why the tree stayed where it fell.

The saw that changed that was a 38.2-cc chainsaw with a 16-inch bar that cost 199 dollars on a shelf. The whole shelf price of the machine that eventually turned this tree into a building was less than what hauling the trunk to somebody else's mill would have cost before a single board was cut — not an exaggeration, a comparison, and it is the comparison the method lives on. The industrial route charges for handling, and handling is most of the route. A saw you already own, on the site where the wood already lies, removes the route and charges only fuel, a chain, and time.

But on the day I am describing, none of that was obvious. What was obvious was a 48-foot trunk that no mill wanted, cut from a tree no buyer would bother with, on ground nobody was being paid to stand on. The saw sat on its shelf because a saw alone does not make the trunk usable. A saw makes cuts. What I did not yet know was which cut made the tree stop being a problem and start being a frame.

Measured properly, the trunk held 318 board feet of solid wood. That figure is not a boast; it is what the tape said, and the way the book counts it is explained in full in Chapter 13. The same trunk, sawn into two-by-fours the way a mill would, keeps about 45 per cent of itself as saleable lumber. Split the way this book will split it, it keeps 87.4 per cent — 1.93 times as much — without a second machine ever touching it. All of that is later. Here it only matters as the reason the question in the next paragraph was worth asking, because the difference between 45 and 87.4 per cent of a 318-board-foot trunk is the difference between a load of firewood and a building.

The tree, measured

The drawing matters more than it looks. Every chapter that follows — the recovery arithmetic, the two sizing methods, the cut list, the fortnight — takes its numbers from this same trunk, and if the book ever disagrees with itself about the tree, the disagreement is a bug, not an editorial position. So the tree gets its own page, drawn to scale, with the three measurements the whole method hangs on.

The butt. 12 inches. The widest point that still counts as trunk; below it is the flare, which splits badly and gets bucked off.

The top. 8 inches, where the straight stem runs out at the first limb. Above it is crown, which is firewood.

The usable length. 48 feet between the two. A trunk tapers from 12 to 8 inches along that run, and everything about how the log gets cut is decided by that taper: buck it into 6-foot sections and each section is close enough to a cylinder that one jig fits them all.

Those are the only three numbers a tree-first build needs from the tree. Not species, not grade, not a timber buyer's walkaround. Butt, top, length. Chapter 22 turns exactly these three into a dome diameter, and the round trip in Chapter 23 proves it is the same calculation a designer runs from the other direction. The drawing here is not decoration. It is the input.

What I actually wanted

It is worth saying what the tree was *for*, because the method is only interesting against the thing it was asked to build.

I wanted a round building, big enough to live in, built alone.

Round, not square: a dome, 21.6 feet across with 365 square feet of floor when the arithmetic was done — a serious room rather than a shed, and the largest size these trees would fill. Chapter 22 shows that 3 trees are what it takes, and that the frame only actually consumes 1.88 of them; the rest of the pile is spares, blocking, floor stock and firewood, and the book says so rather than hiding it.

Built alone, because that is the constraint that shaped every decision after it. A sixteen-foot log section is a two-person lift; a six-foot one is one person's job. So the members are 6 feet, the panels are lifted one at a time, and the whole frame is designed around what a single person can carry, set, and fasten without a second pair of hands. The book never pretends this is the fastest way to build. It is the way that works when the only labour budget is your own, and the fortnight in the middle of this part is what that constraint actually costs.

And from what was on the ground, because the economics only work that way. Buy the timber and the frame costs what timber costs; Chapter 46 does that arithmetic both ways and prints the number that does not help the argument. Here it is enough to say the goal was never to build cheaply in money alone. It was to build with the tree that was already down, one saw, and two weeks — and to find out what that made possible that the lumberyard price could not.

One sentence of provenance, so the book cannot contradict itself later: the windfall above is where the method was found, and its three measurements are the ones every table in this book runs on. The frame the fortnight raises was cut from 3 trees felled for the purpose — Chapter 27 is that story, miss and all — measured to the same three figures. The tree that fell in the storm taught the method; the three that were felled became the building.

That is the book. The rest of it is the tree answering.

Fifteen Middlemen

Why a two-by-four costs what it costs, and how much of that is wood

Between a standing tree and a board on a rack there are, depending on how you slice the chain, something like fifteen separate businesses, machines, and handoffs — and every one of them gets paid. The board does not cost what the tree costs. The board costs what the tree costs *plus fifteen wages, fifteen fuel bills, fifteen margins and fifteen sets of handling*, and almost none of that is wood.

This chapter walks the chain, because the whole wedge method is a way of skipping it, and you cannot skip something intelligently without first knowing what it is.

The claim to hold in your head while reading it: a two-by-four's price is mostly distance and patience. The tree spent a lifetime standing still; the board has to be moved, processed, dried, graded, and waited on, and each of those is a service somebody sells. The mill route runs 14 operations from standing tree to graded board; the wedge route runs 8 from standing tree to member in the frame. The two share only 3 — the fell, the buck and the dry — and every one of the mill's other 11 is a service the wedge never buys. That is not a metaphor. It is the count, and the rest of the book is the consequence of it.

From stump to rack

Walk it once, end to end, and count the hands.

Fell. Somebody drops the tree. Trained, insured, standing in the weather.

Skid. The trunk is dragged to a landing where a truck can reach it.

Haul. A log truck moves it to the mill, paid by the mile and the hour, and the round trip is empty one way.

Scale. Somebody measures and grades the log, because the mill buys by volume and quality, and the measurement is a negotiation.

Saw. The mill breaks it down: slabs off the round, boards out of the cants. This is the step people picture, and it is one step of 14.

Edge. Every board gets its wany edges ripped off square.

Trim. Ends squared to length.

Dry. The boards are stickered and air-dried, or kilned — weeks to months of time, and the kiln burns fuel.

Plane. The rough board is dressed to its finished size. A nominal two-by- four has now become a dressed one-and-a-half by three-and-a-half, and the missing wood is not an accident; it is this step.

Grade. Somebody looks at every board and assigns it a grade, because graded lumber can be sold by phone and ungraded cannot.

Bundle, ship, stock. It is unitised, moved again, and shelved in a yard that rents its ground, its racks and its fork trucks.

Sell. A desk takes an order and a margin.

That is the honest list — 14 operations between the tree and the graded board, and every one of them has fuel, wages, rent and a margin in it. None of them is a scam. Each one exists because the customer for a board is a contractor forty miles from the forest who needs a straight, dry, graded, predictable rectangle *tomorrow*, and the chain delivers exactly that. The chain is good at its job. Its job is just mostly not wood.

Now count the method this book uses, in the same words the diagram uses: fell, buck, then the three splits that make a wedge — halve, quarter, eighth — then dry, jig and cut. 8 operations from standing tree to member ready for the frame. The middle of the mill list — limb, haul, set up, slab, rotate, cant, resaw, edge, trim, sort, plane — is gone, not because it was unfair but because it was unnecessary. In its place are the 5 the wedge actually needs: the three splits, the jig and the end cuts. The wedge does not need the tree to become a rectangle, so it does not need the machinery whose entire purpose is producing rectangles.

The chain, and where the wood goes

The drawing tells the story the walk could not, because the story is not only who gets paid. It is also what disappears.

Every processing step removes material, and the industrial chain removes it at both ends and in the middle. The slabs and edgings off the round log. The planer shavings off the dressed board. The trim cuts. The degrade — boards that crack, warp or stain while drying and get down-graded or scrapped. When the same trunk this book uses is sawn into two-by-fours, roughly 45 per cent of the solid wood survives as saleable lumber. Split into wedges instead, the same log keeps 87.4 per cent of itself — only the kerf is lost. Chapter 13 does that arithmetic in full, including the parts that are less flattering than the headline. Here it belongs in the chain picture for a simpler reason: the waste is not an accident of carelessness. It is the price of the rectangle, paid in wood, at a chain of stations that each exists to pay it.

Look at the diagram with that in mind. The industrial route is long because each station exists to solve a problem the previous station created. The sawmill exists to cut rectangles, so the edger exists to clean their edges, so the planer exists to smooth them, so the kiln exists to dry them flat, so the grader exists to sort what all of that did to the wood. The wedge route is short because it never creates the rectangle problem. There is nothing to edge, nothing to plane flat, and a green wedge that dries is allowed to move a little, because the frame was designed for a stick that is already the shape the tree made.

The price of a 2x4, taken apart

The shelf price this book uses is 8.88 dollars for a sixteen-foot two-by-four — a real sticker price, declared up front in the constants table, because it is a price and prices are looked up, not derived. That table prints in the front matter and is repeated in the master tables at the back.

Now take it apart, and the first thing to notice is how little of it is the tree.

The tree's share of that board, at the stump, is a fraction of the sticker. A sixteen-foot two-by-four is nominally 8.00 square inches by sixteen feet — a little under eleven board feet of wood. At a stumpage price of a few cents a board foot standing, the wood in that board left the forest worth pocket change, and arrived on the rack worth 8.88 dollars. Everything in between — the fell, the haul, the sawing, the drying, the planing, the grading, the shipping, the yard, the desk — is what the price actually buys. The wood is the smallest line on the receipt.

That is not a complaint about the lumber industry. It is a measurement of what the industry is: a logistics and processing business that happens to have trees as its raw material. And it is the exact reason the wedge method works economically. When you stand next to a tree that is already down, with a saw in your hand, you are not competing with the industry's skill. You are simply not buying its

product. The 14 steps are replaced by 8, and the price of the building material collapses to fuel, chain oil, and your own hours.

Chapter 46 does that arithmetic both ways — what the frame costs bought, against what it costs cut — and prints the hourly rate it implies, including the version that flatters the method least. This chapter only needed to establish why the two numbers are so far apart. They are far apart because one of them is a price for wood and the other is a price for 11 services, and the services are what the saw on your shelf makes unnecessary.

Nominal, dressed, and the missing third

A two-by-four is not two inches by four inches. It is sold as if it were, and the difference matters enough that the book keeps both numbers on the page wherever wood is counted.

The nominal board — the name on the ticket — is two by four: 8.00 square inches of section.

The dressed board — the thing you actually carry out of the yard — is one and a half by three and a half: 5.25 square inches. The rest went to the planer.

That is a thirty-five per cent difference in section between what the lumber is called and what it is, and it is not a trick; it is the planing step from the chain above, made visible. The book always says which board it means when it compares, because comparing a wedge against a nominal two-by-four flatters the wedge by a third, and a book that flatters its own material by accident is not worth reading. There is also a third board the industry does not sell and the book does not pretend to: the rough-sawn two-by-four, planer-skipped, at something between the two numbers. The money chapter (Chapter 46) and the head-to-head in Chapter 7 both state their board explicitly, and where the choice matters they print the comparison against both.

Why the book cares this much: because the wedge's honest number — the figure that decides whether the whole idea is worth anything — changes by that thirty-five per cent depending on which board you compare it against. The book's answer, stated once here and used everywhere after: print both, and lead with the one that flatters the method less. Against the nominal board — the name on the ticket, the larger section — the wedge replaces fewer boards and the comparison is conservative; against the dressed board — the smaller thing you would actually have bought — the wedge replaces more boards and the comparison flatters. Chapter 7 prints both numbers side by side and says which is which; the honest headline is the nominal one.

CHAPTER 3 · STORY

The Frankendome

Forty triangles, no two struts alike, and the failure that taught the method

Before the dome this book is about, there was another dome, and it matters that it existed, because every decision in the method that followed was a reaction to what that first one got wrong.

It was built from whatever wood was lying around — offcuts, salvaged sticks, mixed sections, boards that had been other things. Forty triangles, joined edge to edge, each strut whatever shape the pile happened to supply. The idea was deliberately forgiving: do not sort, do not select, do not machine. Let the connector absorb the difference between one stick and the next.

The idea worked, which is the worst thing that can happen to an idea that is secretly bad. The frame stood. It stood for six months, outside, through weather, and it taught more about how a dome behaves than any drawing had. And it was awful: lumpy, twisting, no two joints alike, and impossible to reproduce. Every lesson in this book is downstream of that specific combination — *it worked, and it was awful* — because it proved both halves of the method at once. A triangulated frame will stand on almost any consistent-ish sticks, which is the permission. And nothing about a frame built that way repeats, which is the trap.

This chapter is the story of the trap.

Strut-agnostic, and what that cost

The design rule of the first build was *strut-agnostic*: no stick is special, no stick is measured, any piece of wood of roughly the right length can be any member. Assemble the triangles from the top of the dome down, one at a time, and wherever two edges meet, a folded bracket joins them. Eight screws a bracket. Two bolts a seam. Tighten, move on.

The appeal was real. The pile was free. Nothing needed grading. A stick that was two inches wide where the next was three could still be a strut, because the bracket's flaps could be bent to whatever the wood wanted, and the screws found the wood where the wood was. Building that way feels like

freedom — every decision made at the moment of assembly, by the piece in your hand — and it took the whole build to learn what that freedom cost.

It cost everything that comes after a build. The dome stood, but:

Nothing repeated. No two members were alike, so no operation could be jigged. Every cut, every hole, every bracket bend was a one-off, made to fit the exact stick in hand. The fortieth panel took as much thinking as the first.

The connector did all the work. Because the sticks agreed about nothing, the bracket had to absorb every difference: width, thickness, angle, wander. That is a lot to ask of a folded piece of metal, in forty places, for the life of the building — and Chapter 4 is the afternoon I spent learning exactly how much.

It could not be drawn. There was no cut list, because there was no standard member. The frame existed only as itself; you could not hand the build to anyone, and you could not build a second one.

It was never finished. A frame that is improvised joint by joint is a frame you keep adjusting. It stood, but it never stopped being the thing you had to keep an eye on.

The picture on the next page is that dome. It is in this book because the book's method is best understood as the answer to it, and because it is honest to show the before. A book about a better way to build owes the reader the worse way it is better than.

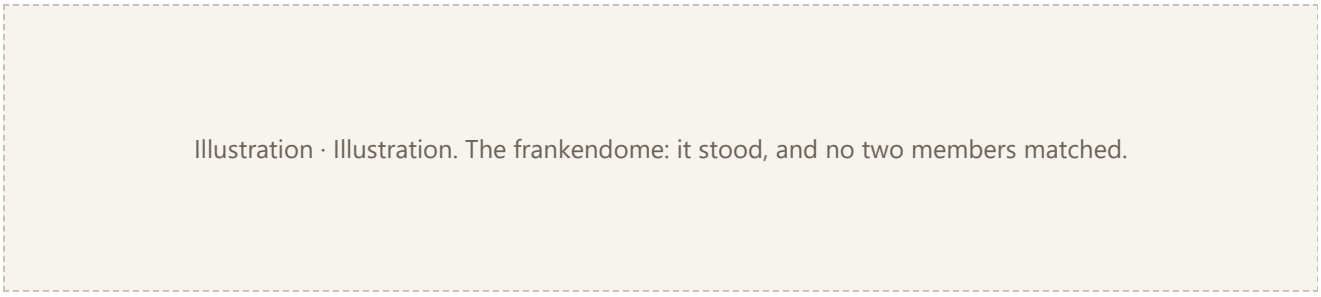


Illustration. The frankendome: it stood, and no two members matched.

The frankendome standing

Look at it. Lumpy where a solved dome is smooth. Every member its own section, its own bow, its own idea of straight. The seams pinch and gap where the sticks disagreed. And yet the shape is recognisably a dome, and it is standing, which is the part that is easy to forget when the tidy version of this story gets told: the thing did work. A frame of forty triangles, joined any-old-how, will carry its

own weight and a skin. Triangulation is that forgiving — Chapter 6 explains why — and the method in this book inherits that forgiveness as its foundation.

What the picture cannot show is the labour inside it. Ten days to stand it, not because the work was hard but because none of it repeated. Every bracket was folded to a different angle. Every screw row was laid out fresh. There was never a second dome; there was no first dome either — only a sequence of one-offs.

The wedge method exists because of what this picture makes obvious: the geometry of a dome does not care what shape its sticks are, but the person building it absolutely does, and the difference between a frame you build once and a method you can build with is whether anything in it repeats.

The ten-piece decagon

One part of the frankendome was genuinely good, and it is worth separating it from the rest, because it was the only piece carried forward.

The base — the ring where the dome meets the ground — was not improvised. It was ten pieces, cut to a plan, joined into a decagon, and it went together in an afternoon and never moved. The contrast with the chaos above it was the whole lesson of the build in miniature: the one part that was drawn before it was cut, the one part where every piece had a name and a number, was the one part that behaved.

That decagon became the footing of the method. The frame this book builds also meets the ground as a ten-sided ring — 10 rim edges, each a member in its own right — and the ring is the one place the book tells you to set out with a string and a check rather than a shrug. Chapter 39 is that ring, given the treatment the frankendome proved it deserved.

The discipline behind that ring came forward too. The frankendome's triangles were joined edge to edge in whatever arrangement the assembly order produced; the method's panels are assembled into the same two triangle shapes every dome has — the same count, the same shapes, the same corners. What changed is not the geometry. It is that in the method, every panel of a given shape is built on the same fixture, from sticks of the same section, with the same holes. The frankendome proved the shape. The method made the shape repeatable.

Why I stopped

The frankendome was not abandoned because it fell down. It was abandoned because of a sentence I could not unthink, after watching the tenth improvised joint go in:

A thing you build twice is only worth doing by hand once.

Nothing in that dome repeated, so nothing could be jigged, so nothing could get faster, and nothing could get more accurate, and no second dome could ever be the same as the first. The strut-agnostic idea had removed the need for precision at the cost of removing the possibility of it. Every bracket was a small negotiation between two sticks that had never met and never would again. That is not a build method. It is a sculpture process that happens to produce a room.

The question the wedge method answers is the question the frankendome left on the table: what would the sticks have to look like for one fixture to build every panel? The answer turned out to be in the next chapter — not *which sticks to buy*, but *which way to open the log*. And the moment that answer landed, the frankendome stopped being a failed building and became what it is in this book: the control group. Same dome. Different pile.

Chapter 5 is where the pile changed.

CHAPTER 4 · STORY

The V-Bracket Afternoon

Trying to join shapes that have nothing in common

Between the frankendome and the wedge there was an afternoon spent trying to make the frankendome's problem go away by making a better bracket. That afternoon matters because it is the step most builders would have stopped at, and because the bracket it produced is a genuinely good answer to the wrong question.

The problem was stated clearly by then: the frame's sticks did not match, so the joint had to. The fix seemed obvious — stop improvising the joints one at a time and design one connector good enough to join any stick to any stick. Do the tolerance in metal, once, instead of in thinking, forty times.

So I sat down and designed the V-bracket: two flat flaps meeting at the angle the panels make, four holes in each flap, the flaps long enough to run half the length of a member so the screws spread along the stick instead of bunching at its end. Fold one from flat stock in a minute on a bench brake, and it will hold any two pieces of wood that arrive at the corner, whatever their sections, whatever their bows.

And it worked. That is the trap, again. A bracket that absorbs every difference works on the first panel and the fortieth. What it costs does not show up until you look at what the bracket has become.

A bracket that does not care

Here is why the V shape is good at its job, because the goodness is real and the method that follows borrows from it.

A V-bracket does not assume a member's width. The flaps are wider than any stick the build will use, so a two-inch board and a three-inch board both land on the same plate, and the screw rows find whatever wood is there. It does not assume a thickness: the bend that forms the V is made to the panel's fold angle at assembly, not stamped to a tolerance at a factory. And it does not assume straightness: the screws are what pull the stick to the plate, and a bowed stick just gets pulled harder.

That is a beautiful property, and it is worth saying what it is in one sentence, because the wedge method is, in a way, this sentence taken to its conclusion: **the connector absorbs the difference, so the wood does not have to be made the same.**

The warning sign was in the same sentence. When the connector absorbs every difference, the connector is now doing all the work of the frame. Every joint is a negotiation, held by screws that are the only thing in the whole structure that know the shape they are enforcing. Eight screws a bracket, two bolts a seam, all of it site-bent, site-drilled, site-tightened — the connector was not a detail. It had become the building.

And a connector that does everything has to be *right* everywhere. The flaps were long, which spread the load, which was honest engineering — and it meant forty-odd brackets, each consuming stock, each needing folding, drilling, fitting, checking. The frankendome had replaced the problem of matching wood with the problem of manufacturing forgiveness, in metal, at forty corners. Same labour, moved.

Four joints that nearly worked

The drawing shows the four joints that happened along the way, because the V-bracket was not arrived at in one step, and each of the earlier ones failed in a way that named a piece of the final answer.

The hub. A flat plate with every member bolting to its own face — the classic geodesic hub. It assumes all members arrive at the same point at the same angle, and they never do when the sticks are salvaged. Every attempt needed shims, washers, longer bolts. The hub is perfect for a kit of identical struts, and useless the moment the struts differ.

The angle plate. A single fold, made to the panel's angle. It joins two sticks in a line but cannot hold the third that closes the triangle, and the third is the one that turns three sticks into structure.

The wrap. A strip of steel wrapped around both members and screwed through. It holds anything, for a while, but it crushes nothing into alignment — the joint stays as crooked as the wood, and every wrap is a one-off that has to be cut to the pair it is joining.

The V-bracket. The survivor, and it failed too — not at holding, but at repeating. It can join anything to anything, which is exactly why no two of them are alike, which is exactly why the frame never gets faster to build.

The pattern across all four is the same. Each joint failed not by being weak but by being asked to solve, in metal, a problem the wood kept re-creating. The hub assumed sameness. The plate assumed a line. The wrap assumed nothing and therefore had to be made for everything. The V accepted all of it and paid in labour. None of them asked the question the next chapter does — not *how do I join whatever arrives*, but *what would have to be true for the sticks to arrive the same.*

The question that ended it

The afternoon ended with the bracket in my hand doing its job, and a question that would not go away. The bracket worked by absorbing difference, and it had taken me all afternoon to make a bracket good at that. But every flap length, every screw, every bend was the cost of a difference that did not need to exist.

So: if the bracket no longer had to absorb any difference — if the sticks arrived at the corner already the same section, already at the same angle, already the same length — what would the joint collapse to?

The answer, worked out later, is Chapter 15: a member end landing on a member side, one key, a few screws, nothing folded, nothing measured twice. But the more important answer arrived that same afternoon, and it is the sentence that turned the project around:

Do not build a joint that tolerates the difference. Build sticks that do not have one.

That is the wedge method's founding sentence. The bracket was a good idea pushed past its natural size; it remains in the method only as a small, fixed part where a fold is genuinely the right tool. What it taught was that the wood was the thing to fix, and the wood was the one thing I had not tried changing — because a stick of wood seemed like something you buy, not something you decide. Chapter 5 is the afternoon that stopped being true.

The bracket attempts earned their page in this book for one more reason: they are the roadmap of every wrong turn a builder can make at a dome's corner. Somebody, somewhere, is designing a hub right now. The book's answer is the drawing above, and the sentence under it: make the sticks match, and the joint stops being the hard part.

CHAPTER 5 · STORY

The Cut That Changed It

Split the trunk like a cake, and the stick is already the right shape

The Cut That Changed It

I want to tell this one without hindsight tidying it up, because the tidy version — *I realised the dome would accept any consistent section, therefore I split the log radially* — is not what happened and it teaches nothing. What happened was that I was tired of brackets.

Half, half, half again

The frankendome had taught me that a frame of forty triangles will stand up even if no two of its sticks match. The V-bracket afternoon had taught me that making a connector tolerant enough to join anything to anything means the connector is now doing all the work, forever, in forty places.

So I had stopped asking how to join dissimilar sticks and started asking the opposite question: what would it take for them not to be dissimilar?

The answer that everybody gives is *mill them*. Square them up, run them through a planer, and now they are all 2x4s. That answer costs a mill, a planer, a shed to keep them in, and about half the tree.

I was sitting on a section of trunk at the time. Not thinking about it. Sitting on it.

And the thing about a round section of trunk is that it is already rotationally symmetric. Every radius is the same as every other radius. If you split it through the middle you get two identical halves, not because you were careful but because the log was round before you touched it. Split those again and you get four. Split those again and you get 8.

Three passes. Eight sticks. Every one the same, to whatever tolerance the tree was round to, which is a tolerance nobody had to hold and nobody had to check.

I did it that afternoon on a piece I had been about to burn.

What became obvious immediately

Four things, in about a minute, and I want to record that they were obvious because the ones in the next section took a week and I do not want to flatter the story.

One section, everywhere. Every stick in the frame has the same cross-section: 3.83 inches across the bark face, 5.00 inches from pith to bark, 9.82 square inches of wood. So one jig fits all of them. One fixture, one set of stops, one procedure repeated a hundred and twenty times. That is what makes this a fortnight instead of a summer.

Depth where you want it. A sector is deep — it runs the whole way from the pith to the bark — and depth is where bending stiffness lives. The stick is not a compromise between what the tree gave and what a mill wanted. It is the deepest member you can cut from that log, which is a strange thing to get for free.

No machine between the tree and the frame. Fell, buck, split, build. The chainsaw does all of it. There is no second machine, no second setup, no second building to keep the second machine in.

Four ways up. This one I noticed while stacking them. The sector is not symmetric about its own long axis — it has a point and a curved back — so you can rotate it four sensible ways in the wall and get four different things: which face the weather sees, which face the next stick lands on, whether the points of a pair aim together or apart. That is Chapter 14, and it is the most genuinely new thing in the method.

And what took another week

Now the part the tidy version leaves out.

I assumed, because the sticks were identical and the panels were identical, that the *joints* would be identical. Forty triangles, all the same, all meeting each other the same way.

They do not.

A dome is a sphere approximated by flat panels, and the angle at which two panels meet is not the same everywhere on it. It cannot be — the sphere curves differently in different places, and a 2V hemisphere has two classes of edge for exactly that reason. So the seam between two panels folds by one amount here and a different amount there, and a key cut to fit one seam will not fit the other.

I did not want this to be true. I spent the better part of a week trying to make it not be true: fudging the panel outlines, trying a compressible gasket, trying to talk myself into the idea that the difference

was small enough to force closed. It is not small enough. Forty panels of *nearly* closing is a dome with a gap in it.

What actually resolved it was measuring instead of arguing. Across all 55 seams there are only **2** fold angles. Not forty. Not a continuum. Two: 18.03 degrees and 22.46.

That was the moment the method became buildable rather than merely elegant. 2 angles means 2 key profiles: cut one of each until it fits, then repeat it 25 and 30 times. The difference between "the seams vary" and "the seams take 2 values" is the difference between a research project and a fortnight, and I lost a week to not knowing which one I was in.

The chart is in Chapter 16, where it belongs.

The bit I got wrong in public

While all this was going on I was saying, out loud, in the project's own published claims, that nothing in the dome needed a mitre.

That is not true and it was never true. The end of each stick that lands on the side of the next one — the butt end — is a compound cut: a mitre and a bevel together. What is true, and what I should have been saying, is that no end is ever mitred *to another end*. Every joint is an end landing on a flat side, which means each stick is cut once, on one end, before it goes anywhere near the assembly, and the other end is left long and sawn off in place.

That is a much better claim than the one I made, and it is a shame I spent several months making the wrong one. Chapter 15 has the geometry, and the corrections live in the chapters where the claims were made, named rather than silently edited.

What it is worth

The number that convinced me it was not just a neat trick: splitting this way keeps 87.4 per cent of the solid wood in the trunk, against about 45 per cent for the same log sawn into two-by-fours the way a mill would. That is 1.93 times the usable wood, from a cheaper process, with one machine instead of four.

Chapter 13 does that arithmetic properly, including the parts of it that are less flattering than they first appeared.

But sitting on that log, what I had was a much simpler observation, and I still think it is the whole book:

A round thing splits into identical pieces for free. A square thing has to be made square, and you pay for that in wood, in machines, and in time. Domes are the buildings that will accept the free pieces.

Triangles Do Not Care What Shape Your Stick Is

Why a triangulated frame can accept a member no mill would sell

Triangles Do Not Care

This chapter is the permission slip for the whole method, so it goes first in the explanation strand and it is short enough to be unfair to nobody.

Start with the sentence the title page made its argument out of: **your skull is not symmetrical either, and it protects the brain just fine.** The frame is the part of this building you are allowed to get wrong — the lumpy, split, self-labelled frame that the frankendome proved will stand. The parts that must be right are the skin and the ground, and the book treats those with the care the title page promised. This chapter is the *reason* the frame is allowed to be lumpy, stated as geometry rather than as reassurance.

The wedge method proposes to build a building's frame out of sticks that are not rectangular — one eighth of a round log, sawn faces, bark back, a point at one edge. No lumberyard sells that. No building code assumes it. And yet the frame stands, because of one property of triangles that is easy to state and takes the rest of this chapter to spend:

A triangle does not care what shape its sticks are. It only cares how long they are, and that they reach each other.

Everything the mill does — square the edges, flatten the faces, dress the section to a standard — exists because rectangles ask their pieces to be beams: flat boards, loaded across their faces, whose stiffness and strength depend on the exact shape of the section. A triangulated frame asks something else entirely of its pieces. It asks them to be *struts*: pushed or pulled along their own length, ends only, never sideways, never flat. For a strut, what matters is how much wood is in the section and how far the load is from pith to bark — not whether the section is a rectangle.

So the central question of the book is not “is a wedge as good as a 2x4?” in general — Chapter 7 runs that comparison honestly and prints the column the wedge loses. The right question is narrower and much more useful: *inside a triangulated frame, where members are loaded along their length, is a wedge a good member?* This chapter builds the frame that question assumes, so the later chapters can answer it.

Axial, not bending

The difference between a strut and a beam is the difference between the two ways a building can load a piece of wood, and most of conventional framing is the second way.

A beam is loaded across its length. A floor joist carries the floor on its narrow face; a rafter carries the roof on its wide one; a stud carries weight down its length but braces against sideways push. In all of them the load is trying to bend the piece, and bending is cruel: the material at the middle of the section does almost nothing, which is why an I-beam has its steel at the top and bottom and air in between. Bending strength lives in the extreme fibres, far from the section’s centre, so the shape of the section — rectangle, and which way round — decides how strong the piece is. This is why a 2x4 on edge is dramatically stiffer than the same 2x4 laid flat, and why the mill bothers to dress boards to a true rectangle at all. A rectangular building is a system for turning wood into beams, and beams need rectangles.

A strut is loaded along its length. Squeeze the ends of a stick toward each other, or pull them apart, and every fibre of the section shares the load equally. Shape stops mattering; the amount of wood matters. A round stick, a square stick, a wedge — for axial load, they differ only in section area (and, for a long thin strut in compression, in the way the section resists buckling, where depth matters more than shape).

Now look at a triangulated frame. Every load that arrives at a joint is carried to the next joint along the members, as push or pull along their lengths. The frame does not bend its members; it moves force through them. That is the entire trick of geodesic structures, and it is why a dome’s struts can be slender for their span, and why the frame stands up without a single piece of sheet material keeping it square. A rectangle needs a diagonal, or a sheet of plywood, or it racks over and lies down. A triangle is already its own diagonal. The drawing on the next page is that sentence.

The same load, two frames

Look at the two frames side by side.

On the left, a rectangle. Push on its top corner the way wind pushes on a wall, and the rectangle does what rectangles do: the corners stop being ninety degrees, the frame leans, and unless a sheet of plywood or a diagonal brace is holding it, it goes over. The joints are doing everything the shape cannot. That is why a conventional wall is a frame *plus* sheathing — the studs are beams, and the sheet is the triangle they never built.

On the right, the same corner, triangulated. Push on it. The load cannot change the angles, because the angles are fixed by the lengths of the three sticks, and a stick's length cannot change without stretching or squashing the stick itself. So the push travels down the members as pure compression and pure tension, along the grain, exactly where wood is strongest and exactly where shape does not matter. Nothing bends. Nothing needs a sheet. The frame is its own bracing, at every corner, forever.

The whole method follows from that picture. Because members are loaded axially, the frame will accept a member of any section that has enough area and enough depth — which is precisely what a split log provides. Because nothing bends, the frame does not ask for the flat, dressed, graded faces a beam economy demands. And because the triangle's angles are fixed by its lengths, the frame enforces its own geometry: get the lengths right and the shape has no choice. The frankendome stood on that fact alone, with no two members matching, and it stands on the same fact with every member identical. Chapter 3 was the demonstration. This chapter is the reason.

What the wedge is good at

If the frame wants area and depth, the wedge should be measured by area and depth — and on a modest log, it beats the store-bought board at both.

A dressed two-by-four carries 5.25 square inches of wood, planed down from its nominal size — a rectangle, because it must be a beam. The wedge this book cuts is 3.83 inches across its bark face and 5.00 inches from its point to its bark, for 9.82 square inches of section. The comparison that matters is in Chapter 7, computed properly on an 8-inch log so the wedge gets no advantage from a fat tree. Here, two qualitative points are enough to set the direction:

Depth is where stiffness lives. In bending, stiffness rises with the *cube* of depth; double the depth and a beam gets eight times stiffer. A wedge runs from the pith to the bark — the full radius of the log. For the same width of stick, that is more depth than any board a mill cuts from the same tree, because the mill's rectangle has to fit inside the circle, and the wedge does not.

Area is where axial strength lives. The wedge is a sector: one eighth of the log's circle. Measured as a strut — which is what the frame asks it to be — it carries more wood in its section than the dressed board the same log would have produced, because it never gave up the corners to the edger

and the planer. Chapter 13 counts those corners; the count is the difference between 87.4 per cent and 45 per cent of the tree.

The honest note, saved for the full table: a wedge is not a better *beam*. Lay it flat and ask it to be a shelf and the missing rectangle matters. The claim is narrower and it is the only one the book makes — in this frame, loaded this way, the wedge is a better use of the tree than the board, and the shape the tree already had is not a defect the mill forgot to remove. It is the feature the frame was waiting for.

One eighth of a tree

The section, stated plainly, because the next chapters argue with it:

A 6-foot section of this book's tree, split into 8 equal sectors, yields sticks of 9.82 square inches each — 3.83 inches across the bark face, 5.00 inches deep from point to bark.

The dressed two-by-four the same tree would have been sold as carries 5.25 square inches. The nominal one — the name on the ticket, which no longer exists in the wood — carries 8.00.

So the wedge is a bigger section than the board the mill would have cut from the same log, in the one measure that matters for a strut, and it cost a chainsaw's fuel instead of a mill, a kiln, a planer and a yard. That single line — 9.82 against 5.25, from the same tree — is the economic engine of the whole book, and everything after this page is the machinery that turns it into a standing frame.

Not a Worse Two-by-Four

The head-to-head nobody runs honestly, run honestly

Not a Worse Two-by-Four

Everybody asks the same question, and they are right to ask it.

Is a split log actually as good as a board?

The honest answer has three parts, and only two of them are flattering. This chapter gives all three, because a method defended only with its good numbers is a method nobody should trust with a roof over their head.

The rectangle is not the tree's idea

Modern construction has trained us to think of wood as something that naturally arrives in rectangles. A tree goes into a sawmill and a stack of 2x4s comes out, and the form is so familiar that it stops looking like a choice.

It is a choice. The rectangle is not a property of wood. It is the output of a manufacturing process designed around a particular kind of building — one made of vertical studs, horizontal plates, flat walls, right-angle corners and rectangular openings. For that building the rectangle is close to perfect, and the industry that produces it is very good indeed at what it does.

A geodesic dome is almost the opposite of that building. Its structural language is triangles, changing face planes, radial geometry and repeated angular joints. Nothing in it is a right angle. Nothing in it is a flat wall.

So the question reopens, and it is worth asking properly rather than assuming the answer in either direction: *for this building, is the rectangle still the right thing to make out of the tree?*

More wood, from a modest tree

Start with the simplest measure: how much wood is actually in the stick.

Take an 8-inch log — not a monster, a tree most people could fell and move without help. Split it 8 ways and each sector holds **6.28 square inches** of wood.

A dressed 2x4 — the board actually on the rack, which is 1.5 x 3.5 inches whatever it says on the label — holds **5.25**.

That is **+20 per cent more wood** in the wedge, from a tree you could carry out of a wood lot in sections.

And it is the conservative version of the claim. A larger log widens the gap sharply, because area grows with the square of the radius while a 2x4 stays a 2x4. The book quotes the small log deliberately: a claim is worth more made at its weakest.

The wedge is also deeper. Pith to bark is 4.00 inches, against 1.5 x 3.5 for the board — and depth is where bending resistance comes from.

And now the number that does not help

Here is where an honest chapter parts company with a sales pitch.

Area is not strength. How much a section resists bending depends not only on how much wood is in it but on *where that wood sits* — material far from the middle does most of the work, material near the middle does almost none.

And a wedge puts a great deal of its wood near the pith, in the narrow part of the pie slice, where it contributes very little.

Run the numbers properly and the comparison comes out like this:

	wedge	2x4 on edge
Wood in the section	**+20%**	—
Resistance to bending (stiffness)	**-1%**	—
Bending strength	**-33%**	—

Read that middle row again. The wedge has +20 per cent more wood in it and is **not stiffer**. It comes out within a per cent of a 2x4 stood on edge — effectively identical, having spent +20 per cent more material to get there.

And the bottom row is worse: in bending *strength* the wedge is -33 per cent against the same board.

That is the number this method's advocates do not usually print, and it is printed here, in a table, near the front.

One consolation, before the argument

There is a caveat to the caveat, and it is a fair one.

The comparison above puts the 2x4 **on edge** — the orientation in which a board is strongest, and the way a carefully framed floor joist is installed. A great deal of real framing does not get that. Sheathing battens, blocking, purlins and plenty of cheerfully improvised structures end up loading boards the flat way, and against a 2x4 laid flat the wedge is **5.4 times stiffer**.

So the honest summary is: worse than a well-oriented board, dramatically better than a badly-oriented one.

That is still not the argument.

Which is the right question

The argument is that *the comparison itself is the wrong shape*.

Asking "is this wedge as good a beam as a 2x4?" assumes that being a beam is the job. In a rectangular building it usually is: a stud, a joist and a rafter all spend their lives resisting bending, largely alone.

A triangulated frame does not work that way. That is the entire point of triangulating it. A properly designed triangle converts an external load into forces running *along* its members — compression down one, tension along another — instead of asking any single member to bend. The network carries the load; the sticks mostly just push and pull.

And for a member being pushed or pulled along its length, the property that matters is **cross-sectional area**. Not section modulus. Area.

Which is the column the wedge wins, by +20 per cent.

So the useful question is not:

> *Is this wedge as good a rectangular beam as a 2x4?*

but:

> *Is this wedge an effective axial member inside this triangulated system?*

And there the answer is straightforwardly yes, with more wood in the section than the board it replaces, from a tree that never went near a mill.

Mostly

One more piece of honesty, because “mostly axial” is doing real work in that paragraph.

Panels are not purely axial. Wind pushes on the skin between the members and snow sits on it, and those loads reach the frame as bending in the individual sticks. It is not the dominant case and it is not nothing, and it is exactly where the wedge’s -33 per cent deficit is a real deficit rather than an academic one.

Two things about that:

- The wedge is deep. 4.00 inches pith to bark is more depth than the board has, and depth is what resists the bending it does see — which is why the stiffness came out level rather than behind despite the shape being worse.
- Panel span is short. The distance between supports in a 2V panel is a few feet, not the fifteen a floor joist crosses, and bending demand falls away fast as spans shorten.

None of which is a substitute for somebody competent checking your particular building for your particular snow load. Chapter 10 says the same thing at more length and names who to ask.

What this chapter does not claim

It is worth fencing the argument, so nobody quotes it doing work it cannot do.

This chapter does **not** claim that a split-log wedge is better than a 2x4 as a wall stud, a floor joist, a roof rafter, a shelf, a header or a piece of furniture. For all of those the board wins, often easily, and

the reasons are in the table above.

It claims something narrower and much more defensible:

> *For a purpose-designed geodesic dome whose geometry, connectors and jigs are built around radially split timber, the wedge is a better use of the tree than first converting that same tree into ordinary dimensional lumber.*

That is the whole claim. It depends on the building being designed around the material from the beginning — which is what the rest of this book is about.

We are not asking an unconventional piece of wood to imitate a 2x4.

We are building something in which it does not have to.

Forty Panels, One Hundred and Twenty Members

The 2V hemisphere, counted

Forty Panels

This chapter names the frame itself, because every number after it — the cut list, the jig, the fortnight, the seams — is a count of something that exists on this frame, and the frame deserves to be introduced before it is counted.

It is a two-frequency geodesic dome: a hemisphere of 40 flat triangular panels, held up by 120 wooden members. There are 65 distinct edges in its geometry, of which 55 are seams where two panels meet and 10 run around the base where the dome meets the ground and join nothing. Those are the four numbers the whole book hangs on, and none of them was chosen. All of them fell out of the geometry — an icosahedron, subdivided once, projected onto a sphere, cut in half at its widest line — and the rest of this chapter shows where they come from, because a count you can retrace is a count you can trust, and a count you cannot is a story.

The frame in the picture is the actual solved dome this book builds — the same meshes the simulator solves, pushed apart along each panel's own normal so you can see that it really is forty separate triangles and not one continuous surface. That separation is not just a drawing trick. It is how the thing gets built, one finished triangle at a time, on the ground, and it is the subject of Chapter 18.

Where 2V comes from

The geometry is older than the dome and worth knowing once, because everything the frame does follows from it.

Start with an icosahedron: twenty identical equilateral triangles, twelve vertices, the roundest of the five Platonic solids. It is already close to a sphere, but its faces are big flat slabs. Subdivide: split every edge in half and connect the midpoints, and each of the twenty faces becomes four smaller triangles — eighty faces in all. Now the move that makes it a dome: push every vertex outward until it lies

exactly on a sphere of the same radius. The triangles stop being identical — those nearer the original vertices stretch differently from those near the midpoints — but they all lie on one spherical surface, which is what "geodesic" means here: the straight lines of the original faces become the chords of a sphere.

Cut that sphere at its equator and keep the top half. What remains is the frame: 40 triangles, because a full subdivided icosahedron has eighty faces and the hemisphere keeps half.

That one operation — subdivide once — is what "2V" names: the frequency two subdivision of the icosahedron. It is the smallest subdivision that produces a useful building, and its gift is the number in Chapter 9: all 65 edges of the hemisphere come in only two lengths, because the subdivision only ever produces two kinds of edge — the original icosahedron edges, now halved, and the new edges between midpoints. Two lengths, not forty, is what makes one person with one jig able to build this at all. Chapter 48 shows what higher frequencies buy and cost; the whole of the rest of this book stays at 2V, because 2V is where the trade stops being worth paying.

The frame, exploded

The plate is the frame with its 40 panels pushed apart along their own normals, each triangle floating a little way out from where it will finally sit. Read it as an inventory before you read it as a picture.

Forty triangles. Two shapes, in a fixed ratio: the triangles that sit around the five-fold points and the triangles that fill between them. Not forty shapes — two — which is the second great simplification of the frame, after the two lengths.

Three members each. Every triangle is built complete, with its own three sticks, its own joints, its own edge. Three times forty is 120: the member count of the whole frame, and the number the tree chapter will chase in wood.

The seams where they will meet. Where two panel edges lie against each other there will be a seam — a key between the two members, a gasket, a line on the outside of the shell. The frame has 55 of them. The 10 edges around the base are different: nothing meets them from outside, so they are rim, not seam, and the book keeps the two counts apart because a builder closing 55 seams must not be told to close 65.

What the frame contains

The table counts the frame two ways and they agree, which is the whole point of printing it.

By panels: 40 panels times 3 members each is 120 members.

By edges: 55 seams each holding two members — one from each neighbouring panel — plus 10 rim edges holding one, is 120 again.

The two routes give the same number because the frame is panelised: every panel owns all three of its edges, so an interior seam is always a sandwich of two members and a key, never one shared strut. That costs wood — 1.85 members for every unique edge — and Chapter 18 is the full accounting of what the extra wood buys. For this chapter the table's job is narrower: the frame is not a number somebody remembered, it is a number the geometry produces, and the geometry produces it twice.

Why hemispheres and not spheres

One honest question deserves an answer before the build chapters start: why is the book about half a sphere, when the geometry naturally makes a whole one?

A hemisphere has a floor. Cut a sphere at its equator and the cut line is a flat circle, lying in a plane, exactly where a building wants its ground. The dome meets the earth along its 10 base edges, all in one plane, and every vertex of the base ring sits on the ground at once. A full sphere has no such line; it touches the ground at a point and the entire weight of the building hangs from that point's anchorage. The hemisphere is the shape a sphere becomes when you give it somewhere to stand.

The rim is the honest part. Those 10 base edges are the one place the geometry does not close on itself — the place where the frame's seams run out. The book treats the rim as its own piece of the build: Chapter 39 gives it its own footing, Chapter 41 puts its doors in it, and Chapter 38 names it as the one place error can accumulate. A hemisphere is not a sphere with the bottom missing. It is the shape that has a bottom, and that is why it can be a building at all.

And the crown is the trade. Half a sphere sacrifices the volume above the equator — the part of the ball nobody could stand in anyway — and keeps everything from the widest line down, which is where the room is. The floor area, the headroom, and the standing-height map are all counted from that choice in Chapter 44. For now the frame simply is what it is: 40 triangles, 120 members, 55 seams, 10 rim edges — the count the geometry gives, printed whole, before anything is built from it.

Two Lengths, Not Forty

The whole reason a dome can be built by one person

Two Lengths

Chapter 8 counted the frame: 40 panels, 120 members, 55 seams. This chapter is about the number inside that count that makes the whole thing buildable by one person with one jig, and it is the smallest simplification in the entire method:

Every one of the dome's 65 edges comes in two lengths. Not forty. Two.

A room of forty different panels sounds like forty different kinds of stick, forty different cuts, forty different chances to be wrong. The 2V geometry quietly removes that nightmare before it starts. Because the frame is an icosahedron subdivided exactly once, its edges inherit only two families: the halves of the original icosahedron edges, and the new edges that connect midpoints. Project them onto the sphere and they stretch, but they stretch into two values, not forty. Two lengths of edge, in a fixed ratio of 1.1308 to one — and everything else in the build method follows from that sentence.

A two-length frame is a frame you can jig. A forty-length frame is a frame you can only improvise, which is exactly the failure Chapter 3 documented with ten days of labour. This chapter states the two lengths, shows the four actual cut lengths that fall out of them, and explains the surprising part: that four is not a mistake and not a regression. Four is what the pinwheel joint does to two, and the reason is worth its own section.

Short and long

The two edge families have names, and the names matter because they travel all the way to the cut list and the jig.

Short. Half of an original icosahedron edge — the older edge, cut in two by the subdivision. It spans 31.72 degrees of the sphere, and its length is 0.5465 times the sphere's radius.

Long. The edge between two midpoints — the new edge the subdivision created, spanning 36.00 degrees, 0.6180 times the radius.

Those two factors are not something a designer chose to keep the saw settings down. They fall out of the icosahedron's coordinates, and four independent routes — straight coordinates, central angles, the law of cosines, and a measured drawing — all land on the same pair, which is the book's idea of proof for a number this important. At this book's dome, the two lengths are 72.00 inches and 62.73 inches before the joint takes its bite: a difference of less than a foot across the whole frame.

The triangle shapes follow the same logic. The hemisphere contains 10 equilateral triangles — three long edges, sitting at the centres of the original icosahedron faces — and 30 isosceles ones, two short sides over a long base, ringing the five-fold points. Two triangle shapes, two edge lengths, one dome. The forty panels of the previous chapter are forty *copies* of two designs, not forty designs.

The member schedule

The table is the bridge from geometry to lumberyard, and it is generated, not remembered: every row comes out of the same solved frame the simulator builds, so the table and the dome cannot disagree.

The headline is the first column of it: the whole frame reduces to 4 cut lengths. The longest member is 72.00 inches; the shortest 62.73. Between them sit the other two, and the spread across all four is the width of one hand. That is the entire vocabulary of the build: four numbers on a board beside the jig, and every stick that passes through the bench is one of them.

Why four and not two, when the frame has only two edge families? Because the pinwheel joint insets every member from the vertex it serves, and the inset depends on which side of the corner a member lands on. A long edge becomes two slightly different long members — one for each side of the triangle it borders — and the same for short. The table prints all four, with their counts, because a cut list that hides the difference produces forty panels that almost fit, which is the most expensive kind of fit there is. Chapter 15 is the joint's geometry in full; here the point is only that two became four *for a reason*, and the reason is printed.

That four-length schedule is what the model prints. What the members *measure* — the master cut list, 3 resultant lengths that make a panel the right size once it is together — is shorter: 72.0, 63.5 and 61.5 inches. The equilateral panels take three of the long member; the isosceles panels take the long at the base, the middle on the left and the short on the right. The build's list is the measured one and the model's is the approximation, and Chapter 24 prints both side by side rather than quietly picking one — a builder needs the master list at the saw and the seating schedule at the bench.

Why the pinwheel splits two into four

The step from two lengths to four is the one place a sceptical reader should slow down, because it looks like the dome is quietly getting more complicated, and the honest answer is that it is — by design, and the design buys more than it costs.

A hub dome cuts every strut to end at the vertex, so two edge families really do mean two stick lengths, and the saw never needs a third setting. The price is the hub itself: a machined connector at every corner, drilled to a tolerance, bolted to the end of every strut, and — as Chapter 4 recorded — useless the moment the sticks stop being factory-identical.

The pinwheel throws the hub away by making every member end land on the *side* of the next member instead of at the corner. That single decision is what lets rough, split, slightly irregular timber frame a dome at all: a butt end on a flat sawn face does not need the precision a hub demands. But it has a geometric consequence, because a member that stops short of the corner stops at a different distance on each side of the triangle. A long edge needs a long-left and a long-right; a short edge needs a short-left and a short-right. Two becomes four.

So the honest ledger reads: the pinwheel costs one extra saw setting per edge family, and in exchange it deletes the hub, the mitre on every end, and the requirement that rough timber behave like milled stock. Four cut lengths instead of two is not the dome getting complicated. It is the entire bill for the joint that makes the method possible — and it is a bill paid once, on a jig, where a fence and a stop make the four lengths as repeatable as two. The next chapter spends that simplicity on the question of what a stick of this shape can actually carry.

Compression, Tension, and One Eighth of a Tree

What a wedge strut can actually carry

One Eighth of a Tree

The last three chapters built the case for the wedge as a shape: a triangulated frame loads its members along their length, and along their length all that matters is how much wood is in the section. This chapter turns the case into numbers — and then, because a book is not an engineer, hands the rest of the question to one.

The unit of the whole method is one eighth of a tree: a 45-degree sector of the trunk, 3.83 inches across its bark face, 5.00 inches from pith to bark, 9.82 square inches of wood. That stick will be pushed and pulled along its grain for the life of the building, and its ends will press sideways into the sides of its neighbours. Those are the two load directions this chapter cares about — along the grain, and across it at the joint — and they behave so differently that the entire honesty of the method depends on keeping them apart.

The chapter ends with the one number the book will actually give you, and the reason it will not give you the number you probably wanted. The two are related: the book can compute geometry, and geometry is what a wedge's bearing area is. What a wedge's wood can *take* is a property of your tree's species, grade and moisture, and any book that prints that number for you is guessing about your tree while pretending to know it. This one does not.

Along the grain, and across it

Wood is not one material. It is two, depending on direction, and the difference is roughly an order of magnitude.

Along the grain — end to end, the direction the tree grew — wood is the material engineers mean when they say pine is strong. A stick squeezed along its length carries load through the fibres themselves, every one of them sharing. A stick pulled along its length is even stronger: the fibres take

tension along their length the way rope does. This is the direction a triangulated frame uses, which is why Chapter 6 could say that shape does not matter and mean it: for push and pull along the member, the section's area is the count that matters, and the wedge has 9.82 square inches of it.

Across the grain — sideways, pressing into the flat of the wood — the same stick is a different material. Load the side of a pine stick and the fibres are asked to squash between their neighbours; they yield far sooner, they dent rather than break, and the load the joint may carry is a fraction of what the same member carries along its length. Every wooden building lives with this fact; it is why a post stands on a plate instead of a point, and why bolts through timber are spread out rather than clustered.

The wedge frame meets the problem head-on, and the design decision is in the next section. Here the point is the discipline: this book's claims about what a wedge can carry are claims about the *along-grain* direction — the direction the frame actually loads — and its numbers at the joint are numbers about *area*, which is geometry and computable, not about permissible pressure, which is not. The safety page further down exists because the two are easy to swap, and swapping them is how dome books get people hurt.

The end-to-side bearing

The pinwheel joint's load path is worth stating precisely, because its elegance is real and its limits are real, and both live in the same sentence.

Every member's butt end — cut once, on the jig, at the one compound angle the frame needs — lands on the flat sawn face of its neighbour. The contact patch is the end of the stick, and the end of a sector is the sector itself: 9.82 square inches, nothing notched away, nothing reduced to a tongue or a dowel. A conventional joint would cut away wood to make the connection and then bolt what is left; the pinwheel does the opposite. The whole section bears, because the whole section is what the log produced, and the joint gets its strength from refusing to throw any of it away.

The bearing patch, stated as geometry: the butt end is a sector of a 10.0-inch circle — the depth of the member, doubled — cut at 45.0 degrees, which is the same sector shape the tree was split into.

So the book's honest number for the joint is 9.82 square inches of end grain against side grain, per member, at every interior corner. The frame has 120 members, each with one loaded butt, and the arithmetic ends there, because the next number — how many pounds that patch may carry — belongs to your tree, not to this book.

What the book *can* say, and does: the bearing is generous by the standards of light framing, the load spreads across the full width of the sector rather than a bolt row, and the key between the members carries the seam's alignment while the wood carries the building. The chapter on the pinwheel (Chapter 15) and the one on the connector (Chapter 17) show how that division of labour is built. This chapter's job was the load path, and it is this: along the grain through the members, across the grain at the butt, area the whole way.

What this book will not tell you

This page is a warning, and it is not decorative.

This book will not tell you what your tree's wood is allowed to carry. Allowable stresses depend on species, grade, moisture content, and the direction of load; they come from published tables tied to specific species grown in specific regions, and they are applied by someone who can look at the actual timber. A pine from a plantation is not a pine from a windbreak, a knot at a bearing face is not a knot mid-span, and green wood carries less than dry wood while it dries. None of that can be looked up by a book that has never seen your log.

It follows that nothing in this book is a structural sign-off. The frame described here is a prototype method, demonstrated by a build, not an engineered standard. Where a load figure matters — a hanging frame, a floor, a roof carrying snow — the book names the number as an engineer's job and leaves it there. Chapter 11 repeats this in its accounting of the costs nobody puts in a dome book, and when this project forgot its own rule — printing a load-ish claim it could not stand behind — it printed the correction in the chapter where the claim was made.

The rule, once, plainly: **compute the geometry, check the code, and ask an engineer about the loads.** Everything this book computes is geometry. Everything an engineer decides is the rest.

The number I will give you

One sidebar, one number, so the distinction above has a concrete object.

The bearing area of one butt end against one side: 9.82 square inches. That is geometry — the sector's area, computed from the tree's own taper — and the book will stand behind it, because it is the same arithmetic that produced the cut list, and it can be checked with a ruler on the stick in your hand.

What that patch is *allowed* to carry, in pounds per square inch, is a property of the species, the grade and the moisture — a table entry, not a computation — and this book will not print one. It will tell you which direction to look (across the grain, at the butt, at the wet case), and it will tell you that the

frame's own redundancy is the reason the method tolerates being built from timber nobody graded: 40 panels mean a member can be watched, and a member can be swapped, which is how the frame was kept honest through its first year.

The book gives you the geometry and the method. The engineer gives you the stamp. Neither one can do the other's job, and this book is built so that the boundary between them is a page you cannot miss.

What a Dome Costs You

The chapter that argues against the book

Every book about a building method owes its reader one chapter that tries to talk them out of it. This is that chapter. It is not here for balance; it is here because the costs it lists are real, they arrive late, and they are the ones the rest of the dome literature leaves out — which is exactly how people end up surprised by them.

The argument of this book is that a wedge dome is a remarkable way to get a strong, cheap, buildable frame out of 3 trees. The argument of this chapter is that a frame is not a house, and between the two sit three costs the frame does nothing about: the room is round, the roof is many angles, and the world of permits and appraisals expects a rectangle. Each one is priced in time and money, each one is worse than the same problem in a box, and none of them is fatal. They are the admission price of the shape, and the honest builder pays it knowingly or pays it twice.

The chapter ends with the cases where the rectangle wins outright, because those exist and the book's method is not a religion. If you read this chapter and still want the dome, the rest of the book is yours. If you read it and do not, the book has just saved you a year.

Furniture does not fit a curve

The first cost shows up on moving day, and it never fully leaves.

A wall that curves is a wall that refuses the rectangle. Beds, dressers, bookshelves, kitchen cabinets, a sofa — the furniture economy is built on straight walls and ninety-degree corners, and against a dome's wall every straight piece leaves a wedge of dead air behind it. That dead wedge is real floor you paid for and cannot use, in every direction, at standing height. The honest floor number for a dome is smaller than the plan-area number, and Chapter 44 computes the standing-height map so the loss is visible before you build rather than after you move in.

The fixes are real and none of them is free. A pony wall under the dome buys back vertical wall at the edge — Chapter 53 prices it, and it is the cheapest square footage in the building — but it is still a cost

the box never pays. Built-in storage shaped to the curve costs custom work. Loose furniture loses its footprint to the geometry, piece by piece. And the centre of a round room, where the headroom is highest, is also where the room most wants its furniture to be — which is how a round room ends up furnished as a ring around an empty middle, with the good space in the middle spent on walking.

None of this is hidden. It is just not in the frame's price, and a frame's price is what dome books quote. Count the round-room tax as real: a fraction of the floor, a custom-furniture bill, or both, from the first night you sleep in it.

Roofing a compound surface

The second cost is the one that decides whether the dome survives its third winter, and it is the reason dome builders talk about skin more than they talk about anything else.

A dome's roof is one surface made of many flat panels meeting at 55 interior seams, every one of them a line where water can get in. A shingled rectangle sheds water downhill in sheets; a dome sheds it down forty facets and fifty-five valleys, and the valleys are joints. The frame chapters show the seam doing its job with a key and a gasket, and the weather chapter (Chapter 42) says the number out loud — 55 of them — because the honest way to think about a dome's roof is not "one roof" but "55 small roofs, all meeting".

What that costs: the seam system is not optional and not trivial. The key must fit its fold angle, the gasket must take up the tolerance, and the whole assembly must still shed water after the wood has moved through a year of drying. The skin options are priced in Chapter 40, and the pattern across them is the same — the cheap skin costs more later, and the good skin costs now. A rectangle with standing-seam metal and two valleys is a solved problem with a hundred years of practice behind it; a compound surface is a solved problem only if you do the work this book spends its middle chapters describing. The roof is not where the dome is weak. It is where the dome is *particular*, and particular costs attention, which costs time, which is the currency a two-week frame is supposed to save.

Permits, appraisals and resale

The third cost never touches the building and still moves the price.

Permits assume a rectangle. The building department's checklists, the prescriptive code tables, the inspector's mental library — all of them are written around conventional framing, and a hubless dome of split wedges is not in any of them. That does not make it unpermittable; it makes it *unusual*, and unusual means the plan review takes longer, the official may want an engineer's letter (Chapter 10

tells you what the engineer's letter does and does not cover), and the fee schedule may have no line for you. Somewhere, the system will charge for being surprised.

Appraisals assume a comparable. An appraiser prices a house by finding three similar recent sales; a dome has no comparables, so the value comes out of the method instead of the market, and the number is whatever the method defends. That does not make the dome worth less — it makes its worth *harder to finance*, and financing difficulty is a real cost for anyone who is not paying cash.

Resale is where the two meet. The buyer's lender wants the same appraisal the seller's did, the buyer's inspector has never seen a wedge, and the pool of buyers who want a round house is smaller than the pool who want a rectangular one. The network chapters of the Complete Edition — this book no longer contains them — exist because this project's answer to resale is to make the dome *movable* — value it as a building that can leave its pad — but that is a proposal, not a market, and those chapters say so. The honest line for this chapter: a dome you build yourself and keep is not exposed to any of this. A dome you build hoping to sell like a house is exposed to all of it.

When to build a rectangle

The closing test, stated as plainly as the method allows.

Build a rectangle when the building is a shed, a garage, a workshop, or a storage room — anything where the square shape, the standard sheet goods, the off-the-shelf door and the appraiser's familiarity are worth more than the dome's savings. The catalogue this project prices shows the border: the cheapest and dearest per-square-foot numbers in it are 7 and 68 dollars, and the box shed it costs beside the dome shed comes out at 12,426 dollars against 9,444 — close enough that the shed decision is about doors and shelving, not about the frame. Build a rectangle when the site, the code, or the lender is already fighting you, and the fight would eat the savings. Build a rectangle when you need the work done by a crew whose language is studs and trusses, not jigs and seams.

Build the dome when the floor is round and wants to be, when the frame must be cheap enough to build twice, when the tree is already down, or when the building is a home and the home is yours. The finished-home comparison in the same catalogue — 84,983 dollars against 87,378 — shows the dome's frame advantage persisting into the finished house, and Chapter 46 takes that number apart line by line, including the lines that do not help.

The rectangle is a fine building. It is the right building in most cases. This book exists for the cases where it is not, and the method it describes is the strongest version of the alternative — measured against the rectangle's own numbers, printed here, in the chapter that was supposed to talk you out of it.

Two prices for the same frame

The chapter has spent its length on costs that have nothing to do with the frame: furniture, roofing, permits, appraisals. Close it with the frame's own two prices, because a reader who has met them in different chapters is entitled to see them on one page.

the frame, two ways	dollars	what is in it
Cut it yourself	492	the fortnight's receipts: fuel, chain oil, hardware, fasteners
Buy the equivalent	9,798	the same frame in store-bought timber, at the national rate

The difference — 9,306 dollars — is what the build keeps, and Chapter 46 turns it into a rate on hours, including the version that flatters the method least.

The product pricing — the same building made and sold as an article, with margin, warranty and overhead — is deliberately not in this paperback. It lives in the Complete Edition on the site, alongside the catalogue of styles and the kit rungs.

And the skin is the line that changes. The watertight skin over this shell is 730 square feet of membrane, and it is the one part of the building that cannot be made at home and cannot be skipped. Every price above includes it; a reader comparing a dome to a shed and finding the dome more expensive has usually just found the skin.

Split Like a Cake

Three passes, eight sectors, and why eight

This is the chapter with the actual cut in it — the one motion the whole method hangs on, reduced to a procedure you could hand to somebody at the log.

The idea was introduced in Chapter 5 as a discovery. Here it is a technique. A round section of trunk, bucked to 6 feet, gets split along its length into 8 equal sectors — like a cake — by seven cuts, all of them straight rips through the pith. Every stick that comes off is the same shape: a 45-degree wedge, 3.83 inches across its bark face, 5.00 inches deep from point to bark, 9.82 square inches of section. There is no squaring, no planing, no second machine. The saw that felled the tree makes the members.

The rest of the chapter is the order of the cuts, the reason the order matters, and the reason it is eight and not six or twelve. The chapter is short because the operation is. That is the point, and it is worth remembering when the pile is knee-deep: the method's entire conversion step is seven cuts per section, and each cut is a straight line.

The three passes

The procedure, in order. The order is not tradition; it is what keeps the section stable while you cut it, and a cut made out of order costs you a pinched bar and a ruined pass.

Pass one: halve it through the pith. Stand the section on end or lay it in a cradle, aim the bar down the centre of the end grain, and rip from one end to the other. You now have two halves, each with one flat sawn face and one round bark face. Do this pass first, every time, because it is the only cut in the whole sequence where the section is still round and wants to roll; once it is halved, every piece has a flat to sit on, and everything after is stable.

Passes two and three: quarter each half. Lay each half on its flat face and rip it into two quarters, again through the pith line. Four quarters, each 45 times two degrees wide, each with one

flat and one round face still on it. The flat sits on the cradle; the saw follows the centre line; nothing moves.

Passes four through seven: eighths. Each quarter gets one more rip, from the remaining flat face through to the bark, and becomes two eighths — the final sector. Seven cuts in total: one halving cut, two quartering cuts, four eighth cuts. 8 sticks, all the same sector, from a section of trunk that no longer exists.

Two things make the sequence forgiving rather than fussy. First, every cut after the first is made on a piece that already has a flat face to sit on, so the workholding is the piece itself. Second, you are never measuring an angle: the cut list is "through the middle", repeated, and the tree supplied the symmetry. A sector is 45 degrees because a circle divided in half three times is; nobody held a protractor to it.

Three cuts, eight sticks

The drawing shows the sequence end-on: one circle, then two halves, then four quarters, then eight sectors, with every cut a diameter. Read it as a production plan rather than a diagram, because the sequence is what makes the work fast.

The first observation is that every pass doubles the number of pieces — one becomes two becomes four becomes 8. The second is that the work per piece falls as the pieces multiply: the halving cut is the whole width of the log, the quartering cuts are half that each, the eighth cuts a quarter of the log's width each. The third is that the sequence never asks a decision. Nothing in it requires choosing where the next stick comes from or how thick it should be; the geometry decided all of that when the log was round. A procedure with no decisions is a procedure you can repeat forty times without drifting, and a procedure that doubles its output per pass is a procedure whose rhythm pays you as you go.

Chapter 29 puts the sequence on a clock — 7.5 sticks an hour, sustained, once the cradle and the rhythm are settled — and Chapter 33 shows what happens to the pile after. Here the sequence stands alone, because it is the one thing in the method that cannot be substituted: no mill, no planer, no router table will produce this member, because the member *is* the cut.

One log, two, four, eight

The production hierarchy deserves its own paragraph, because it is the part of the method that looks obvious and is quietly load-bearing.

One log becomes two halves becomes four quarters becomes 8 members. Every major cut doubles the piece count, which means the whole tree converts in three layers of work: one cut at the top of the tree, two in the middle, four at the bottom. That shape — wide at the bottom, narrow at the top — is exactly what one person with one saw can exploit. The slow cuts are few and wide; the fast cuts are many and shallow. And because the pieces at each layer are identical to their siblings, the workholding at each layer is one cradle, one stop, one posture, repeated.

This is also the shape a guide rail, a cradle or a splitting fixture would be built around, if you ever stop doing it freehand — the geometry already knows where every cut goes, so a fixture that knows it too is just the sequence, made of steel. The book keeps the hand version, because the hand version is the whole point: the method exists so that a person with a chainsaw and 3 trees needs nothing else.

Why eight and not six or twelve

The question gets asked, usually at the first split pile: why eight? The answer has three parts, and each one is a hard constraint.

The sector angle has to leave the seam key workable. A sector is 45 degrees, and the seam between two panels folds by 18.03 to 22.46 degrees depending on where it sits — Chapter 16 prints the full schedule. The gap the key has to fill is the difference between the sector angle and the fold, and at eight splits that difference is a real, makable, compressible thing. Split the log into six and the sector is sixty degrees, the key becomes a structural wedge in its own right, and the seam starts deciding the geometry instead of the other way round. Split it into twelve and the sector is thirty degrees — a sliver, hard to handle and hard to keep straight over six feet.

A sixth is too fat to handle alone. Six sectors from a 12-inch butt are heavy, wide pieces with a sixty-degree point; they are awkward to cradle, awkward to carry, and awkward to fasten. Eight lands the member at 3.83 inches across and 9.82 square inches of section — the size a single person moves, lifts and sets all day, which is the constraint Chapter 28 keeps coming back to.

Eight is the halving count. Two, four, eight — the sequence above only needs cuts “through the middle”, never a measured angle. Six needs a sixty-degree first cut, held by eye or by fixture; twelve needs a thirty-degree refinement. Eight is the finest split you can reach with nothing but halving, and halving is the only cut a saw can aim by symmetry alone.

The table on the next page runs the alternatives side by side so the choice is visible rather than asserted.

What other split counts give you

The table is the arithmetic of the choice: six, eight, ten, twelve, each with its sector width, depth and section. Eight is the reference row, and the reason is not that eight is magic — it is that eight is the row where the key fits, the stick is carryable, and the cut list is still “halve it”. Every other row trades one of those three away, and the table makes you look at which one.

One more note for the pile, because it will come up: the split count is a *build decision*, set before the first section is opened, and it travels. The member width it produces sets the pinwheel inset, which sets the cut list, which sets the jig. So the decision is made once, from this table, and every number downstream follows. Change it later and you are not adjusting a cut; you are starting the frame again. So the table gets its own page, and the choice gets its own warning: read the row, then commit.

Eighty-Seven Percent

The recovery arithmetic, with the kerf paid for

> ***This chapter corrects something.*** *The 88.4 per cent recovery figure was computed for a 15-inch tree; this book's smaller 12-inch tree keeps 87.4 per cent, and the chapter prints both numbers rather than quietly picking the flattering one.*

Eighty-Seven Percent

The headline number of the whole method, stated once with its caveats attached, because the caveats are where the number lives:

Split the way this book splits, and a trunk keeps 87.4 per cent of its solid wood as structural members. Sawn the way a mill saws, the same trunk keeps about 45. That is 1.93 times the usable wood, from a cheaper process, with one machine instead of four.

The caveats, in the same breath: the 87.4 counts *split wood only* — nothing planed, nothing dried, nothing graded; the sawn comparison is the honest one for two-by-fours packed in a circle, which is flattering to neither side; and the 88.4 per cent figure this project published earlier belongs to a bigger tree than this book's. This book's tree is 12 inches at the butt and keeps 87.4 per cent; the earlier claim's tree is 15 inches and keeps 88.4. Both numbers are correct, for different trees, and the last page of this chapter prints both rather than quietly picking the flattering one.

The rest of the chapter is the arithmetic, so the headline stops being a claim and becomes a thing you could re-derive with the saw's kerf in your hand.

What a mill loses, and to what

A sawmill's job is to turn a round log into rectangular boards, and everything it loses is the price of the rectangle.

The slabs. A circle does not contain its rectangle. The two rounded sides of the log — the slab wood — cannot become boards of any width, so they come off first, and they are the biggest single loss. On a small log they are a huge fraction of the circle; the fatter the log, the smaller the fraction, which is why mills pay per board foot and prefer the big trees.

The edgings. Even the boards cut from the heart of the log carry wany edges — bark, roundness — and must be edged square. More wood off.

The trim. Boards are sold in even lengths, square at both ends. Every board loses its trim.

The kerf. Every cut turns wood into sawdust. A mill's band or circle blade is thinner than a chainsaw's, but a mill makes far more cuts per log, and the kerf is paid on every one.

Drying degrade. The boards are stickered, stacked and dried, and a fraction of them check, cup, twist or stain past their grade. The wood was there; the board is not.

The grade fall. Of what remains, some boards grade lower than the rest, and low-grade lumber sells for less — a loss in value even when nothing is lost in volume.

Add the six together and the famous range appears: a log sawn into two-by-fours keeps roughly half of itself as saleable lumber. This book's tree, packed honestly with true two-by-fours, keeps 45 per cent. None of that is incompetence. The mill is extremely good at a job whose product is a rectangle, and the waste is the price of the rectangle — paid in wood, at every station of the chain Chapter 2 walked.

What the wedge loses

Now convert the same trunk the wedge way, and count the losses again.

The bark. Lost either way; it was never wood. Not counted against either method.

The kerf — and that is nearly all. The wedge makes seven cuts per section, through the full diameter. Each cut is a 0.25-inch kerf — a chainsaw's kerf, fatter than a mill's blade — and across 8 sections the kerf bill comes to 40 board feet of the trunk's 318. That is the honest, paid price: 40 board feet of sawdust.

Nothing else. No slab, because the sector uses the whole radius. No edging, because nothing needs a square edge. No trim beyond the bucking. No drying degrade to speak of — a wedge is allowed to move as it dries, and Chapter 49 explains why a split moves less than a board. No grade fall, because the book grades its own pile by eye and uses the rejects for blocking.

So the ledger closes: 318 board feet in the trunk, minus 40 of kerf, leaves 278 board feet of structural member — 87.4 per cent — at 4.35 board feet a stick, which is what the worked build in Chapter 24 turns into a cut list. The wedge does not beat the mill at milling. It beats the mill at *this* — making a structural member for a frame that never asked for rectangles — and the beat is the difference between 87.4 and 45 per cent of the same tree.

Both conversions, same log

The drawing shows the same round section twice. On the left, the mill's version: rectangles packed into the circle, slabs and edgings falling away to the outside, the honest count of what a two-by-four economy keeps. On the right, the wedge's version: the circle divided into 8 equal sectors, and every sector kept whole.

The picture makes the argument the numbers cannot. The mill's losses are the *outside* of the circle — the part farthest from any rectangle. The wedge's only loss is the thin lines of the kerf itself, running through the middle. The mill throws away the corners; the wedge keeps the corners and throws away the cut lines. The corners of a round log are not bad wood. They are perfectly good wood of the wrong shape, and "wrong shape" is a judgement the rectangle made, not the tree. Chapter 6 is the reason the frame is allowed to disagree, and this drawing is the disagreement, priced in board feet.

Section by section

The table carries the comparison down the whole trunk, one bucked section per row, both conversions, because a taper means the answer is not one number but a run of them — and the run is what you actually cut.

Each 6-foot section of the trunk is a slightly smaller circle than the one before it. The butt section packs more rectangles and yields fatter sectors; the top section packs fewer and yields slimmer ones. The table shows both columns shrinking as the taper runs, and the recovery staying where the geometry put it: the wedge's advantage is not a lucky average, it is a property that holds section by section, from the 12-inch butt to the 8-inch top. Read it as a check on the headline rather than as a second claim: if any row disagreed with the story above, the story would be the thing to fix.

Where 88 became 87

This section exists because an earlier published claim of this project said 88.4 per cent and this book prints 87.4, and a reader who notices the gap deserves the explanation rather than the silent edit.

The earlier published recovery figure — 88.4 per cent — was computed for the earlier claim's reference tree: 15 inches at the butt, 60 feet usable, the big plantation pine. A fatter log puts more wood inside its circle and less into its slabs, so the same split-and-kerf arithmetic keeps more of it: 88.4 per cent.

This book's tree is smaller — 12 inches at the butt, 48 feet usable, the windfall of Chapter 1 — and the same arithmetic, run on the smaller circle, keeps 87.4 per cent. The kerf is the same 0.25-inch chain; the circle it eats is simply a smaller fraction of a smaller pie.

Neither number is wrong, and neither was ever a lie. They are two recoveries for two different trees, and the earlier claim named its tree while this book named a different one. The book's discipline — stated on this page, and kept on every correction page like this one — is that a correction names the error, prints both numbers, and leaves the arithmetic visible. That is what this page is: 88.4 for the pine the earlier claim priced, 87.4 for the tree this book builds, and the reason they differ is the diameter, printed in the table above, one row per section.

Four Ways Up

One stick, four orientations, four different buildings

This chapter is the most original idea in the method, and it is the one nobody sees coming, because it concerns a property most building methods do not have: the stick has a front.

A two-by-four is symmetric enough that nobody asks which way up it goes. A wedge is not. It has a point and a curved bark back, two flat sawn faces meeting at a 45-degree angle, and it is never the same stick twice. Which way its point aims — into the room, out to the weather, along the seam — changes what the frame does with it, without changing a single dimension. One stick, cut once, can be four different members, and the four buildings that result are four different answers to the same geometry.

The chapter walks the four orientations and ends with the table that chooses between them, read straight from the simulator's own seam readings. The default — the one the rest of the book builds in — is named and defended. The other three are not mistakes; they are the method's spare capacity, and each one buys something specific at a specific price.

Rotating about its own axis

Hold the wedge out in front of you, point up, bark toward your chest. Rotate it about its own length — the axis from butt to head. At every quarter turn it presents a different face to the room and a different edge to the weather, and nothing about its cut list changes. That freedom is the whole idea: the frame's geometry cares where the stick's *surfaces* sit, not where the stick's *shape* came from, so the builder gets to choose the surface, free.

The four orientations come from the two decisions the rotation makes. First, where does the point aim — toward the centre of the dome or toward the centre of its own panel? Second, do the two sticks flanking a seam aim their points together or apart? Point *dome in* means every wedge points at the middle of the building, bark to the weather. Point *panel in* means each stick aims into its own triangle's middle, the pair beside a seam pointing apart. Point *dome out* turns every wedge backward,

bark to the room, point at the sky. Point *panel out* turns the pair toward each other across the seam. Four ways up, same stick.

What changes between them is real, and it is measurable: the two sawn faces open by a different angle, the key has to be a different shape, and the key's own base moves from 2.80 inches across to 11.67 in the widest arrangement. The next four sections take them one at a time; the plate shows all four.

All four, side by side

The plate is the whole chapter in one look, and it is not a sketch: each pane is a seam **cut across, through the solver's own meshes**, so what you are looking at is the joint, not an illustration of it. Two members, one from each panel, and the key in gold between them. In every pane the weather is up.

Read the panes for the two things that change. First, *which face of the stick the key bears on*: in the first pane the key sits between the two narrow sawn faces, a slim spline; in the second and fourth it bears on the broad sides and becomes a wide shim; in the third the wood is turned around, so the key is fitted from the inside. Second, *how much key*: the width printed under each pane is the base of the key as the solver slices it, and it is not the same number twice.

The four arrangements are not four aesthetic choices. They are four key-cutting choices and four weather choices, and the table at the end of the chapter prices them. First, each in its own words.

Point dome in

The default. Every wedge points at the centre of the building, bark out, sawn faces lying along the seams. This is the orientation the rest of the book builds in, for three reasons that stack.

The key gets flat faces. A seam's key sits between the two sticks' sawn faces — the only faces the saw ever made flat — so it bears on machined wood rather than on bark. At this orientation those two faces stand 26.97 degrees apart, the key is a narrow taper 2.80 inches across its base, and it bears 6.0 inches along the seam. That is the joint every chapter before this one has been drawing.

Bark faces the weather. The curved back of every wedge is the building's outside face: one continuous, shed-ready, bark-clad surface. Chapter 42 builds the rain strategy on exactly this — water runs down bark, off the seam's shoulder, to the rim.

The stick tells you which way it goes. This is the quiet one, and builders feel it at the top of a ladder. Bark out, pith in: the stick is self-labelling. Nobody marks “this side out” on a hundred and twenty members, because the tree already did — the bark is the outside, the point is the inside, and a wrongly rotated stick announces itself the moment you try to lay the key against it.

And one geometric detail that turns out to matter at the bench: the key is **outboard of the two points**. Cut a seam across and the members’ points are the innermost thing in it, with the key sitting in the wedge of space between them and the meeting line of the two bark faces — so the key is reached, fitted and checked from *outside* the shell. That is the practical reason this orientation is the one the book builds in, and the next one down the page loses it.

The cost of point dome in is the ridged interior: the room’s wall is a field of points. Acceptable in a frame you will sheath; Chapter 44 decides how to live with it.

Point panel in

Each stick aims its point into the middle of its own triangle, so the pair beside a seam point *apart* — away from each other, one into each neighbouring panel’s centre.

What it buys: the key moves off the narrow sawn edges and onto the broad sides of the pair, and it stops being a spline and becomes a broad shim. Cut the same seam across and the two faces now open 153.03 degrees, with the key 11.67 inches across its base — 4.2 times the width of the default’s key. For a seam asked to carry more than alignment, that is real contact area rather than a line of it, and the simulator keeps the orientation as a first-class option for exactly that.

What it costs: the bark faces lean away from the weather and into the seam’s shadow, so the outside surface is no longer a clean field of bark; the wide key is more timber per seam than the default; and the self-labelling gets subtler, because “point toward the middle of your panel” is an instruction you must remember, not one the stick shouts.

Point dome out

Every wedge reversed: bark to the room, point to the weather. The building’s outside becomes a field of ridges — forty panels of them — and its inside becomes the smooth curved field of bark.

What it buys is mostly weather. The pointed outside sheds water the way a ridged roof does, each ridge its own drip line, and the seams sit in the valleys between ridges instead of on the crests — a configuration some roof builders prefer on principle, because the joint lives where the water is already heading.

What it costs is the thing the previous section praised most. The key still sits between the two sawn faces — the geometry does not change what the wood is — but the *meeting line* moves to the other side: the bark faces now converge on the inside of the shell, the key's base is 6.27 inches across (against 2.80 in the default) with the faces 63.03 degrees apart, and the key now lies *inboard* of the two points. Fitted from the room side, not from the weather side. Everything about the joint still works; the person fitting it is now indoors, on a ladder, in a small round room, which is a worse place to be laying a key than the outside of a roof. And the self-labelling inverts: pith out, bark in, which reads wrong to every instinct a woodworker has.

Point dome out is the orientation to remember exists and not choose — unless the building's skin is a ridged shingle system designed for it, which is a real design and not this book's.

Point panel out

The fourth rotation: the pair beside a seam aim their points *at each other, across the seam* — each stick's point reaching over the joint toward its neighbour's back, bark faces turning in toward the panel's middles.

What it buys: the room side of every panel is a field of the flat sawn faces — the smoothest interior the four orientations offer — and the key is broad, like the panel-in orientation: 11.67 inches across its base with the faces 153.03 degrees apart. What it costs: the pair's points are toward each other across the seam, so the weather face of the building is partly sawn face — the face the finish chapters warn you never to leave exposed — and the two sticks' broad faces, not their edges, are what the key bears against, which is more timber per seam than any other orientation. Chapter 49 returns to this when it discusses char and oil, because the one orientation that puts sawn face outdoors is the one this book does not build.

Choosing one

The table is the decision, and it is read straight from the simulator's own seam readings rather than from adjectives. Each row: one orientation, its contact geometry, its weather face, its inside face, and the verdict the solved model gives it.

The row that matters is the default's, and the reason it wins is not that it is perfect — it is that it is the only orientation where every job lands on the face built for it. The key gets the sawn faces. The weather gets the bark. The room gets the points it can sheath over. And the stick labels itself, so the hundred and twentieth member is placed as correctly as the first.

The other three rows are the method's spare capacity: point panel in when the seam must be stiffer, point dome out when the skin is designed for ridges, point panel out almost never. The table keeps all four because a choice made from the table is a decision, and a choice made from habit is a gamble — and the orientation, once the first panel is built, is the one decision in the method you cannot take back.

The Pinwheel

Three sticks, one triangle, and not one mitre

> ***This chapter corrects something.*** *The 'no mitres anywhere' claim, corrected here in full.*

The Pinwheel

This is the joint that makes the whole method buildable by one person, and it is worth stating in one sentence before the geometry spends it:

No member end ever meets another member end. Every member's butt end lands on the flat side of the next member, one cut per stick, and the corner is empty.

That sentence deletes, in order: the hub, the mitre at every end, the compound-angle saw settings, and the requirement that rough split timber behave like milled stock. What it costs — and it does cost — is a slightly smaller panel than the triangle it sits in, and a cut list four lengths long instead of two. Chapter 9 already counted that bill. This chapter shows the joint that issues it.

The name is descriptive. Look at a corner of the built panel: the three members overlap like the blades of a pinwheel, each butt end lying on its neighbour's side, chasing each other around the triangle, none of them meeting at the vertex. The vertex — where a hubbed dome puts its connector — is empty air. The frame does not need anything there, because the triangle is closed by the three overlaps, and the overlaps are held by screws through flat wood into flat wood.

Why mitres were the enemy

The conventional way to join three struts at a corner is to make them meet: each end cut to the right compound angle so the three ends kiss at a point, then a hub or a plate holds the kiss together. It is the geometry of every geodesic kit ever sold, and for machined aluminium struts in a factory it is fine.

For this method it is fatal, for three reasons that compound.

A mitre is two angles at once. The end of a dome strut has to match both the panel's fold and the member's position around the corner — a mitre and a bevel together. Cut one slightly wrong and the struts no longer meet; cut forty panels' worth slightly differently wrong and the dome has a gap for every mood.

Every stick needs two of them. Both ends of every member would be mitred — 120 members, two ends each, each end its own setting — against a stick that is rough, tapered, green and moving. Precision joinery on wood that never agreed to be precise is not a building method; it is a bet against the material.

The corner has to be perfect or nothing is. When three ends meet at a point, all three must arrive exactly, or the hub cannot close. The tolerance is concentrated in one place, which is the worst place for a frame built from a log: every error in the whole build queues up at the vertex and waits.

The pinwheel sidesteps all three by moving the joint. The ends do not meet, so they do not have to match. Each end lands on a flat side, and a butt end on a flat side needs the one compound cut — cut once, off the jig. The overlap is where the tolerance lives, and an overlap is a joint that can be off by half an inch and still hold, because the screws have a whole side of wood to find.

Each end lands on a side, not a point

The pinwheel, in the way that finally makes it obvious, because the obvious version is what you will teach somebody at the bench.

Take three sticks. Lay the first two in the triangle's shape, and instead of cutting them to meet at the corner, let each one run *past* the corner along the side of the other. Now the first stick's end rests on the second stick's side, not its end. Lay the third stick the same way: its end rests on the first's side, and the second's end rests on the third's. Three ends, three sides, one closed triangle, and the actual corner of the triangle — the point where the lines would meet — has nothing in it at all.

That is the whole joint. Every member gets its one compound butt cut and is left long on its head end, and the overlap is resolved at assembly. The result reads as a pinwheel because each stick's butt chases the next stick's side around the corner, and the chase closes the triangle without any stick ever touching the vertex.

Why this survives rough timber: the bearing patch is the whole end of a stick — the full sector of the butt against the neighbour's side, 9.82 square inches of flat-to-flat contact, as Chapter 10 measured — and the screws run through the overlap at right angles to the grain, where wood holds best. Nothing in the joint depends on an angle being cut accurately, because the only cut in the whole assembly is

the one compound butt cut — made once, off the jig — and it is the one cut a jig can make identical forty times.

One panel, taken apart

The drawing explodes one panel and labels the three overlaps, so the sentence above has a picture to check against.

The butt end. The one machined cut on the member: a compound cut of mitre and bevel, made once, off the jig, with the stick located on its sawn faces — Chapter 34 is that cut in full.

The bearing side. The neighbour's sawn face, where the butt lands. Flat, because the saw made it flat, and wide, because the whole sector is wide.

The head end. Deliberately long. It runs past its own bearing by the overfit allowance — 6 inches in this build — and gets sawn off in place against the jig's fence, which is how the panel ends up true without anybody measuring a stick to a tape. Chapter 35 is that trick, and it is the quiet reason the method forgives.

The empty corner. The vertex itself, untouched. In a hubbed dome this is where the money is — the machined connector, the bolt, the precision. Here it is air, and the air is doing its job by not being asked to do anything.

Read the exploded drawing once and the build order falls out of it: butt cuts first, on the jig; panels assembled flat, overlaps screwed; heads flushed; panel lifted. Nothing met anything. Nothing had to.

What the inset costs

The honest price of the pinwheel, because the joint's elegance is real and its bill is real, and the book prints both.

The triangle's *edge* — the geometric line from vertex to vertex — is longer than the member that fills it, because the member stops short of the corner to land on its neighbour's side. The difference is the inset: the distance each stick pulls back from the vertex so its butt lands on the next stick's side instead of on the next stick's point. Every member length in the cut list is the edge length minus its share of the inset, which is why the cut list has four lengths where the geometry has two — the inset differs between the long and short edges, and between the two sides of each.

The inset also shrinks the panel. The frame's triangles, drawn vertex to vertex, are bigger than the wood that fills them; the panel is the triangle minus three strips around its edge. That strip is not waste — it is the bearing, the overlap, the screw row — but it is real, and it is why the member schedule in Chapter 24 is the number it is rather than the bare chord factors.

And the inset scales with the *stick*, not the dome. A wider member pulls back further; a bigger dome changes nothing. That single fact is why the method survives scaling — Chapter 48 takes the same stick to bigger and smaller domes — and it is paid here, in the cut list, once.

At the bench the inset is not paid in arithmetic at all. It is already in the members: the master cut list of Chapter 24 — 72.0, 63.5 and 61.5 inches — is what the sticks measure once they have been cut to make a panel the right size, so the inset is spent at the saw and never re-derived at the bench. The model's own schedule prints the inset as a fourth length; the build's list is the measured three, and the difference between the two is a model of the joint rather than a measurement of it.

The mitre claim, corrected

The correction, named in full, because this project said it in public and the correction belongs in public too.

An earlier published claim of this project said that no cut in the dome is a mitre. That was wrong, and it was corrected before this book existed.

The true statement is narrower and better. No member end is ever mitred *to another member end*, because the pinwheel never lets two ends meet. But the butt end itself is not a plain square crosscut: it carries the bevel that matches the neighbour's side and the angle that matches the panel's fold — a compound cut, made once, on one end, off the jig, with the same setting for every stick of a class. The whole dome reduces to a handful of saw settings — the bevel is the same everywhere, and the mitre takes a few fixed values — and that is the claim worth making: not "no mitres", but "no mitre is ever measured, and no two ends ever meet".

The corrections printed in the chapters of this part carry the rest of the list. This page is one entry in it: the claim, the error, the corrected version, and the arithmetic that decides which is which — the same treatment the book gives every number it has ever got wrong.

The Seam and Its Key

Where two panels meet, and the angle nobody wants to hear about

This is the least glamorous chapter in the book, and it is the one that decides whether the dome closes.

The frame's 40 panels do not simply sit next to each other. They meet at 55 interior lines, and at every one of those lines two flat triangles fold against each other at an angle that the sphere, not the builder, chose. Between the two members in each seam sits a key — a strip of wood or compressible material cut to the fold — and the key is where the whole tolerance budget of the method finally lands. Get the keys right and the dome closes like a lid. Get them wrong and forty panels of *nearly* closing is a dome with a gap in it, which is what Chapter 5 spent a week learning the hard way.

The chapter has one unwelcome fact to deliver first — the fold angle is not constant — and then two honest ways to answer it, a table of every seam in the dome, and the third option that sidesteps the whole question. It is short, because once the fact is accepted the answers are small. The week was spent on the accepting.

The fold angle is not constant

Say it early, because everything after depends on it: **the angle at which two panels meet is not the same everywhere on the dome.**

It cannot be. A dome is a sphere approximated by flat triangles, and a sphere curves differently in different places. The panels around the five-fold points of the underlying icosahedron meet at one angle; the panels along the belt between them meet at another. A 2V hemisphere has exactly two fold angles — 18.03 degrees and 22.46 degrees in this build — and that is the good news hidden inside the bad one. Not forty angles. Not a continuum. There are exactly 2. The bad news is that 2 is not one, and a key cut to one of them will not seat in the other.

The schedule on the next pages counts them: 25 seams at the tighter angle, 30 at the wider, 55 in all. The difference between the two — under five degrees — is small enough that every instinct says to

ignore it, and large enough that forcing it closed puts the frame in permanent tension, the keys in permanent shear, and the builder in the position the frankendome already visited. The honest answers accept the difference and spend a little labour on it. The dishonest one — "it is nearly the same" — costs the dome.

And here is the one fact that makes the method's peace with it, stated plainly rather than dressed as a theorem: each fold angle plus its leftover gap sums to exactly the sector the tree was split into — 45 degrees. That is true by definition, because the gap is defined as whatever is left of the 45-degree sector after the fold. The gap the key fills beside a 18.03-degree fold is the rest of 45; beside a 22.46-degree fold it is the rest again. The sector you split out of the tree is, exactly, the seam budget the dome needs. Nothing in the method was tuned to make that true; it is a definition, not a discovery, and it is why the key problem has an answer at all.

Raw trapezoid, or shaved flat

Two honest ways to make the key fit the two folds, and the book prices both.

The raw trapezoid. Leave the split faces of the members exactly as the saw left them — each sits 22.5 degrees off its member's centreline — and cut the key to fill the V they make. Cut the seam across and the two sawn faces stand 26.97 degrees apart, with the members' points as the innermost wood and the key lying in the space between them and the meeting line of the two bark faces: a taper whose base is 2.80 inches across, at the inside, and whose tip reaches the apex at the outside. It bears 6.0 inches along the seam. The cost is a key that is not one part but two — one profile per fold angle — and a stock pile sorted by profile. The gain is that the member is never touched by a plane, and every piece of the joint stays exactly what the saw made. The figure below is that section, cut out of the solver's own meshes.

The shaved flat. Plane a narrow land along each member's two sawn faces until they sit parallel to the seam, then a flat key fills each seam — 1.12 inches across as the solver models it, in 2 profiles, one per fold angle. The cost is planing — 0.48 inches at the deepest, which is 10 per cent of the member's depth, over a land only as wide as the key's bearing — and the labour of it, which Chapter 17 works through completely, because the number is smaller than the instinct expects and "smaller than expected" is exactly the claim this book verifies before it repeats.

Both answers are correct; they are two ways to spend the same small labour, and the figure shows both sections side by side so the choice is a comparison rather than a preference. The book's build uses the shaved-flat answer, because 2 profiles is a small number for the pile to be sorted into, and

because a planed land gives the key a true bearing. The tapered key remains the answer for a builder with no plane and no patience for one.

One seam, both ways

The drawing shows the same seam twice. On the left, the raw split faces and the tapered key between them — the key's two edges cut to different slopes, each matching the face it bears on. On the right, the same two members with a narrow land planed flat on each, and a parallel-sided key dropped in the gap — one of the 2 profiles, matched to its fold angle.

The drawing is the decision, and the decision is smaller than it looks. The tapered key moves the work into the key stock — two profiles, sorted and matched. The shaved land moves the work into the members — the same two profiles, one plane setup, repeated at every seam. Either way the work exists; the book simply chooses to pay it where the sorting cannot go wrong. Chapter 33 shows the fixture that makes the shaved land repeatable, and the seam schedule below shows the two angles it is compensating for.

Every seam in the dome

The table is the whole seam budget on one page: all 55 seams, their fold angles sorted, the two extremes named, and the key profile each one takes. It is generated from the solved dome, not surveyed from the build, so it is the plan the frame was drawn to — and the plan the build should be checked against.

Read it as a work order rather than a curiosity. 25 of one profile, 30 of the other, and the check that closes the chapter: when the last panel is in, the number of keys placed should equal 55, the number of profiles used should equal 2, and the dome should close without being forced anywhere. If any of those checks fails, the frame is not the frame this book drew, and Chapter 38 is where the book tells you where the error went.

Hose, key or nothing

The third option, and the one that ends the chapter, because sometimes the honest answer to an angle is a material that does not care about it.

The compressible seam. A gasket — rubber, foam, a length of hose — fills the gap by deforming instead of by matching. It absorbs the fold difference, the drying movement, and the build's own tolerance in one object, and it seals while it does it. What it cannot do is carry structural load the way

a timber key can, which is why the frame's seams are built with a timber key *plus* the gasket where the shell must be weathertight — the key locates the geometry, the gasket seals the gap, and neither is asked to do the other's job.

And the honest nothing. The frame, left bare, closes without any key at all: the butt ends bear directly on the sawn faces, and the seam is a gap the shell chapters will have to deal with. It is not a building; it is a stage of one. But it is a legitimate stage, and building the frame bare first — checking the fold schedule against the real sticks before any key stock is cut — is the cheapest way to find the error the table above was designed to prevent.

The key, then, is the method in miniature: a small part, cut to the two numbers the geometry actually produces, repeated until the pile is gone, and checked against the count when the dome closes. The least glamorous chapter in the book. The one that decides whether it closes.

Let the Connector Hold the Angle

Keep the wood, machine the small part

Let the Connector Hold the Angle

There is a principle underneath this whole method, and it is worth stating on its own because it is the thing that makes rough natural timber buildable to a tolerance at all.

> *The tree supplies the mass. The jig supplies the repeatability. The connector supplies the precision.*

Three jobs. Three different things doing them. Nothing asked to do two.

Precision is expensive per cubic inch

The instinct with an irregular material is to machine it until it is regular. Plane the faces, square the ends, bring everything to a dimension, and then build with confidence.

That instinct is why dimensional lumber exists, and inside a mill it is the right instinct — a mill can afford to hold a tolerance because it is holding it on thousands of pieces with a machine built for nothing else.

A person with a chainsaw cannot afford it, and the reason is not skill. It is that **precision costs roughly in proportion to how much material you have to be precise about**. A long, heavy, awkward member is expensive to hold, to reference, to feed and to correct. A small block is cheap to hold and cheap to remake if you get it wrong.

So there is a choice about *where to put the accuracy*, and the two answers lead to very different afternoons:

One route asks for accuracy 120 times, on the largest and most awkward pieces in the build, with one chance at each. The other asks for it 2 times, on the smallest pieces, and then repeats the result 55

times.

Both end with a dome that closes.

This is not a way of avoiding accuracy. Nothing here is less accurate than the alternative — the seams end up at the same angles either way. It is a question of *where the accuracy is cheapest to buy*, and the answer is: in the small part.

The wedge starts close to useful

There is a piece of luck in this, and it is the reason the wedge suits a dome rather than merely tolerating one.

A rectangular board arrives with two faces at ninety degrees to each other. Ninety degrees appears nowhere in a geodesic dome. Every angular relationship in the frame has to be created from nothing, by cutting.

A split sector does not arrive square. It arrives with two sawn faces that each sit **22.5 degrees** off the member's own centreline, because that is half of one 45-degree sector.

And the seams of this dome fold by 18.03 and 22.46 degrees.

Those are not the same numbers. The wedge does not magically arrive at the right angle — that would be too good, and this book does not do too good. But it arrives *within a few degrees* of what the joint wants, rather than sixty-odd degrees away.

The natural shape of the tree becomes part of the joint system instead of an obstacle that first has to be removed. That is the difference between a material that suits a structure and one that merely survives it.

Two ways to close the difference

The gap between the face you have and the face the joint wants can be closed from either side.

Shape the key. Leave both sawn faces exactly as split and cut a tapered key that fills whatever is left. No wood is machined at all. The cost is that the key is not the same shape at every seam, so there is more than one key profile to make.

Shape the wood. Plane both faces parallel and drop in a plain rectangular key, cut in 2 profiles — one per fold angle — for the whole dome. The cost is a machining pass on every stick.

Chapter 16 covers what each does to the seam. This chapter is about what the second one costs, because it sounds much worse than it is.

What planing the faces actually costs

Here is the arithmetic, and it is the most reassuring table in the book.

The raw split face sits at 22.5 degrees. A flat key wants a face at half the seam's fold angle. The difference between those two is everything that has to come off.

Over a two-inch mating land — you do not need the whole face to bear, only a band of it — the deepest cut anywhere in the dome is **0.48 inches**.

On a member 5.00 inches deep, that is 10 per cent of the member's depth.

So even the "machine the wood" option does not convert a wedge into something resembling dimensional lumber. It removes a narrow sliver from a narrow band and leaves the overwhelming majority of the member exactly as the saw left it. You are still not making boards. You are making one precise interface on an otherwise raw stick.

That is worth knowing before you decide, because the shaved-flat route is often written off as a betrayal of the whole idea. It is not. It is a 10 per cent concession that buys you flat keys instead of tapered ones.

Or do not machine anything at all

There is a third answer, and in some builds it is the best one: make the connector out of something that does not need to fit exactly, because it deforms.

A rubber or closed-cell spline, a length of hose, a compressed gasket — any of these fills a *range* rather than a value. Feed it a seam anywhere between the two fold angles and it takes up the difference by squashing.

What a compliant key can do, all at once:

- absorb the angular difference,
- absorb manufacturing variation in the members,
- seal the seam against weather,
- isolate one panel from another acoustically and against vibration,

- hold a preload so the joint stays tight as the wood moves.

What it cannot do is carry the load a timber key carries. A compliant seam is a weather joint and a tolerance joint; it is not a structural one. So the choice depends on a question you should answer before you buy anything: **is this seam holding the building up, or holding the weather out?**

In a frame where each panel is a complete, self-braced triangle — which is what Chapter 18 is about — a great deal of the shell's stiffness lives inside the panels rather than across the seams, and that is what makes the compliant option viable at all.

Why this is a manufacturing idea, not a dodge

It is worth defending this properly, because "let the connector take up the difference" can sound like an excuse for sloppy work.

It is the opposite. It is how repeatable production works everywhere.

A machine shop does not hold a tolerance on every surface of every part; it holds it on the surfaces that locate and mate, and leaves the rest rough because rough is cheaper and does no harm. An engine block is machined to a few thousandths where the bearings sit and left as cast everywhere else. Cabinetmakers scribe one edge to a wall and leave the other three alone.

The principle is always the same: **decide which surfaces do work, and spend your accuracy there.**

In this dome, the surfaces that do work are the mating faces of the seam and the ends where one member bears on another's side. Everything else — the bark, the taper, the natural bend, the exact diameter — can vary, and the jig is designed so that it can.

The tree is not being asked to be accurate. It is being asked to be strong, which it already is.

Why Every Panel Keeps Its Own Edge

One hundred and twenty members for sixty-five edges, and what the difference buys

One hundred and twenty members for sixty-five edges, and what the difference buys

If you have been adding up as you read, something has not been adding up.

The arithmetic that does not look right

The frame has 40 triangular panels. Each panel has 3 members. So:

> $40 \times 3 = \mathbf{120\ members}$

But the shell only has **65 edges**. Count the lines on a drawing of the dome and that is what you get: 55 interior edges where two panels meet, plus 10 around the open bottom where a panel meets nothing.

So there are 55 more members than there are edges for them to be.

That is not an error. It is the single biggest deliberate decision in this frame, and it costs **1.85 times** the timber a conventional geodesic skeleton would use.

It is worth being clear about the size of that, because it is not a rounding difference. If you built the same dome with one shared strut per edge you would need 65 members instead of 120. The panelised frame uses almost twice as much wood in the frame.

Here is why it is worth it anyway.

What is actually at a seam

In a classical geodesic frame, an edge is one strut, and the two triangles either side of it both fasten to that same strut. Efficient, and the standard way it is done.

In this frame, each panel is built complete — three of its own members, closed into its own triangle — before it goes anywhere near its neighbours. So where two panels meet there are two members, one belonging to each, with the key between them.

Panel A's member. The key. Panel B's member.

The duplicated edge is not lumber somebody forgot to remove. It is the price of a panel being a *thing* rather than a position in an assembly.

What the extra wood buys

Six things, and the first three are why the fortnight in Part One is a fortnight rather than a season.

A panel can be built flat. All three of its members exist and belong to it, so the whole triangle can be assembled on a bench, at waist height, on the ground, in the dry. Nothing about building it requires any other panel to exist yet. In a shared-strut frame you cannot finish a triangle without the struts its neighbours also use, which is why those domes tend to get assembled in the air, strut by strut, at the top of a ladder.

A panel can be checked before it goes up. It has a measurable shape while it is still somewhere you can measure it. A panel that came out wrong is firewood and half an hour; a panel discovered wrong when it is the thirty-eighth thing lifted into place is a much longer day.

A panel can be finished before it goes up. Skin it, seal it, insulate it, run a cable through it, paint it — all on the ground. This is a genuine transformation of the work: it turns awkward three-dimensional construction at height into repetitive two-dimensional production at a bench, which is manufacturing rather than improvisation.

The seam gets room for a gasket. Two members with a gap between them is a place a key, a spline or a compressible seal can live. One shared strut has no such space — the panels either side bear directly on it, and weatherproofing that joint is a separate problem solved with tape and hope. Chapter 17 is about what can be done with that space, and Chapter 16 about what has to be.

A panel can be taken out again. Undo one seam's fasteners and a panel lifts out, leaving its neighbours standing and complete. Nothing else in the shell depended on that panel's members. Replace a rotted one in year fifteen, cut a window into one in year three, take three out to get a bathtub in — these become ordinary jobs rather than partial demolitions.

The frame gains redundancy. Every interior edge has two members in it. One splitting, checking or being damaged does not leave that edge unsupported. It is not a designed redundancy and nobody

should rely on it, but it is real.

The check that has to close

Two independent ways of counting the members have to give the same answer, and this is worth doing yourself once, because it is the check that tells you the frame in your head is the frame on the ground.

Count panel by panel. Every panel has 3 members.

> $40 \times 3 = 120$

Count edge by edge. Every interior seam holds two members, one from each panel. Every rim edge holds one, because there is nothing on the other side of it.

> $(55 \times 2) + 10 = 120$

Both give 120. The software in this project runs exactly this check every time the book is built, and refuses to print anything if the two disagree — because if they ever did, one of the two pictures would be of a building that does not exist, and there would be no way to know which.

When to share a strut instead

This chapter should not pretend the choice is obvious in every case.

Share the strut if:

- **Timber is your scarcest resource.** The panelised frame spends 1.85 members for every unique edge, and if you have one tree where this book's build had 3, that bill decides the question.
- **You have a crew and scaffolding.** The panelised method's main advantage is that one person can do it on the ground. With four people and a lift, in-air assembly is not the ordeal it is alone.
- **The dome is small.** Below a certain size the members are short enough to handle comfortably at height, and the panel machinery is overhead you do not need.
- **You want the lightest possible frame.** For a temporary structure, a greenhouse, or anything that has to be carried in, the weight of the duplicated members is a real cost.

Keep the panel if:

- **You are building alone**, which is what this method was designed for.
- **You want to finish panels on the ground**, which is most of the reason the build fits in a fortnight.
- **You want a weatherable seam**, because the gap is where the seal goes.
- **You expect the building to change**, because panels come out.

This book chose the panel, and every number in it follows from that choice. Chapter 22's 3 trees are 3 trees precisely *because* the frame duplicates its edges. A shared-strut dome of the same diameter would come out of about one.

That is the trade, stated plainly: **roughly twice the wood, for a building one person can make on the ground and take apart again.**

Bends, Knots and Small Units

Why six-foot pieces forgive what sixteen-foot boards do not

Small Units

Every chapter so far has argued from what the method saves — the mill, the haul, the middlemen. This chapter argues from what it *tolerates*, which is the quieter advantage and, for a builder working with whatever grows, the more useful one.

The method's smallest unit is 6 feet: a bucked section, then a wedge, then a member. That smallness changes what counts as a defect. In a stick-frame economy, a bend or a knot ruins a sixteen-foot board and gets charged to the whole load. Here, a bend costs you a section, a knot costs you a stick, and the pile around them is untouched. The method does not need straight trees; it needs short straight *runs*, and a forest full of trees that are "nearly straight for six feet" is a forest full of building material nobody else will pay for.

That is the whole chapter: the defect economy of small units. It is why the method can grade its own stock by eye, why it can build from storm debris and clearing waste, and why the wood it uses never has to travel further than the ground it fell on. Chapter 2 counted the chain the wood avoids. This chapter counts the *wood* the chain avoids — the crooked, knotty, low-grade trees that never get on the truck in the first place.

A bend costs you a section, not a tree

Take the crookedest trunk you have — the one a mill would wave off the landing without pricing. The method looks at it differently.

A bend does not make a tree useless. It makes the tree into two straight logs with a kink between them. Buck the trunk on either side of the bend — cut the kink out as firewood — and what remains is two sections of straight-ish timber, each long enough to split into 8 members. The bend cost you a foot of trunk and a minute of saw time. In a mill's accounting the same bend costs the whole log its

grade, because a mill needs its boards straight over their full length, and a kink in the middle of a sixteen-foot board is a board with no market.

The reason is the unit size. A mill's product is long boards; a bend anywhere in the length ruins the board. This method's product is a 6-foot member; a bend only ruins the member it falls inside. Cut around it and keep cutting. A trunk with three bends yields four good sections and three short firewood rounds, and all four sections are full-value frame stock. The same tree, sawn, would have produced a pile of shorts worth the haul to the dump.

And because the frame's members are short, they can be *short of* straight in a way long boards cannot. A member with a slow bow still sits in the jig, still lands its butt on its neighbour's side, still closes the panel — the pinwheel's overlap takes up what the eye can see, and the panel comes out flat because the jig, not the stick, decides where the corners are. Chapter 30 is that sentence built in wood.

Knots, and where they are allowed

The second defect economy rule: a knot is not one problem, it is two, and only one of them matters here.

A knot mid-span is a stress riser. In a beam — a joist, a rafter — a knot in the middle of the span is where the bending stress concentrates, and a big knot there can decide the board's grade by itself. Even in this frame, a knot in the middle of a member is the worst place for it, because a compression member buckles at its weakest section, and the weakest section is the one with the knot. So the mid-span rule still applies: grade the pile, and keep the heaviest knots out of the middle third of compression members.

A knot at a bearing face is a joint problem. Where one member's butt lands on the next member's side, the load spreads across the whole contact patch — 9.82 square inches of it, as Chapter 10 measured. A small knot inside that patch is a dent in the bearing, not a hinge; the load goes around it through the wood beside it. A knot *at* the contact line, on the other hand, can split the face the screws are pulling against. So the bearing rule: sort the knotty ends away from the butt, and let the head end carry them — the head end gets sawn off anyway, and Chapter 35 makes that offcut free.

The sorting takes an afternoon and a crayon, and it is the difference between "the pile has knots" and "the pile has a plan". The next section gives the procedure.

One bent trunk, salvaged

The drawing shows one crooked trunk twice. On the left, the mill's view: a log that will be rejected or down-graded, because no board can be sawn straight out of it. On the right, this method's view: the same trunk bucked on either side of its bends into straight sections, each section split into eighths, the kinks themselves stacked as firewood.

The two pictures are the same tree. The difference is not in the wood — it is in what each system demands of a length of wood. The mill demands sixteen straight feet, because its product is a board of sixteen straight feet. The method demands 6 usable feet at a time, because its product is a member of 6. The tree that fails the first demand and passes the second is not a worse tree. It is the same tree, measured against a different product — and the trees that fail the mill's demand are exactly the trees that are free.

Grade your own stock

The procedure, because "sort the pile" is advice until it is a list.

One: the bend pass. Walk the pile and mark any member with a visible bow over its length. A slow bow — under a thumb's width over six feet — goes in the general pile; the jig will hold it straight. A hard bow goes to blocking, bracing and stakes, where a curve costs nothing.

Two: the knot pass. Mark the middle third of every member as it comes off the split, and reject to the blocking pile any member with a knot cluster in that third. A single small knot mid-span stays; a knot cluster does not.

Three: the bearing pass. Look at the butt end of every member — the end that will land on a neighbour's side. A knot within a hand's width of the butt face sends the member to the other orientation: flip it, so the knotty end becomes the head end and the clean end becomes the butt. Only if both ends are knotty does the member leave the frame pile.

Four: the pith pass. A member that split along the pith — a crack running down its point — goes to blocking. It will keep splitting under load, and a member with a crack in it has stopped being wood you can trust.

The passes take an afternoon for the whole pile, and they are the entire quality system. No kiln, no grader, no certificate — four passes, a crayon, and the same eyes that will build with the wood. The frame's redundancy does the rest: 40 panels mean a suspect member is a member you can watch, and a watched member that moves is a member you can swap, which is the honest engineering behind "grade your own stock".

Trees nobody else wants

The economic consequence of tolerating variation: the pool of building material widens, and the price of the widened pool is nothing.

Storm debris. Clearing waste. The windfall of Chapter 1. Low-value sawlogs no mill will pay haulage on. A tree too bent or too short for a mill is 8 good sticks a section here, and it is often already lying on the site. Which removes the haul, the mill fee and the markup, and replaces them with fuel and bar oil — the ledger Chapter 46 closes to the dollar.

The point is not that crooked trees are as good as straight ones. The point is that this method's unit is small enough that *a usable straight run* is the requirement, not *a straight tree*, and the world is full of usable straight runs inside trees nobody wants. A builder who can use them builds from the one timber supply whose price never rises: the wood that is already on the ground where the building goes.

And the journey it does not take

Close the chapter where Chapter 2 opened, because the two arguments are the same one seen from different ends.

The industrial route moves wood through 14 operations and a chain of hands: forest, logger, mill, kiln, distributor, retailer, site. This method's route is 8: tree, cutting station, drying stack, jig. Every removed stage is fuel, time and a margin — and every removed stage is also a *filter*, because each middleman selects for the wood worth hauling and grading and selling. The crooked tree never survives the first filter. The bend costs it its ride.

The method's answer is to build where the wood is, from what the filters rejected, in units small enough that the rejection never happens. The chain argument was about money. This one is about material — the same 8 steps, counted as distance instead of cost, and the same conclusion: the best wood is the wood that never had to travel, and the method is the only route that can use it.

Which Way Round Are You Working?

Choose your method before you touch anything

Which Way Round

There are two ways to size a wedge dome, and the first thing you have to do is work out which one you are actually in. Not which one you would like to be in. Which one you are in.

If you get this wrong you will spend a week doing arithmetic that describes a building you cannot build, and you will not find out until the day you go to cut, when the longest stick the frame wants turns out to be eight inches longer than the longest stick you own.

Everything in the next five chapters follows from this one decision, so it gets its own short chapter and no other content.

Design first, or tree first

Design first is the way everybody expects to work. You know what you want: a room of a certain size, a ceiling you can stand under, a footprint that fits between the trees you are not cutting down. You start from the dome and you work backwards to the woodpile. The output is a shopping list — this many sticks, this long, and therefore this much trunk.

Tree first is the way this book was actually written. You have trees. They are the trees you have: a certain height, a certain taper, a certain amount of straight length between the butt flare and the first serious bend. You start from what a section of that trunk can be cut into and you work forwards to the dome. The output is not a shopping list. It is a diameter, and you find it out rather than choosing it.

Both are honest. Neither is better. They answer different questions and they run in opposite directions, and the entire content of Chapters 21 and 22 is those two directions worked through in full.

The honest test

Here is the test, and it is uncomfortable.

Are the trees already down, or already chosen?

If they are — if you are standing next to a windfall, or you have marked two pines at the back of the property, or somebody has offered you a load — then you are in Method B, and picking a diameter is not a decision you get to make. It is a wish. You may still write a number on a piece of paper. The trees will not read it.

If the trees are genuinely not chosen — you will buy or fell whatever the design calls for, and there is enough standing timber that a foot either way on the bucking length costs you nothing — then you are in Method A, and you should design the dome you want.

Almost everybody thinks they are in Method A. Most people are in Method B. The tell is whether you would actually go and get a different tree if the arithmetic asked for one. If the answer is "well, I'd probably make it work with these," you are in Method B and you may as well admit it on page one rather than on cutting day.

What both methods share

Underneath the two directions there is one relationship, and it is worth understanding before you use either method, because it is the thing that makes both of them exact rather than approximate.

In a 2V dome of radius R, every panel is a triangle whose corners sit on the sphere. If the members simply ran corner to corner you could scale them straight off the radius: double the dome, double the stick. But they do not run corner to corner. They pinwheel — each member's end lands on the *side* of the next one rather than on a shared point — and the amount each member is set back from the true corner depends on **how wide the member is**, not on how big the dome is.

A wider stick sits back further. And the width of your stick comes from the diameter of your log, which has nothing whatsoever to do with the dome.

So member length is not proportional to radius. It is radius *minus* a correction that stays the same size while the dome grows. Which means:

- Going from radius to member length is a calculation.
- Going from member length back to radius is that same calculation, solved backwards.

Neither direction is a rule of thumb, a table, or a fudge factor. They are one function and its inverse, and Chapter 23 exists solely to run them against each other and show that the difference comes out to zero — to the last decimal the computer keeps.

Before you go on

Whichever method you are in, you need three measurements before either chapter is any use to you:

1. **The diameter of your log at the middle of a section.** Not at the butt — the butt is the fattest part and it flatters you. The middle, because that is what the average stick will be.
2. **The straight length you can get out of the trunk.** Between the flare and the first bend that would spoil a section.
3. **The longest piece you are willing to carry on your own.** This is a real constraint and it does more to set the dome's size than anything you will read in a book. A six-foot green pine wedge is heavy. An eight-foot one is a two-person lift, and if there is only one of you, that is the end of the discussion.

Write those three numbers down. Then turn to Chapter 21 if you are designing first, or Chapter 22 if the trees are already on the ground.

And then, whichever one you used, read Chapter 23 and check it with the other one. That takes ten minutes and it is the cheapest insurance in the book.

Method A: From the Dome You Want

Radius in, cut list out

Method A

You know what you want. Now find out what it costs in trunk.

This is the method to use when the timber is genuinely not decided — when you will fell, buy or scrounge whatever the design asks for. It runs from the building to the woodpile, which is the direction every other construction book works in, and it is the more comfortable of the two.

It is also the one that will tell you, politely and in advance, when the dome you have in mind needs more trees than you have.

The procedure

1. Start from floor, not radius

Nobody wants a radius. They want a room.

Decide the floor area you want to stand on and convert:

> *radius (feet) = √(floor ÷ π)*

300 square feet gives a radius of 9.77 feet — call it 117 inches. A 2V hemisphere's standing height at the centre *is* its radius, so that same number is also your ceiling.

This is the first place a dome surprises people. A 300-square-foot dome is 9.77 feet tall in the middle, which is more house than the floor area suggests, and the volume is doing something a rectangle of the same footprint cannot.

It is also the first place a dome disappoints people, because not all of that floor has headroom over it. Chapter 44 has the map.

2. Measure the stock you expect to have

Even in Method A you cannot skip this. The pinwheel setback depends on how wide your members are, and you cannot compute a member length without it.

If you have not felled anything yet, use the diameter you expect to be cutting. A 12-inch butt tapering to 8 gives sectors about 3.83 inches wide at mid-trunk, and that is the figure this book's tables assume.

If your logs will be fatter, your members will be wider, your setback will be larger, and every length in your cut list will come out shorter than the tables here. The tables are computed at this book's stock; recompute for yours.

3. Solve the member lengths

For a chosen radius, the layout gives you 4 member classes. This is the calculation Chapter 20 described, run in the forward direction: radius in, lengths out.

At 300 square feet the longest member comes out at 64.5 inches and the shortest at 56.1.

4. Round the bucking length up

Your longest member is 64.5 inches. Buck to 6 feet — 72 inches.

You cannot buck to 64.5 inches. Well, you can, but you should not: a section cut to exactly the longest member leaves no allowance for the head-end overfit (Chapter 35), no allowance for the butt cut (Chapter 34), and nothing at all for the end of a log being split or shaken.

Round up to the next whole foot and take the waste. It is the cheapest line item in the build.

This does mean the design-first path throws away more wood than the tree-first path, because the tree-first path chooses a dome that uses the whole section. That is a real cost of getting to pick your diameter, and it is worth saying out loud rather than discovering in the offcut pile.

5. Count sections, then trunk, then trees

The frame needs 120 members. Each section gives 8, so:

> *sections = 120 ÷ 8 = 15*

> *trunk needed = 15 × 6 ft = 90 feet*
>
> *trees = 90 ÷ 48 = 1.88*

So: **fell 2.**

Notice the number. Designing a 300-square-foot dome from scratch and sizing the trees to it lands on 2 whole trees — the size of dome ordinary pines make, and the reason this method produces houses of roughly this size over and over.

6. Check it against the tree you can actually get

Then go and look at the trees.

If the tallest straight trunk you can find is thirty feet rather than 48, your 1.88 trees becomes three. If your logs are much thinner at the butt than 12 inches, your members are narrower, your sticks are longer for the same dome — and thinner, which Chapter 10 has opinions about.

This is where Method A hands over to Method B, and where Chapter 23's round trip earns its place.

Where this method surprises people

Member length does not scale with the dome. Double the radius and the members do not double. They grow by slightly less, because the pinwheel setback stays the same size while the triangle grows around it.

This is good news at large sizes and bad news at small ones. A very small dome has members so short relative to their own width that the setback eats a serious fraction of them, and below a certain size the geometry stops making sense at all — a stick that is nearly as wide as it is long is not a strut, it is a block. Chapter 48 says where that line is.

Height is not a free variable. In a plain hemisphere, height equals radius. If you want a wide floor and a ceiling taller than half that width, you cannot have both from a hemisphere; you need a riser wall under it (Chapter 41) or a different fraction of a sphere, which is a different book.

The trees number is rarely a whole one. You will get 1.88, or 2.4, or 3.1. Round up and accept the leftover, or shorten the bucking length and see whether the arithmetic lands better. There is no third option that gets you a partial tree.

The lookup table

If you would rather not do any of this, the table below is the whole method precomputed at this book's stock section. Find your floor area on the left and read across.

Two warnings about using it:

1. It assumes members 3.83 inches wide. Different logs, different table.
2. The `trees` column assumes 48 feet of usable trunk per tree. If your trees are shorter — and most are — scale it.

Then check it the other way

You now have a cut list. Before you cut anything, take the longest member off that list, hand it to Method B, and see what dome comes back.

It should be the dome you started with. If it is not, you have made an arithmetic slip or changed an assumption halfway through — a different gasket thickness, a different split count, a different stock width.

That check is Chapter 23, it takes ten minutes, and it is the reason this book has two methods rather than one method and a warning.

Method B: From the Tree You Have

Section length in, dome out

Method B

This is the method the book is named after, and it is the one that produced the dome in the photographs. It starts from a tree lying on the ground and ends with a number you did not choose.

That last part is the whole character of the method. In Method A you decide how big the building is and the woodpile has to comply. Here the woodpile decides, and your job is to find out what it decided.

The procedure

1. Measure the trunk

Three numbers: diameter at the butt of the usable length, diameter at the top of it, and the straight length between them.

“Usable” means the part with no bend serious enough to spoil a section and no branch stub big enough to spoil a stick. It is almost always shorter than the tree. On this book’s tree it was 48 feet, out of a considerably taller pine — the butt flare went in the fire and the top went for kindling.

The tree this book is written around measures 12 inches at the butt of that length and 8 at the top.

2. Pick a bucking length you can carry

This is the decision, and it is a physical one rather than a mathematical one. Go and pick up a green section of the length you are considering. If you cannot get it onto a sawhorse on your own, it is too long, and no amount of wanting a bigger dome will change that on day six.

This book bucks to 6 feet.

Everything downstream comes from this number. Choose it honestly.

3. Count your sections

Divide the usable length by the bucking length and throw away the remainder. 48 feet at 6 feet a section is 8 sections, with 0.0 feet left over.

If the remainder is large it is worth going back to step 2 and trying a different bucking length. A foot of offcut is firewood; four feet is a stick you did not get.

4. Multiply by eight

Each section splits into 8 sectors of 45 degrees each, and — this is the part that surprises people who have milled lumber — **each sector is a finished structural member**. Not a blank to be trimmed. Not stock to be dimensioned. A member.

8 sections × 8 sectors = **64 struts from one tree.**

Three trees: **192 struts.**

The frame needs **120**.

That is the arithmetic the whole book is built on, and it is not a slogan. It is 72 spare — sixty per cent more than the frame needs — which is roughly the right margin for a stick that splits along the pith, a butt cut that goes wrong, and the one panel everybody builds backwards.

5. Find your member's width

The dome does not care how long your sticks are until it knows how *wide* they are, because the pinwheel setback depends on width.

The width of a sector is the chord across its bark face, measured at the diameter you are cutting. Use the diameter at the **middle** of the trunk, not the butt: half your sections come from thinner wood than the butt suggests, and sizing the layout off the fat end gives you a dome that does not quite close at the top.

At this tree's mid diameter the sector is 3.83 inches wide and 5.00 inches deep, pith to bark — a cross-section of 9.82 square inches.

For scale: that is 1.23 nominal two-by-fours' worth of wood, or 1.87 of the dressed boards you would actually find on a rack. Chapter 46 is about which of those numbers to believe.

6. Solve for the dome

Now the arithmetic. The longest member the frame may contain is the longest stick you have, which is your bucking length: 6 feet.

Find the radius whose longest pinwheel member is exactly that.

This is not a division. As Chapter 20 explained, member length is radius less a setback that does not scale, so the relationship has to be solved rather than scaled. The software in this project does it by bisection, and you can do it by trying radii until the longest member comes out right — it converges in about six guesses.

The answer for this tree:

Radius	129.33 in
Diameter	**21.6 ft**
Height at the centre	10.8 ft
Floor	**365 sq ft**

7. Read your cut list

The frame's 120 members come in 4 lengths, running from 62.73 inches to 72.00 inches. Thirty of each.

Note that the longest is exactly your stock length, to the last decimal place. That is not a coincidence — step 6 solved for it — and it has a consequence the book states plainly rather than smoothing over: the longest members use the whole section, so they are the one class cut dead to length, with no overfit allowance. Every other member is shorter than the section, and the difference is where the 6-inch head and the 8-inch butt allowance come from. The longest class gets neither, which is why the bench treats those sticks with the most care — and why the design-first path of Chapter 21 bucks long instead. Chapter 35 is about that.

You do not get to choose the diameter

This is the part people resist.

You measured a tree, you picked a length you could lift, and out came 21.6 feet. Nobody consulted you. If you wanted twenty-four feet you cannot have twenty-four feet — not from these trees, bucked

this way, without doing something else that costs you elsewhere.

You have three ways out and they are all in Chapter 25:

- **Buck longer.** A longer section gives a bigger dome, out of fewer and heavier pieces, and at some point even 3 trees stop being enough. The lookup table later in this chapter shows exactly where that line is.
- **Get another tree.** Fine, if you have one. It is the honest answer and it is not a failure.
- **Accept the dome the trees gave you.** Which is what happened here.

What you must not do is design for the dome you wanted and cut for the tree you have. The frame will not close, and it will not tell you until the last ring.

A note on the third tree

3 trees is 192 struts for a 120-member frame, so the frame strictly consumes 1.88 trees. The third tree — and a slice of the second — is not needed for the frame itself, and the leftover is not waste — it is the spares, the blocking, the floor's stock and the firewood, plus the next dome's first cut list.

But it does mean the honest headline is not "one dome from three trees." It is "one dome from a bit under two trees, plus a third tree because a pile with no spares is a pile that assumes nothing knots, splits, or gets cut wrong."

Which is a worse title.

The Round Trip

Proving the two methods are one calculation

The Round Trip

Two chapters ago there were two methods. This chapter is here to show that there is really only one, held at opposite ends, and to give you a ten-minute check that catches almost every arithmetic mistake you can make with either.

The reason this matters is not elegance. It is trust.

A book can tell you that a 6-foot stick gives a 21.6-foot dome and you have no way to know whether that came out of a solver or out of somebody's memory of a similar dome. Rules of thumb in building books are usually the latter, and they are usually close enough to be dangerous. So rather than ask you to take it on faith, here is the check, and here is the actual number it produces.

Run it forwards, then backwards

Take any radius. Compute the longest member the frame would contain. Now throw the radius away, and using *only* that member length, compute the radius that a dome would need in order for that to be its longest member.

If the two methods are one calculation, you land exactly where you started.

- Start: a radius, any radius.
- Forwards (Method A): radius → longest member.
- Backwards (Method B): longest member → radius.
- Compare.

The residual

Here is the check run on six domes, from small to large:

The worst disagreement across all six is **zero to the last decimal the computer keeps**.

For scale: the kerf of the chain on the saw this book was built with is 0.25 of an inch. The disagreement between the two methods is smaller than the width of the cut by a factor with a lot of zeroes in it. It is smaller than the seasonal movement of the wood. It is smaller than anything you could measure with a tape, a story pole, or a laser.

It is, in other words, zero for every purpose a person with a chainsaw has.

That number is not asserted here — it is the check actually run, and it comes back to zero to the last decimal the computer keeps. If the two methods ever stopped agreeing, the difference would show up here as an obvious number rather than a quiet one.

What would break it

The round trip closes because both directions are solving the same relationship with the same assumptions. Change an assumption between the two and it will not close — and this is exactly what makes it a useful check, because those are the mistakes people actually make.

Change the gasket thickness. The seam key sits between panels and every member length accounts for it. Design at 0.75 inches, then compute your bucking with a half-inch key in mind, and the two answers diverge. The gap will be small and it will be in the wrong direction — your sticks will be very slightly too long, forty times over, and the dome will fight you at the top.

Change the split count. Eight sectors gives a member 3.83 inches wide. Six gives a wider one, twelve a narrower one, and the pinwheel setback moves with it. If you designed for eighths and then split a few logs into sixths because they were fat, those members are not interchangeable with the rest, and the round trip on the fat ones will not close against the layout.

Change the stock diameter. Same problem, from the other side. Sizing the layout off the butt diameter and cutting most of your sections from thinner wood up the trunk means the members you actually get are narrower than the ones you designed. Use the mid-trunk diameter, as Chapter 22 step 5 insists.

Mix up feet and inches. Says itself. It happens. The round trip catches it immediately, because the recovered radius comes back off by a factor of twelve rather than by a thousandth of an inch.

How to actually do the check

You do not need software.

1. Finish whichever method you used. Write down your radius and your longest member.
2. Take the *other* method's procedure from Chapter 21 or 22.
3. Feed it the number you got out, and see if you get back the number you put in.

If you are within a quarter of an inch on the radius, you have not made a mistake worth worrying about. If you are out by more than an inch, stop — something in the middle changed and you should find out what before you cut a hundred and twenty of anything.

Why this chapter exists at all

Because the mistake this is designed to catch is not a mistake in the arithmetic. It is a mistake in the *assumptions*, and those are invisible.

Arithmetic errors announce themselves: a number is obviously wrong, or the units are absurd, or the answer is negative. Assumption drift does not. Everything looks reasonable at every step. You get a cut list of plausible lengths and you cut a hundred and twenty plausible sticks, and the first thing that tells you is the frame, on day fourteen, when the last ring is a half inch short all the way round and there is nothing to be done about it.

Ten minutes with the other method, before anything is cut, is the cheapest insurance in this book.

The Worked Build

Every number for this book's dome, in one place

This chapter is the whole build on five pages: the tree, the dome it makes, the cut list, the seam schedule, and the frame they describe. A builder who wants to work from nothing else can work from this chapter alone — every number in it comes from the same solved model the rest of the book quotes, and the chapter exists so that the numbers have one place to live where they cannot contradict each other.

The two sizing methods of Chapters 21 and 22 converge here. Method A says: pick the floor, and the arithmetic hands back the tree. Method B says: measure the tree, and the arithmetic hands back the dome. The worked build is Method B run to the end on the measurements Chapter 1 took from the windfall, matched by the three trees Chapter 27 fells — bucked, split, and solved into the frame whose numbers follow.

Nothing on these pages was chosen for the pages. It was computed, and the computation is the same one the simulator runs, so the book and the model cannot disagree about the dome.

The tree

The input, measured before any cutting: 12 inches at the butt, 8 at the top, 48 feet of usable trunk, bucked into 8 sections of 6 feet, each section split into 8 wedges. One tree yields 64 struts; 3 trees yield 192, against a frame that needs 120. The recovery arithmetic — 318 board feet in, 278 out, 87.4 per cent kept — is Chapter 13 in full; here it is only the row that feeds the rest of the chapter.

The dome

The output: the solved hemisphere that 3 trees of this taper will fill. 21.6 feet across, 10.8 feet to the crown, 365 square feet of floor. The longest member is 72.00 inches; the shortest 62.73; the whole frame stands on 673 feet of timber. The strict count says the frame actually consumes 1.88 trees, and the book says the rest of the pile is spares, blocking, floor stock and firewood rather than pretending the arithmetic is tidier than it is.

The cut list

All 120 members, by class, length and count, straight from the pinwheel layout. 4 distinct lengths cover the whole frame — the two edge classes of Chapter 9, split into four by the pinwheel's inset of Chapter 15. The table is the jig's job description: every stick that crosses the bench is one of these rows, and nothing else.

Beside it sits the shorter list the build actually came out with — **the master cut list**, 3 resultant lengths: **72.0 inches**, **63.5** and **61.5**. Resultant means *finished*: cut a member to one of these and the panel it goes into is the size the dome wants, with nothing left over to trim. Every equilateral panel takes three of the long member. Every isosceles panel takes the long member as its base, the middle length on the left and the short one on the right.

That orientation is a bench rule rather than a preference, and the shape shows why. Lay an isosceles panel with its base nearest you and the 63.5-inch left member is the longer of the two sides: it reaches over to the top point, which sits off the middle of the base because the two sides are not equal. The 61.5-inch right side runs up to meet it, and it is that shortest member the base butts up to — the base's end lands on the side of the short one, which is the pinwheel's chase closing the triangle. Build a panel with the two sides swapped and it does not close: two inches over five feet is not something the eye catches, so the sides are picked from their two marked stacks, in order, every time.

The two lists differ, and the book prints both rather than picking one. The table above is the pinwheel model's seated schedule, which represents the inset as a fourth length; the master list is what the saw cut and what the panels measured when they closed. The model is a model of the joint; the master list is a report of the build, and where the two disagree, the tape on the finished panel is the authority.

The seam schedule

Every seam, its fold angle and its key: 25 at 18.03 degrees, 30 at 22.46, 55 in all, the two profiles of Chapter 16. The schedule is generated from the solved dome, so the build's check is simple: keys placed equals 55, profiles used equals 2, and the dome closes without being forced.

The dome this describes

The frame itself — the simulator's own solved meshes, in the book's print palette, exactly the dome the four tables specify. If the chapter above is the arithmetic, this page is what the arithmetic is *of*: 40 panels, 120 members, 55 seams, 21.6 feet across, standing.



Illustration. The worked dome: 21.6 feet across, 365 square feet of floor.

When the Numbers Say No

Reading a result that means do not build this

The two sizing methods of the previous chapters end in numbers, and sometimes the numbers say no. This chapter is the permission to listen to them, and the procedure for what to do after.

A book about a build method is at its most useful when it tells you when *not* to build — before the tree is down, before the pile is split, before the jig is built. The failures this method can produce are not dramatic; they are quiet, and they arrive as arithmetic. Two trees that give one hundred and twelve struts when the frame needs one hundred and twenty. A member too heavy for the person who has to lift it. A stick too slender to trust. Each one is a sentence the numbers say, and each one has a standard answer — sometimes a smaller dome, sometimes a fourth tree, sometimes a different bucking length, and sometimes, honestly, not this build at all.

The chapter is short and it is howto, because the decision it teaches is the one that saves the most: reading a bad result as information instead of as a setback, while the fix is still cheap.

Not enough struts

The classic case, and the one the title of this book invites. 3 trees, bucked and split, give 64 struts each — 192 in all, against a frame that needs 120. The arithmetic works out with 72 to spare. But run the same method on trees that are shorter, thinner or bucked longer than this book's, and the count shrinks: 3 trees can come out short of the frame's count, and then the method says no.

Read the shortfall as a measurement of how far off the trees are, not as a verdict on the plan. The frame's count — 120 — is fixed by the geometry; it does not negotiate. The trees' count is the variable, and it is a product of three choices: the usable length you bucked, the section length you chose, and the number of trees you felled. Any one of the three can be changed, and the section below — "Three ways out" — runs the arithmetic for each. The one answer the chapter forbids is the silent one: one hundred and twelve struts do not become one hundred and twenty by cutting more carefully. The count is not a quality issue; it is a quantity issue, and only a quantity fixes it.

The spare margin deserves one honest note, because it is easy to read wrong. This book's 72 spare struts are not generosity; they are the rejection allowance for the grading passes of Chapter 19 — the bent, the knotty, the pith-split that leave the pile. A build planned with zero spares is a build that assumes every stick passes, and no pile does. So the real threshold is not "one hundred and twenty struts"; it is "one hundred and twenty, plus the rejection rate your own pile actually showed you". If two trees give you exactly one hundred and twenty, the numbers are not saying yes. They are saying *tight*, and tight is a no with a schedule.

Too big to lift alone

The second way the numbers say no has nothing to do with counts. A 6-foot green wedge is heavy — the book's own members run 9.82 square inches of wet pine — and the whole method is built around one person carrying, setting and fastening every piece alone. A dome sized by the tree can produce members past the point one person handles, and then the numbers say no in your back instead of in the ledger.

The honest test is not the table, it is the afternoon: pick up the heaviest member the design produces, green and wet, and carry it the distance the build will ask. If it can be carried, set and held with one hand while the other drives a screw, the design passes. If it cannot, the fix is not a stronger builder — it is a shorter bucking length, which makes every member lighter, or a second pair of hands, which Chapter 50 treats as its own design change. The arithmetic follows the carry: shorter sections mean more sections, more splits, and the same frame from lighter pieces — and the tree count re-solves with the new section length, which is the "shorter bucking" row of the section below.

The limit that matters here has a name: the declared longest member one person will handle is 6 feet, and a dome whose members run past that is a two-person build by definition. The sizing methods do not know about your back. You do, and the numbers wait for you to tell them.

Stock too small in section

The third no, and the one that hides inside the first two: the trees are long enough and the count works out, but the sticks are slender — a small butt diameter, split eight ways, yields a sector too thin to trust.

The check is against the comparison Chapter 7 runs: a wedge from a very small log can fall below the dressed board it is supposed to replace, and then the method's central claim quietly inverts. The crossover is real and it is diameter-dependent — below some butt size, the wedge is the wrong answer and the honest move is to buy the boards. The chapter on sections prints the comparison; here the

procedure is the thing: compute your own tree's sector area before committing, compare it against the dressed two-by-four's 5.25 square inches, and if your tree loses the comparison, the numbers are saying no to *this tree*, not to the method.

The options are the same three, with a twist. Shorter bucking does not fatten a stick; it only shortens one. An extra tree adds count, not section. The real fixes for a slender section are a thicker tree, a coarser split — six sectors instead of eight, the row Chapter 12 priced — or a different member entirely: a hollow, laminated or manufactured wedge instead of a solid split one. Those alternatives exist precisely because some trees say no.

Three ways out

The three standard exits, with the arithmetic each one runs, so a "no" from the numbers always lands on a next step.

Shorter bucking. Cut the sections shorter and every member gets lighter and shorter; the dome the tree sizes shrinks accordingly, and the strut count per tree rises. Run Method B again with the new section length — Chapter 22 is a procedure, and a procedure re-runs. The cost is a smaller floor and more sections to split; the gain is a build one person can carry.

A fourth tree. Add a tree and the count problem disappears — four trees of this book's size give far more than 120 — while the member weight and section stay exactly as designed. The cost is one more fell and one more buck, and Chapter 46 shows what an extra tree actually adds in fuel and hours: not nothing, and rarely the thing that kills the build.

A smaller dome. Solve Method A downward — Chapter 21 runs in either direction — until the tree you have is the tree the dome needs. The floor shrinks with the radius, and the shrink is where the honesty of this chapter lives: a dome you can actually build is better than the dome whose numbers said no.

And the fourth option, stated because a howto chapter owes it: **do not build this.** Not every tree, not every site, not every back should become a dome. The method's own arithmetic will say so when it is true, and the builder who hears it has lost an afternoon of sums instead of a fortnight of felling. The numbers saying no are the cheapest form of the no there is.

CHAPTER 26 · STORY

Day Zero: The Saw

A hundred and ninety-nine dollars and a sixteen-inch bar

Day Zero

Before the fortnight there was a day zero, and day zero was the saw.

The whole method claims to need one machine, and the claim deserves to be looked at honestly before the felling starts. The saw this build used was the smallest one on the shelf: a 38.2-cc consumer saw with a 16-inch bar, 199 dollars, bought new. Not a professional felling saw, not a mill, not a rental — the same saw a homeowner buys for firewood. If the method works with this saw, it works with almost any saw; if it secretly needed more, better to find out on day zero than on day nine.

This chapter assesses the tool the way the book assesses everything else: what it can do, what it cannot, what the counterfactual would have bought, and the setup and safety that decide which of the two lists you live on. The fortnight's days each open with a sentence of their own; this one opens the whole clock, because every day after it is this saw's day, and the saw is the only tool in the book.

The smallest saw on the shelf

What 38.2 cubic centimetres and a 16-inch bar can actually do, because the honest list is shorter than the advertised one and more useful.

It fells a modest tree. A 12-inch pine is well inside what this saw drops, as long as the notch and the hinge are cut the way Chapter 27 describes and not the way impatience does. The saw is not the limit at felling; the technique is.

It rips a six-foot section along the grain. This is the real test, and the saw passes it with a margin — a rip through wet pine is slow work, the chain must be sharp and the bar must stay cool, but the 0.25-inch kerf through a 6-foot section is a normal afternoon's work, not a stunt. Chapter 29 is the whole ripping ledger, and its honest finding is that the saw is adequate and the arm is the bottleneck.

It cannot be hurried. A small saw has a small engine, a small oil pump and a small appetite for abuse. Buried to the dogs in green pine, it will pull its chain off the bar, overheat, and dull twice as fast as a professional saw doing the same cut. The method works *with* this saw by working at its pace — 7.5 struts an hour, sustained, is that pace, and the pace is the method.

What it cannot do at all: mill a board, rip a full-length log in one pass, or survive being used like a bigger saw. None of those is needed. The saw's job in this method is exactly three cuts — fell, buck, split — and every one of them is within this machine's honest envelope. The method was not adapted to a small saw; it was designed so a small saw could do it, which is a different and stronger claim.

What I would have bought instead

The honest counterfactual, because a build book that never admits its tool was marginal is not being read by somebody who owns the same saw.

A bigger saw — a professional 60-cc class with a longer bar — would have made the ripping faster, the felling safer-feeling, and the afternoons shorter. It would have cost several times the 199 dollars, and it would not have changed the method by one step. The split count, the bucking length, the jig, the pinwheel — none of them are set by the saw's size. They are set by the tree and the back, and a bigger saw changes neither.

What a bigger saw would have changed is the *margins*: fewer sharpening stops, a cooler chain, a faster recovery when a cut went wide. Those are real, and the book says so rather than romanticising the small saw. But the method's claim was never "this saw is the best tool". It was "this saw is *enough*", and enough is a fact you can verify at the store, in the wood, on day zero — which is the day this chapter is.

If you own a bigger saw, use it. If you own this one, the method works with it. If you own neither, the cheapest saw that will hold a 16-inch bar is the one this book's numbers were produced with.

Setting it up

The setup, in order, because a chainsaw set up wrong is a saw that punishes the setup, not the wood.

The chain. New or sharpened, tensioned so it snaps back against the bar but still pulls around by hand. A loose chain comes off; a tight chain burns the bar. Check it cold, then again after the first cut, because a warm chain is longer than a cold one.

The ripping chain. Ripping with a crosscut chain is possible and slow. A ripping chain — filed to a shallow angle — cuts with the grain instead of fighting it, and it is the single cheapest upgrade the fortnight has. If the budget allows one extra chain, make it a rip chain, and keep it for the splitting days.

Bar oil, always full. Ripping runs the bar buried for minutes at a time, which is exactly the condition that cooks a dry bar. The oil cap is the only gauge that matters on day five through eight; check it every tank of fuel.

The file. A round file matched to the chain, used every few tanks, by hand, to the chain's own angle. Chapter 29 counts what a dull chain costs in kerf and in arms; here the rule is only: sharp is the state the saw arrives in, and you are the one who keeps it there.

Fuel, mixed right. Two-stroke mix at the ratio on the saw, mixed fresh, not last season's. A saw that starts first pull every morning does so because the fuel is right — that is a fuel-quality fact before it is a mechanical one.

Chaps, helmet, and never alone

The safety page, placed here on purpose: the saw starts on day zero, so the rules start on day zero.

Chaps. Chainsaw-protective trousers, worn for every cut, no exceptions for a quick trim. They are the difference between a bruise and an ambulance, and they cost less than the saw.

Helmet, visor and ear protection. A falling branch, a kicked chain, a saw at full throttle for hours — all three are ordinary events in this fortnight, and all three are aimed at your head.

Never alone. The single rule that overrides every other rule in the book: do not run a chainsaw with nobody in earshot. The felling chapter says it again, the ripping chapter assumes it, and the story chapters record the one day it was nearly broken. A phone is not a second person. A second person is a second person.

The book's warning from the front matter stands unrepeated here but not unrenewed: felling kills people who have done it a hundred times, a chainsaw is the most dangerous tool most people ever hold, and nothing in this book is worth what a silent afternoon with a saw can cost. Day zero is the day the tool arrives. The respect it deserves arrives the same day, or not at all.

Days One and Two: Felling

Putting 3 trees on the ground where you want them

Felling

The fortnight's first two days are its most dangerous, and the book does not soften the sentence: felling is the operation that kills people who have done it a hundred times, and these two days are felling.

The job is simple to state — put 3 trees on the ground, where the bucking and ripping can reach them — and every word after that sentence is the reason it is not simple to do. The tree decides where it falls; the feller only influences the decision, through the notch, the hinge and the back cut. This chapter is the influence, in order: the safety that comes before the first cut, the two cuts that steer the tree, the drawing of the hinge that does the steering, the limbing that follows the fall, and the story half — what actually happened on this build's second tree, because the story half is where the chapter earns its place.

Two days for 3 trees is not padding. Each tree is felled, limbed and made safe before the next is touched, and the second day exists because the first tree teaches something the next one needs.

Read this before the notch

The safety page, first, because everything after it assumes it.

The escape route. Before the saw starts, walk the route away from the tree — back and to the side, opposite the planned fall — and clear it. The route is not "away from the tree"; it is a specific, walked, cleared line, and it is the thing you use the moment the tree moves. A tree falls in seconds and the escape takes seconds; the route being pre-walked is the entire margin.

The hinge. The strip of uncut wood between the notch and the back cut is the hinge, and it is the steering. It keeps the tree attached while it starts to fall, aims it along the notch, and holds the butt

down so the trunk cannot kick back. Cut through it — the back cut carried all the way to the notch — and the tree is free to go anywhere, which is how the word “uncontrolled” enters the report.

Barber chair. A leaning tree, cut badly, can split up its own trunk as it falls — the butt tearing open like a hinged lid, fast, at head height. It happens when the hinge fails before the tree is committed. The prevention is the hinge’s size and the order of the cuts, and the respect is: never stand behind a felling tree, not for one second, not to watch.

Widow-makers. Dead branches hung in the crown, freed by the fall or by vibration, that come down before, with, or after the tree. Look up before the first cut. Look up again before the back cut. The crown decides some of what happens next, and it has been deciding since before you arrived.

Two people, minimum. The front matter’s rule, restated here where it binds hardest: never fell alone. The second person is the lookout, the caller, and the one who goes for help — and their one job during the fall is to be out of the fall.

The notch and the back cut

The procedure, in the order that keeps the hinge whole.

Face the fall. Stand where the tree should go. The notch opens in that direction, and the tree will follow the notch unless the hinge says otherwise — so the first decision is the fall line, made from the tree’s lean, the wind and the landing zone, in that order of authority.

Cut the notch. Two cuts meet to take a wedge out of the trunk on the fall side: the first cut slants down into the tree, the second comes in flat from below to meet it, and the wedge falls out. The notch’s depth is about a fifth of the trunk’s diameter — deep enough that the tree has somewhere to rotate into, shallow enough that the hinge behind it is still the width of the trunk. On this book’s 12-inch trees the notch is a hand’s span, and the two cuts must meet cleanly; a notch with a step in it steers a tree sideways.

Cut the back. From the far side, slightly above the notch’s flat floor — a step, not a level cut — the back cut runs toward the notch and stops short of it. The wood left between — the hinge — is a strip about fist-thin, even across the trunk. The tree is now standing on its hinge, committed but held, and it will fall into the notch as the hinge lets go.

Leave. The moment the tree moves, the saw comes out of the cut and the feller takes the escape route, eyes on the crown, without turning a back on the fall. The hinge finishes the job; the feller’s job from here is distance.

The hinge dimension is the whole art. Too thick, and the tree sits on the saw and falls late or not at all; too thin, and the hinge breaks early and the tree falls wherever its weight wants. The even, fist-thin strip is the difference between a fall and an event.

The hinge, drawn

The drawing shows the hinge doing its three jobs at once, because the jobs are the explanation for everything above.

It aims. The hinge is the uncut wood across the back of the notch, and it is flexible only in one direction — along the fall line. As the tree tips into the notch, the hinge bends like a door hinge, and the tree can only go where the hinge bends. Cut the hinge uneven and the tree steers toward the thick side; cut it straight and the notch's direction is the tree's direction.

It holds. While the tree tips, the hinge keeps the butt attached to the stump. That attachment is what stops the trunk from kicking back off the stump as the crown hits the ground — the butt stays pinned, the tree hinges down like a gate, and the fall is one motion instead of a throw.

It lets go. At the end of the fall the hinge's fibres tear, and the tree separates from the stump cleanly. The timing of that tear is set by the hinge's thickness — the feller's one real control over the whole event, and the reason the dimension is measured, not guessed.

Limbing without pinching

The tree is down. The next job is the crown, and it is the part where the saw gets pinched, because a downed trunk is a compressed spring in disguise.

The trunk lying on the ground is bent slightly by its own weight: the top side is in *compression* — the fibres squashed — and the bottom side in *tension* — the fibres stretched. Cut from the top down into that bend and the kerf closes on the bar as the compression settles — pinched. Cut from the bottom up — into the tension side — and the kerf opens as the tension relaxes — safe. The rule: **read the bend, cut the tension side first, or cut from the outside of the bend inward.**

Limb the trunk in sections, cutting the branches off flush with the trunk but never into it — the bucking days want clean wood, not wounded wood. Work from the butt toward the top, on the uphill side of the trunk, and expect every branch to move when its tension releases. A limb under a fallen trunk is loaded wood; the moment the saw frees it, it springs. Stand where the spring is not.

By the end of day one the first tree is a clean, bare trunk on level ground, the crown stacked where the firewood will be cut from it later, and the saw has done the most dangerous work it will do all fortnight.

What actually happened

The story half, recorded the way this book records everything that went wrong: named, not tidied.

The first tree went exactly where it was aimed. The notch was clean, the hinge held, and the trunk landed inside a step of the planned line, which the plan treated as a triumph and the book records as luck doing its job: the tree was straight, the wind was nothing, and the lean agreed with the aim.

The second tree did not go where it was supposed to.

It was the lean. The tree had grown with a slight bow toward the open ground, and the feller — one person, reading the trunk by eye — cut the notch for where the tree *should* fall and the back cut for where the hinge would let it. The hinge steered; the lean overruled; the tree fell a quarter-turn off the aim, taking a young neighbour down with it and landing with its crown in the brush instead of on the clearing.

Nobody was hurt. The escape route was where it was supposed to be, and the miss happened in slow, visible degrees rather than in one event — which is exactly what the hinge, even a mis-aimed one, is for. The lesson cost a morning: the second tree had to be worked out of the brush, the neighbour's remains cut clear, and the fall line for the *next* tree chosen by the lean alone, which is the rule the chapter opened with and the feller had half-ignored.

The book keeps the story because the story is the point of the chapter. The notch, the hinge, the escape route — they are not technique for a straight tree on a calm day. They are the margin for the tree that is not straight and the day that is not calm, and on day two the margin was spent exactly as designed: the tree fell wrong, and the feller walked away. Two down, one to go. The fortnight had begun.

Days Three and Four: Bucking

Cutting the trunk into the only length that matters

Bucking

The trees are down. Days three and four turn each bare trunk into 8 sections of 6 feet — the only length in the whole method, and the number that sizes everything downstream.

Bucking looks like the simple day of the fortnight and is not, because every section after it is the *whole* building in embryo: the section length sets the member length, the member length sets the dome, and a bucking error made on day three arrives on day fourteen as a frame that will not close. The chapter's job is the one measurement, repeated, without drift — plus the two unglamorous subjects that decide whether the day is a one-person job: where the sections start, and how a green six-foot round moves.

48 feet of trunk, cut into 8 sections, offcut 0.0 feet. That is the whole day, counted. The rest is doing it the same every time.

Marking and cutting to length

The procedure, built around the one fact that makes bucking hard: the trunk tapers, so a tape laid along the bark measures around a slope and over bumps, and drifts.

Mark with a stick, not a tape. Cut one 6-foot stick — a story pole — from anything straight, and mark every section with it. The stick does not stretch, does not creep, and reads the same at the butt and at the top. It is the jig before the jig: the length made physical, once, instead of re-read forty times.

Mark a line around, not a tick. One tick on the bark disappears into the cut. Mark the full circumference with the saw's paint or a lumber crayon, rolled around the trunk with a square laid flat, so the cut is guided from any side. A crooked crosscut is a section with sloped ends, and a sloped end is a wedge with one short face — the error that hides until assembly.

Cut the tension side first. The downed trunk is bent by its own weight, exactly as the limbing chapter warned: the top side is in compression and the underside in tension. Cut down from the top — into the compression side — and the kerf closes on the bar; read the bend, cut the side in tension first, or cut from the outside of the bend inward, and let each section's weight drop it free the moment the cut goes through. A section that hangs by a thread of bark is a section that will fall when it wants to; finish the cut.

Cut in the air, not into the ground. The bar through the trunk and into the dirt dulls the chain in one cut. Roll a short log under the cut line, or cut most of the way through and roll the trunk to finish from above. The chain's sharpness budget belongs to the ripping days, not to the ground.

Check the count. 8 sections per tree, counted twice — once as cut, once stacked. The count is the chapter's whole deliverable, and a missing section is a missing 8 members, which the frame will notice before you do.

Where to start, and why not at the butt

The instinct is to start at the butt and measure up. The method says start further in, and the reason is the taper's fault.

The butt end of a felled tree is not cylinder — it is flare. The root swell runs up the trunk a foot or two, and a section cut from the flare is an oval, its rings running wild, its first split a guess about where the pith actually is. So the bucking starts where the flare ends: stand the tape against the trunk, find where the diameter stops falling fast, and make that the first cut. The flare itself becomes firewood — the offcut of the plan below — and every section after it is true round, which is what the split count assumes.

The same rule finds the start around bends. The felling chapter left the trunk with its kinks in it, and bucking is where the kinks come out: cut the sections on the straight runs, leave the bent lengths between them as firewood, and count only straight sections toward the 8. The trunk does not owe you its bends; the plan knows they exist and pays them in firewood.

The top end, too, gets trimmed: the last section before the crown is where the diameter is smallest and the knots thickest, and if the crown side of it looks like a branch knot rather than a log, it joins the firewood pile. The usable length the tree chapter measured was never the whole trunk; it was the straight, flare-free, crown-free run, and bucking is the day the distinction becomes a pile of rounds.

Eight sections from one trunk

The drawing is the day's plan at a glance: 48 feet of trunk, laid out into 8 sections of 6, the offcut — 0.0 feet of flare and crown — marked off and consigned to the firewood pile.

Read it as the book's first real inventory rather than a diagram. Every section in the drawing is 8 future members; the whole trunk is 64 struts; three trunks are 192, against a frame that needs 120. The bucking day is where those numbers stop being the tree chapter's arithmetic and start being a stack of wood you can walk around. If the stack does not match the drawing — a section short, a count off, an offcut bigger than the plan — day three is the day to find out, when the fix is another section or another tree, not another month.

Moving six-foot green sections alone

The unglamorous logistics, because the method's one-person claim lives or dies on this afternoon, not on the geometry.

A 6-foot section of green pine is a wet, round, heavy thing — the butt sections near the flare can be a full day's load on their own. The method does not pretend otherwise; it moves them the way they want to be moved: **roll them**. A round section rolls; the only lift is the short end-over-end to clear a branch or a bank, and the ground does the carrying. Lay the bucking line out so the sections can roll to the splitting station — downhill is free, level is fine, uphill is the day's only real fight — and the "moving" half of bucking disappears into gravity.

For the sections that must lift: never a dead lift of the whole section. Roll it onto a low block, lever the end with a limb, or cut it into two shorter sections and accept the split-count change — the section length is a plan, not a law, and a back is not a substitute for a winch. The peavey, if one is on hand, turns the roll into a one-handed job; without it, the wedge and the lever do.

The real measure of day four is not the pile. It is whether day five can start without lifting a single section twice — the splitting station laid out, the sections staged where the ripping will happen, and the fortnight's only heavy lifting behind you. The bucking days end with 8 sections a tree, stacked dry-side-up, counted, and the saw's chain still sharp for the work that actually makes the members.

Days Five to Eight: Ripping

One hundred and twenty struts, thirty to an afternoon

Ripping

Days five through eight are the bulk of the labour and the heart of the method: every section of the three trunks, split along its length into 8 wedges, until the frame's 120 members exist as a pile. Three trees become a building on these four days; everything before was preparation and everything after is assembly.

The arithmetic sets the pace and the pace is honest: 7.5 struts an hour, sustained, means 16 hours of ripping — 4 afternoons — for the whole frame. That is the number the fortnight's reputation rests on, and it was measured on this build, not promised. The chapter is the measurement made into a procedure: the three passes at the log, the cradle that holds a round section still, the pith you aim at, the rhythm that sustains the rate, and the chain sharpness that the whole four days secretly depend on.

The three passes, at the log

The procedure of Chapter 12, now with the saw actually in hand, at production pace.

Stage the section. Every section gets stood on end or laid in the cradle before the saw starts — never ripped on the ground where it can roll, never half-rolled between passes. The staging is the first third of the work, and a section staged wrong costs more time than a cut made slow.

Pass one: halve it. Aim down the pith line on the end grain and rip the full length. The saw walks itself through a straight section if the bar starts true; the feller's job is to start true and let the saw cut — a ripping saw is steered with small corrections, not pushed. The two halves fall apart with flat faces down.

Passes two and three: quarter, then eighth. Each half sits on its flat face in the cradle and gets halved again; each quarter the same. Seven cuts per section, all of them "through the middle", and the

pile of wedges grows by 8 with every section finished.

Stack as you go. The wedges go straight onto stickers, sawn faces down, bark up, where the drying chapter wants them — not thrown on a pile where the day's own moisture steams them. A wedge stacked right on day five is a straight member on day thirty; the stacking is part of the cut, not the cleanup after it.

Count the pile against the plan. 64 struts a tree, 192 for the three, 120 in the frame, 72 spare for the grading rejects. The count is checked at the end of every afternoon, because a shortfall found on day eight is a fourth tree; a shortfall found on day thirteen is a problem with no good name.

Holding a round section still

The first pass is the hard one for a reason the second and third do not have: a round section wants to roll, and a rolling log under a ripping saw is where pinched bars and ruined cuts are made.

The cradle is the answer, and it does not need to be furniture. Two stakes driven either side of the section, a pair of wedges under it, or a simple V of scrap timber along its length — anything that stops the roll and leaves the top of the section at a working height. The cradle's job is negative space: it holds the log where the saw is not. Once the first pass is made, the halves sit on their flat faces and the cradle's work is mostly done — flat stock on a flat bench is stable by itself, which is the second reason the halving pass goes first, every time.

The wedges earn their place on the *first* pass and again at the end. Driven under a section, they level it, lift it off the dirt, and take the weight so the saw's kerf stays open. A section that settles as you cut closes the kerf behind the bar — the saw bogs, the cut wanders, and the pass is ruined in its last foot. The wedge set under the cut line is the cheapest insurance in the method.

Reading the pith

Every rip aims at the centre of the section, and the centre of a real section is not a point — it is a small, dark, slightly off-centre dot in the end grain, and the first skill of the ripping days is finding it.

The pith is the tree's first year: a soft spot, usually darker than the rings around it, not always exactly centred. The split geometry does not care that it is slightly off — the sectors of a slightly off-centre split are nearly identical, and "nearly" is what the method budgets for. What matters is that every pass goes *through* the pith, so every sector keeps a point that is honest wood, not a rag of pith off to one side. A sector whose point misses the pith by an inch is a member with a soft, split-prone edge exactly where a member most needs honest wood.

On a section that is not round — the oval butt, the flat-sided trunk — the pith still rules. Aim at the centre of the rings, not the centre of the shape; the rings are the tree's own record of where its middle is, and they are right more often than the outline is. The first pass is started from a stance where the pith is visible the whole way — sighting down the bar, checking every few feet — because a pass that starts an inch off arrives a foot off, and a wedge is six feet long.

The cutting rhythm

The chart is the claim and the proof together: 7.5 struts an hour, sustained, for 4 hours, afternoon after afternoon. It is not a sprint rate; it is the rate the body can hold while the saw cools between cuts, the pile is stacked as it grows, and the count is kept.

Read the rhythm as a sequence of stations rather than a series of cuts. Stage, halve, quarter, eighth, stack, count — one circuit per section, the same motions in the same order, the saw never waiting on a decision. The chart's flatness is the point: a sustained rate is a rate with no thinking in it, and the method was arranged so the thinking happened once, in these chapters, and the ripping days just repeat. The afternoons that beat the rate are the ones where something was skipped — a staging, a stacking — and the debt shows up the next morning as a pile to sort and a chain to sharpen before the first cut.

Chain, file, and the cost of a dull cut

The four days' secret dependency: a ripping chain is either sharp or it is spending your arms and your fuel, and the difference is visible in the pile.

A dull chain does not stop cutting. It starts *pushing* — the saw walks off the line, the kerf widens into a groove, and the bar works hot while the feller works harder. The kerf is the honest ledger: every cut is a 0.25-inch-wide line of sawdust, and a dull chain turns that line wide and ragged, eating the recovery figure Chapter 13 computed, one cut at a time. The file is the fix and the cadence is the rule: a few strokes per tooth, every few tanks, *before* the pushing starts, never after. Touch-up filing at the bench costs minutes; a chain sharpened only when it fails costs the afternoon it fails in.

The practical cadence on this build: file at the midday fuel stop, whether or not the chain asks. The ripping days are the saw's hardest work in the whole method — bar buried, grain sideways, wet wood — and the chain that starts the afternoon sharp finishes it dull. The file is the one tool in the fortnight that is used *more* than the saw.

What this is worth an hour

The sidebar's job is to put the four days' labour in the only currency that survives the fortnight, and then to put the caveat beside it.

Splitting your own struts is work, and the work has a rate. The full accounting — the strut valued three ways, the disagreement between the ways kept in the table, and the rate computed against the dressed board that is the honest comparison — is its own subject. Here the summary, once: an hour at the log produces members worth more than the hour's alternative at a wage, *if and only if* the alternative was going to be buying that timber — and the substitution value is not money anybody has been paid. It buys groceries only if the timber was going to be bought anyway.

The ripping days feel like the part of the fortnight where the money is made, and in substitution terms they are — the frame's timber is the single largest line the build avoids paying. But the rate is a rate on *your* hours, priced against *your* alternative, and the full accounting is careful to say which parts of the comparison flatter the method and which do not. The ripping chapter's own verdict is narrower: the four days are the price of the timber, the price is real, and it is paid in afternoons instead of invoices. Whether that trade is worth it is the full accounting's question, asked properly, with the numbers up.

Day Nine: The Jig

One flat bench that makes forty panels the same

The Jig

Day nine is the day that pays for the other thirteen, and it is a single day spent building a single thing: one flat bench on which every one of the frame's 40 panels will be assembled, identical to the last.

The jig is where the method stops being a pile of good ideas and becomes a production line. The wedges are cut, and they are nearly alike — nearly, because the tree was nearly round, the saw nearly straight, the eye nearly true. The jig does not need them to be exact; it needs them to be *held* the same way while the exact parts — the butt cut, the head length, the screw positions — are established by the fixture instead of by the eye. A jig is a decision made once and repeated forty times in wood, and the rest of the book's build chapters are that sentence, carried out.

The full reference for the fixture — every stage, labelled, straight from the simulator — is Chapter 33. This chapter is the story of the day it got built, and the argument for why a day is the right price.

Why a jig and not a tape

The temptation on day nine is to skip the bench and measure the panels one at a time: lay three sticks in a triangle, tape them to length, screw them. It works — for the first panel. The question is what the fortieth panel looks like.

A tape measures each panel against an ideal that lives in the builder's head, and no two measurements of a rough stick agree with each other. The error compounds quietly: a sixteenth here, a thirty-second there, and the panels drift apart in ways that only announce themselves at raising, when two neighbours built a week apart refuse to share a seam. The frankendome — Chapter 3 — was forty panels built exactly this way, and its lesson was the whole reason this book exists: **nothing repeated, so nothing could be checked against the last one.**

A jig inverts the logic. The fixture holds the *position* of every stick — the triangle's shape, the cut lines, the screw locations — as physical stops, and the builder's job becomes loading the stick against the stop. The measurement happens once, when the bench is built; after that, the panels are not measured at all, they are *located*, and locating is an operation that cannot drift, because the stop does not move. Panel forty is built the same way panel one was, and the only difference between them is the sticks — which is exactly the difference the method is designed to tolerate.

Repeatability beats accuracy, when there are forty of something. A jig that is half a degree wrong builds forty panels that agree with each other, and the frame closes; a tape that is read forty times builds forty panels that disagree, and the frame does not. Day nine buys the agreement.

Building it

The bench, in order, kept deliberately simple — the jig's only requirement is flatness, and everything else is stops.

The bench. A flat surface bigger than the largest panel: two sawhorses and a sheet of plywood are enough, levelled and braced so it cannot rock. Flat is the one property that matters, because the panel's truth is measured against the bench; a bench with a twist in it builds a twist into every panel that lies on it.

The triangle. Lay out one panel full size on the bench — the actual member lengths from the cut list of Chapter 24, the actual gasket allowance — and screw down stops along all three outside edges: the triangle's outline, made of scrap, exactly where the members will lie. The stops are the jig's first truth, and they are set once, from the drawing, with a square and a story stick — the tape's last appearance in the whole fortnight.

The fences. Inside the triangle, two more lines of stops mark where the head ends will be sawn off and where the butt ends land: the inset of Chapter 15, built as wood instead of remembered as a number. The fence is the cut line, physical, and the saw runs along it the way a saw always wishes it could.

The cut planes. Where the saw must cut, the jig provides the plane: a straight edge the bar rides against, set at the butt's bevel and the head's square. Chapter 34 gives the angles; the jig holds them; the builder never sights one.

The clamps. Whatever holds a wedge still against the stops while the screws go in — quick clamps, wedges, a foot on a pedal. The member must arrive at its stops and stay; how it is persuaded is the builder's business.

The bench takes a day, and the day is not wasted on accuracy — it is spent on *position*, which is the cheaper thing to be right about. A stop screwed in the wrong place is moved in a minute. A panel built in the wrong place is discovered at raising, in front of a crowd of its neighbours.

The jig, complete

The plate is the finished fixture with a panel loaded on it, at the moment the long heads are being sawn off against the fence — the stage of the work where the jig's whole argument is visible at once.

Look at what is holding the truth. The sticks are rough — bark backs, saw marks, the ordinary drift of split timber. The triangle they lie in is exact, because the stops are. The butt ends are where the fence says, the heads run long past the cut plane, and the saw is about to make the only cuts in the whole panel that have to be right — and they will be right, because the plane the saw rides is the plane the bench made, not the plane a hand held.

That is the day-nine bargain, paid in one afternoon and collected for the rest of the fortnight: from tomorrow, nobody measures anything. Chapter 31 is the production run the bench was built for, and the run is the proof — 40 panels, one after another, each one located, cut and screwed in the same eight motions, and each one leaving the bench the same panel.

Days Ten to Twelve: Forty Panels

The same twelve motions, forty times

Forty Panels

Days ten through twelve are the production run the jig was built for: 40 panels, one after another, on one bench, from the pile of wedges the ripping days left.

The run is where the method's promise becomes arithmetic. One panel is eight motions — load, locate, butt, head, drill, screw, flush, out — and forty panels of eight motions is a production line with one worker, one station, and no waiting. The days go wrong only when the motions stop being the same: a stick sorted into the wrong class, a marking that washed off in the rain, a panel started before the jig was checked. This chapter is the procedure, the sorting, and the two honest notes that make the run real — the fourth panel is the first good one, and the variant count is smaller than the pile makes you fear.

One panel, start to finish

The eight motions, in order, as they will run forty times. Chapter 33 names the same stages from the simulator's side; this is the version your hands do.

Check the bench. Stops tight, cut planes clean, clamps where the hand expects them. Thirty seconds, every morning, before the first panel — the jig's truth is the day's truth.

Sort the three sticks. One long member and one short for the sides, one more of whichever the triangle asks, drawn from the marked piles of the sorting pass below. The sticks arrive at the bench already decided; the bench never decides.

And for an isosceles panel, *which* side each stick goes on is fixed: **the longest — 72.0 inches — is the base, the middle length (63.5) is the left side, and the shortest (61.5) is the right.** Lay the base nearest you and the left member is the longer of the two: it reaches over to the top point, and the short right side runs up to meet it — and it is that short side the base butts up to. An equilateral

panel is three long members and has no such rule to remember, which is most of why the run is quick: 10 of the panels are the easy kind, and the 30 isosceles ones carry the one instruction that must not be swapped. At the bench the two sides look almost identical — the difference is two inches over five feet — so the rule is applied by picking from the marked piles in order, never by eye.

Load the first side. Butt against the butt stop, sawn faces down on the bench, head running long past the head fence. The stop does the locating; the hand only presses.

Butt-cut if it is due. Some classes carry their butt angle from the saw; some are cut at the bench. Either way the cut is made against the plane the bench holds, never freehand. Chapter 34 is that cut in full.

Load the second and third sides. Each lands against its own stops, butts to butts, and the triangle assembles itself under the clamps — three sticks, three overlaps, the pinwheel of Chapter 15 appearing without anybody arranging it.

Drill and screw the overlaps. Through the outer stick into the inner, at the spread the connector chapter established, pulling the overlap closed. The screws are the only fastening in the whole panel, and the jig's holes — or the fence's positions — make them land the same on panel forty as on panel one.

Flush the heads. The three heads, still long, are sawn off against the head fence: one plane, three cuts, and the panel is exactly the triangle the bench describes. Chapter 35 is why the heads were long in the first place; here it is simply the last motion of the panel.

Lift it out. The panel comes off the bench as one finished thing — 3 members, their own edges, ready to be checked, marked and stacked. That is the whole panel. Eight motions, measured in minutes once the rhythm is in.

Sorting the pile

The run only stays a run if the pile does not get to decide anything at the bench, so the sorting pass happens once, before the production days, with a marking scheme that survives the weather.

Sort by class, not by stick. The four member classes of Chapter 9 each get their own stack, marked at the cut — a crayon code on the sawn face, one stripe per class, one colour per tree. The bench pulls from stacks; it never chooses between sticks.

Mark the faces, not the ends. The ends get cut off. The sawn face is where the mark lives through every operation that follows, and it is the face that lies *down* on the bench, so the mark also reads as

an orientation cue: mark up, bark up, wrong way round is visible before the clamp closes.

Mark with what survives rain. Lumber crayon, not pencil. Pencil washes out of wet pine in one night; the crayon is the marking scheme's entire weatherproofing, and the fortnight has already had one rain.

Run the grading passes as you sort. The four passes of Chapter 19 — bend, knot, bearing, pith — are the same walk as the sorting pass, and doing them together means every stick reaches its stack already judged. The 72 spares the tree chapter promised exist as the reject pile, and the reject pile is the margin the run spends when a stick misbehaves at the bench.

The fourth panel is the first good one

The honest note, stated early so nobody panics on day ten.

The first panel of the run is a rehearsal. The motions are new, the clamps are in the wrong places, and the result is a panel you would not hang a roof on. The second panel is faster and nearly right. The third finds the rhythm and the fourth is the first panel that comes off the bench exactly the way the last one will — and the run's real output is not forty panels but forty-three: the first three are practice, the forty good panels that follow are the frame's forty, and the three learners stay behind as the jig's test pieces and spares.

The book says this plainly because the curve is the plan, not a failure of it. The method's whole claim is that the motions repeat, and repetition is what turns the first three panels from mistakes into tuition. The rejected first panels are not waste; they are the cheapest practice the build will ever get, and they end their lives as the jig's own test pieces — the panels you cut into, shim, and check the saw against, so the bench's truth is probed by the very thing it built.

Panel variants

The table closes the chapter with the count that keeps the run sane: how many genuinely different panels the dome contains.

Not forty. The frame's 40 panels are two shapes — 10 equilateral, 30 isosceles, the counts of Chapter 8 — and each shape is built from the same four member classes, so the run is two panel variants in a repeating pattern, not forty originals. The table is generated from the solved dome, and its job at the bench is the same as every table in this book: the pile looks like chaos, the count says otherwise, and the count is right.

By the evening of day twelve the run is done: 40 panels, stacked by variant, the reject pile spent, the bench quiet. The fortnight has two days left, and they are the days the photographs are taken on — the raising, where forty flat triangles become the building the next chapter describes.

Days Thirteen and Fourteen: Raising

Forty panels become one building

Raising

The last two days of the fortnight are the ones the photographs are taken on, and the ones the book is most careful about, because raising is where a frame that was built flat, checked flat and trusted flat suddenly has to stand in three dimensions — and where every error the earlier chapters let through comes home at once.

The job: 40 finished panels become one standing hemisphere, in an order that lets one person do it, held up by temporary props until the shell closes and holds itself — and then, in this book's build, winched off the ground entirely: three cables from the apex hanger to saddles on two standing trees, the brake winch doing the hoisting, and the cabin hanging where the trees hold it, with no footing under it at all. The order matters more than the effort. A dome assembled in the wrong order needs a crane and a crew; assembled bottom ring first, course by course, it needs one person, a ladder, props and a winch — the same economy the whole method has been spending since the tree.

The chapter ends with the honest inventory of what a hanging frame is not, because day fourteen feels like a finish line and the book has one more part to go.

Bottom ring first

The assembly order, which is the whole secret of a one-person raising.

Set the ring. The ten rim panels go down first, laid out flat on level blocks — or on the footing of Chapter 39, for a build that stays on the ground; this book's build uses blocks, because its ground work comes last. The ring is the only part of the frame that touches the ground, and it is the one part that must be right before anything rises off it — its 10 edges set the circle, and the circle sets everything above. Chapter 38 names the ring as the one place error can accumulate, and the check is done here, at the start, when a wrong ring is still a morning's fix: measure the ring across its widest points, both ways, and do not build up until the two numbers agree.

Course by course. The next ring of panels sits on the first, each new panel's base edge meeting the top edge of the panels below it — the seam, the key, the screws, the same joint the bench built, now vertical. Work around the dome, not up one side of it: a dome grows in levels, and a level that closes before the next begins is the prop that holds the next level. The order is not optional; a half-raised dome is a sail, and the wind has opinions.

Key and screw as you go. Every seam gets its key before the neighbouring panel arrives — a seam closed later is a seam closed from a ladder, reaching into a gap that has already tightened. The keys are the schedule of Chapter 16, and the schedule is followed in place, panel by panel, the count climbing toward 55.

Check the ring, again. At every course, one measurement across the open top tells you whether the dome is coming up round or egg-shaped. The check costs a minute and it is the difference between a raising and a rescue; the dome that fails it today gets propped today, not discovered tomorrow.

Propping, and what holds it up before it is closed

The photographs show the finished frame. The raising itself is a building held up by sticks, and the props are the chapter nobody writes.

A dome does not carry itself until the top closes. Until the last panel ties the structure together, every course above the ring is leaning its weight on the course below it, and the courses below are leaning on props — lengths of scrap set against the inside, butted to the ground, holding the shell at its current height. The props are the temporary structure the method admits it needs: not scaffolding, not a crew, just the same wood the frame is made of, pressed into service as furniture.

The prop rule is simple and absolute: **a panel goes up, a prop goes under it, and a prop never comes out until the course above it has closed.** The dome rises prop by prop, and the props travel — pulled from the lower course once the ring above it is screwed closed, carried up, reset. It looks like chaos in the middle and it is not: every panel in a properly propped raising has a load path to the ground at every moment, which is exactly what the finished frame will have, temporarily, in wood.

The one danger the props exist to prevent is the panel that leans out. A panel set at the wrong angle, or a course started before the course below it closed, will try to fall *outward* — and a falling panel takes its neighbours' seams with it, opening the dome like a flower in reverse. The prop is the answer and the order is the prevention; together they make the raising a sequence of small, held, stable moments instead of one long gamble.

Closing the top

The last panel is the hardest, and it is worth saying why before the raising reaches it.

Every seam closed so far was closed against air — the next panel could be offered up, adjusted, persuaded. The last panel closes against a *hole*: every one of its three edges lands on a seam whose other side is already fixed, and the panel must fit the hole the whole dome has been building toward for two days. If the ring was set right, the courses keyed right and the schedule followed, the hole is exactly the panel — the method has been aiming at this moment since the first butt cut. If anything drifted, the drift lives here, in the last triangle of sky.

The order of the last panel's edges matters more than its screws. Offer the panel up gently, key the lowest seam first, then one side, and let the last seam close under the panel's own weight — a final seam that needs forcing is the dome telling you where the error went, and the answer is never more force; it is a shim behind a key, or a prop relieved, or, honestly, a panel taken down and the neighbour adjusted. The method's tolerance budget — Chapter 38 — was built for exactly this conversation, and the conversation is short: the frame closes, or it does not, and it almost always closes.

Then the screws go in, the props come out, and the frame stands on nothing but its own shape — until the winch takes it.

The winch, and the trees

This book's build does not leave the frame on the blocks. It hangs it, which is the step the title page promised and this section reports.

Three cables run from the apex hanger — the mast's lifting ring, or a temporary sling through the crown on a first build — to saddles on two standing trees: two legs to one tree, one to the other, because a tripod of cables on two anchors is what keeps a hanging load from spinning. The brake winch does the hoisting, one slow notch at a time, the frame lifting off its props the way a boat lifts off its trailer, and the moment the props fall free the cabin hangs — the trees holding the whole weight of it, the rim swinging a few feet above the ground. And the rule is plain, stated here because it is the one sentence in this chapter nobody gets to skip: nobody stands under a hanging load, not for one second. The thing overhead was yesterday a pile of wedges; today it is a building, and until it is set and secured everyone stays clear of the ground beneath it.

Say it plainly, the way the floating-dome chapter does: this is a tested construction and a design possibility, not an engineered rating. The trees, the saddles, the cables, the mast and the winch all carry real loads, and none of those ratings is in this book — the engineer's sentence from Chapter 10

binds here harder than anywhere else in the build. The saddles are saddles, not holes: straps that spread the load over bark, so the trees carry the cabin the way they carried their own crowns.

What hanging buys is the deferral the book has been promising: the ground is now *available* — no footing, no ring beam, no pad is owed, because the cabin does not touch the earth. The floor clamps to the mast when it is bought, the skin goes on in the air or on a lowered frame, and the foundation, when it comes, is the last decision of all (Chapter 39). The winch did not just raise the cabin. It unscheduled the ground.

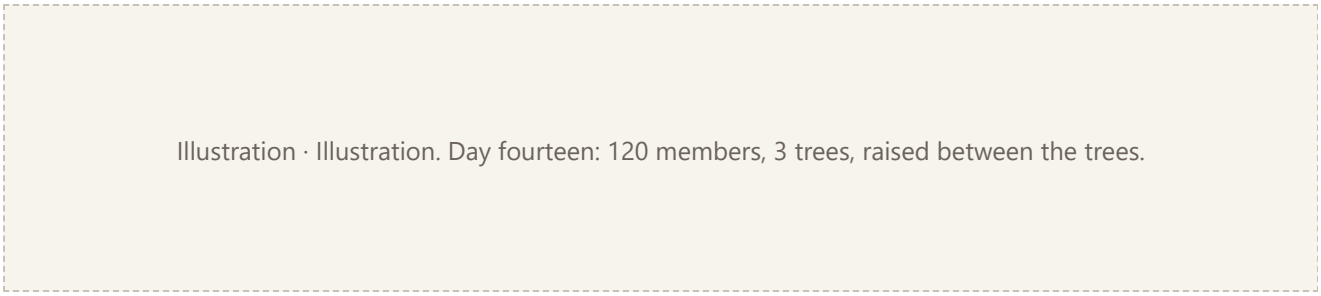


Illustration. Day fourteen: 120 members, 3 trees, raised between the trees.

Standing

The plate is day fourteen: 120 members, 3 trees, a fortnight — the frame this book has been describing since the first chapter, standing where the tree once stood, before the winch took it up between the other two.

Look at it for what the book wants you to see. No hubs. No mitred corners. Forty panels that each remember the bench. The seams running in their two angles, the keys seated, the rim on the blocks. It is the same object the simulator solved before the first tree was felled, and the fact that a solved model and a fortnight's labour can produce the same standing thing — and then hang it — is the method's whole argument, in one picture.

And the next section is the book's own correction of this photograph, because the photograph lies in the way all finished things lie.

What was not done

The honest inventory, written while the frame is still warm from the raising, because day fourteen feels like an end and is not one.

The hanging frame is **not a shell**. It has no skin, no weather barrier, no insulation — the seams are open to the sky's weather, and the first real rain will find the book's weakest chapters and test them. The skin and systems chapters that follow — from 40 to 45 — are the work that turns this photograph into shelter, and it is a season's work, not a fortnight's. The skin can go on in the air or on a lowered frame; the order is the builder's, which is the point.

It is **not a home**. No floor but the air under it, no door but the gap where a panel is missing, no power, no water, no heat. The frame is the part of the building the fortnight was sized for; the rest of the building is sized in months, and the book's later parts are honest about that calendar.

And it is **not finished** — it is *standing*, or rather *hanging*, which is the fortnight's actual deliverable and no more. The dome will move as its green wood dries, the keys will settle, and the frame the builder checks on day thirty is not quite the frame of day fourteen. That settling is a real event in the build's first year, and this book's habit is to name it, measure it where it can be measured, and leave it in the frame rather than tune it out.

So the fortnight ends the way it began — with an honest sentence. 3 trees, one saw, fourteen days, 37.5 hands-on hours: the frame is up, and up between the trees. The rest of the book is what the frame makes possible, and the first thing it makes possible is the next chapter, where the method's reference pages begin.

One Bench, Forty Panels

What the jig is and what each part of it does

One Bench

This is the reference chapter for the fixture itself: what the jig is, part by part, straight from the simulator that models it, so the bench on your floor and the bench in the model cannot disagree about what a jig is.

Chapter 30 told the story of the day it was built. This chapter is the thing, cold: the twelve stages a panel passes through on it, the labelled drawing of the fixture before any wood is on it, and the procedure for laying the whole thing out full size on a real floor with nothing but string and a pencil. Come back to this part while you are working, not while you are reading — the book's reference chapters are written for the bench, and this one lives there.

The twelve stages

The table is the jig's whole working life, generated from the simulator's own stage list: every stage a panel passes through between the raw pile and the finished triangle, named and ordered.

Read it as a checklist, not as a description. The twelve stages are the production run of Chapter 31, expanded to the resolution the bench actually works at — where each member is located, what gets cut when, what gets screwed when. The order is not optional in the way a recipe's order is optional; the head overfit of Chapter 35 only works because the flush cut happens at the stage the table says, after the panel is already a panel and before it leaves the bench. A stage skipped is an error invented, and the table exists so the skip is visible.

The jig, labelled

The drawing is the fixture before any wood is on it: the bench, the triangle, the rails, the fences, the cut planes — every component called out, each with its one job.

The bench is the flat truth everything else is measured from. **The triangle** is the panel's outline, built in stops around the outside. **The rails** carry the saw where a cut must be made. **The fences** are the lines the butts land against and the heads are cut against. **The cut planes** hold the angles of Chapter 34 so no hand has to. Between them they do the one thing a jig must do: they replace every decision with a position, and the position does not move.

The drawing comes from the simulator's own jig model — the same geometry the fabrication package exports — so the picture and the fixture are the same object, drawn twice. If the bench you build disagrees with this drawing, one of you is wrong, and the simulator is not the likely one.

Setting it out on a real floor

The full-size layout, because the jig's accuracy is bought on this afternoon and nothing after it.

Start with the longest member. Draw the first side of the triangle full size, straight off the cut list of Chapter 24 — the actual length, gasket allowance included, marked on the bench surface with a pencil and checked twice. This line is the triangle's truth; the other two sides are located from it.

Swing the arcs. From each end of the first line, swing an arc at the length of its neighbouring side — string and a pencil, the oldest layout tool there is, and still the straightest for distances bigger than a square. Where the two arcs cross is the third corner. The triangle is now on the floor, full size, and the only measured numbers in it were the three member lengths.

Check the angles against the drawing. Before any stop is screwed down, compare the laid-out triangle against the panel drawing of Chapter 37: the base angles, the heights, the relation between the long and short sides. A layout that matches the drawing within a pencil line is the jig's foundation; a layout that does not is the cheapest error the whole build will ever catch.

Screw the stops. The outline becomes physical: stops along all three outside edges, then the fences inside for the butts and the heads, then the cut planes. Each stop is set against the pencil line, not against the previous stop — a stop set from a stop compounds its neighbour's error, and the whole point of the layout is that every position is measured from the drawing once.

Prove it with the first panel. The first panel off the bench is the test: lay it back over the layout lines, and if its edges fall on the pencil, the jig is true. The fourth-panel rule of Chapter 31 applies here too — the first panel proves the bench, and everything after trusts it.

Locate from the sawn faces, never the bark

The last section of the reference, and the principle that makes a jig work on natural timber at all.

Bark is irregular. No two sticks share it, it compresses under a clamp, and it is the one surface of the wedge the saw never touched. The two sawn faces are the opposite: flat, because the saw made them flat, and at the same angle on every stick, because every stick is the same sector of the same circle. So the fixture locates the member on its sawn faces, and only on its sawn faces. The bark floats in an oversized clearance and is never asked to locate anything.

The rule reads as fussiness and is the whole tolerance budget of the method in one sentence: precision where precision matters, and tolerance where the tree varies. The sawn faces are the precision; the bark is the tree varying; and the jig that confuses them builds forty panels, each as individual as the log it came from — which is the frankendome, again, wearing a better bench.

Chapter 38 follows the same principle out into the whole build. This chapter ends it where it starts: one bench, two datum faces, forty identical panels.

The Butt Cut

The one compound angle, made before anything is assembled

Name it honestly at the top, because this project got it wrong in public once: the butt cut is a compound cut. A mitre and a bevel together, on one end of every member, made once, off the jig, before anything is assembled. The claim this book makes is not "no mitres" — Chapter 15 has the full correction of that claim. The true claim is better: no member end ever meets another member end, and the one compound cut the frame does need is made once per stick, against a fixed setup, instead of twice per stick against a guess.

This reference chapter is that cut: where its angle comes from, how few distinct setups the whole dome actually needs, and the procedure for making it with the allowance still attached. It is the shortest of the jig chapters and the one the correction in Chapter 15 points at most, because it is where the book's honesty and the saw's geometry finally agree.

The angle its neighbour presents

The butt end of a member is not cut square to the stick's length. It is cut to match the two facts about the neighbour it will land on: the neighbour's side leans away from the panel's plane by the sector's angle, and the panel itself folds toward its neighbour by the seam's angle. The cut that lands flat against both is a compound — one angle in the plane of the panel, one across it — and that compound is the butt cut.

Where the angles come from, precisely: the member's sawn face sits 22.5 degrees off the member's centreline, so the butt must present that angle to sit flat on its neighbour's face. The panel's fold adds the seam's angle on top — the 18.03 and 22.46 degrees of Chapter 16. Together they make the one cut the dome cannot avoid, and the reason it cannot be avoided is the same reason the seam key exists: the sphere curves, the panels fold, and the wood that meets that fold must be shaped for it.

The good news, stated as a count: the whole dome needs only a few distinct setups, not one per member. The bevel is the same everywhere, and the mitre takes only the values the member classes demand. The table on the next page is the complete list, generated from the simulator's own setup

schedule — and the fact that the list is short is the second half of the correction the pinwheel chapter opened: the compound cut is real, and it is *few*, and "few, fixed, and made on the jig" is the honest version of the claim this project got wrong in public.

Butt cut setups

The table is the entire saw-setting vocabulary of the build: every distinct setup, straight from the simulator's own list, with the member classes each one serves.

Read it as the answer to the question the sceptic is owed: if the butt is a compound cut, how many compounds are we talking about? The answer is the table's length — small enough to write on the wall above the saw, fixed enough that a mistake is a failure to read, not a failure to calculate. The saw is set once per setup, the class is cut through, and the setting is never re-derived at the bench. That is the discipline the whole method runs on: measure the angles once, on the drawing, and let the jig hold them forever.

Making it

The procedure, at the saw, with the allowance still attached — because the butt cut is never made on a stick that is already its final length.

Leave the allowance. The member arrives at the butt cut with the spare stock of Chapter 35 still on its head end and the butt allowance — 8 inches of sacrificial stock — still on its butt end. The allowance is not sloppiness; it is the room the cut needs to be made *against the jig* rather than against a tape, and it is the reason a mis-aimed cut is a do-over instead of a ruined member.

Set the fixture, not the saw alone. The butt cut is made with the stick located on its sawn faces in the jig's cut plane — the saw's own angle settings do the compounding, but the *position* of the cut along the stick comes from the stop, and the stop is the jig's. A butt cut made freehand is a compound angle guessed twice; a butt cut made in the fixture is a compound angle held once.

Cut, check, and only then cut the class. The first stick of a setup is the proof: cut it, offer it to its neighbour, and read the contact. A butt that sits flat across the neighbour's face is the setup's certificate; a butt that rocks on one edge sends the saw setting back one degree and the proof is repeated. Only when the first stick passes does the class get cut — the same rule as the jig's first panel, applied to the saw.

Keep the allowance until the member leaves the bench. The butt allowance is removed at the stage the jig schedule says, not at the saw. Between the butt cut and the bench, the allowance is still

protecting the cut — and the head overfit is still protecting the length. The two spare ends are the method's crumple zones, and Chapter 38 is the accounting of everything they absorb.

One cut, made once, on one end, off the jig. The butt cut is the whole of the frame's precision joinery, and it fits in a table, a fixture, and an afternoon — which is what "no two ends ever meet" was always worth.

The Head Overfit

Deliberately cutting it too long

The head overfit is the idea that makes the whole method tolerant, and it is the single most counter-instinctive rule in the book: **cut every member deliberately too long.**

A builder trained on dimensional lumber works measure-cut-fit: measure the stick, cut it to the number, fit it into the building. Rough timber will not survive that order. The stick is not the shape the tape assumed — it bows, it tapers, its butt lands where the neighbour's face happens to be — and a stick cut to the plan's length before it meets the building is a stick that arrives wrong. So the method reverses the order: cut long, locate in the jig, and let the fixture — not the tape — decide where the end actually falls. The overfit is the deliberate slack that makes the reversal possible, and this chapter is its economics: where the error goes, how much to leave, and what the tolerance costs in wood.

Where error goes

The core argument, in one image: the head end of every member runs 6 inches past its final position, and those inches are where the build's error goes to die.

Follow the sequence. The member is cut long at the split. It goes into the jig with its butt located against the stop — the one position the member must hold — and its head running free, past the head fence, into the overfit zone. The panel is assembled: the overlaps screwed, the triangle pulled tight, the neighbours defining the panel's truth by *being* the panel. Only then — with everything located, clamped and true — does the saw run along the head fence and take the overfit off. The cut is made in place, against the panel the member actually lives in, and the member ends up exactly the length the *assembly* required, not the length the tape predicted.

Every error the earlier steps could have introduced — a butt that landed a quarter inch deep, a neighbour whose face sits proud, the ordinary drift of split timber — was absorbed before the cut, because the cut is the last operation and everything before it had room to be wrong. The overfit is the method's crumple zone: 6 inches of deliberate waste on every member, in exchange for every member being final-cut against reality.

The sidebar rule that pairs with it, stated once here and used everywhere after: **locate first, cut second.** The fixture establishes the cut; the cut never establishes the location. The maxim is the opposite of how most people are taught to work with a tape, and it is the sentence that makes rough timber buildable at all.

Long, then flush

The drawing shows the same panel twice: first with all three heads running long past the fence — the overfit still attached, the panel already a panel — and then after the flush cut, the heads sawn down to the fence line, the panel exactly the triangle the bench describes.

The two states are the chapter's whole teaching. In the first, the panel is *assembled but not decided*: everything about it is final except its outside edges, which still carry the slack. In the second, the slack is gone and the edges are true — true to the bench, not to a measurement, which is the only truth a panel's edges can have that its neighbours will agree with.

Notice where the saw is in the drawing: riding the fence, the bar flat, cutting each head against the same fence plane. The three heads sit at three different corners of the triangle, so the cut is made one member at a time — 3 cuts to the one plane — and it is the only operation in the whole panel where accuracy and *finality* meet. Everything before it was reversible; after it, the panel is committed. The overfit is what bought the reversibility.

How much to leave

The overfit's size is a build decision, and the book declares its own: 6 inches. The number comes from what the overfit must absorb, and the honest way to size your own is to count the variation in your own pile.

The overfit must cover: the drift in your splitting, the taper of your sections, the seating of your butts against the stops, and the settling of the sticks under the clamps. If the worst stick in your pile wanders half an inch from the plan's ideal, a half-inch overfit leaves the good sticks wasting half an inch and the bad sticks with nothing to spare. So measure the pile, not the plan: the overfit is set to the *spread* of your stock, plus a margin for the day you are tired. 6 inches was this build's spread plus margin; a tighter pile could run less, a looser one more.

The trade is one-dimensional: more overfit costs wood, less overfit costs do-overs. The table below prices the wood; the do-overs price themselves the first time a member arrives at the head fence already short, which is the one failure the overfit cannot fix — because the overfit *is* the fix, and a stick cut short never had any.

Offcut per dome

The table is the tolerance's bill, and the book prints it the way it prints every bill: in full, with the units the builder actually pays in.

Every member sheds its overfit into the offcut pile: the flush cuts of Chapter 36, one per member, 120 members, adding up to the figure in the table — board feet of perfectly good wood, cut off and set aside. The honest note is that the offcut is not waste in the moral sense: it is blocking, bracing, wedges and firewood, and the method was sized for it from the start — the tree's spare struts and the offcut allowance are the same budget, counted in different units. But it is wood the frame does not stand on, and the table says how much, because a tolerance that is not priced is a tolerance that will quietly eat the savings it was bought with.

The overfit's ledger closes where the chapter began: cut long, locate, assemble, flush. 6 inches of slack per member, a table's worth of offcut per dome, and in exchange — a frame whose every member is final-cut against the panel it actually lives in. That is the cheapest accuracy there is, and the next chapter is the cut that collects it.

Flush-Cutting in Place

The cut that makes the panel true

Flush-Cutting

The flush cut is the final operation on the jig and the moment the panel becomes itself: the three overfit heads of Chapter 35, sawn off against the fence, in one plane, so the assembled triangle ends up exactly the triangle the bench describes.

It is a small cut and the book gives it a chapter anyway, because the order of operations it completes is the method's entire tolerance argument, and because the cut is the one place a hurried builder can undo, in thirty seconds, the 6 inches of slack the whole design paid for. The chapter has two sections: how the cut is made, and why it cannot be made any earlier — the second being the reason the first is written at all.

Against the fence

The procedure, with the saw held the way it has to be held — against the fence, never freehand.

The panel stays clamped. The flush cut is made on the assembled, screwed, clamped panel — the panel does not come off the bench for the cut, because the bench is what makes the cut true. The clamps hold the members against their stops; the saw rides the fence; the two truths together are the cut.

The bar rides the plane, flat. The saw is laid on the fence's cut plane, bar flat to the bench, and drawn through each head in turn — one member at a time, never all three in one motion, because the three heads sit at three corners of the triangle. The saw does not sight anything; it follows the plane the bench already holds, which is the only way a hand-held saw makes a square cut at all. And a chainsaw note, worth its own line: with the bar flat to the bench, the nose and the upper quadrant of the tip are the kickback zone — keep the tip clear of the fence, the stops and the clamps.

Cut all three heads on the same plane. The three heads stick past the fence by different amounts — the panel is a triangle, and its overfits land where they land — but the *cut line* is one plane, and all three cuts are made to it. That single plane is the panel's final edge geometry, and a panel whose three heads were cut to three planes is a panel whose neighbours will measure the difference.

Do not finish the stick, finish the panel. The temptation at the saw is to clean each head as it is cut — a shave here, a touch-up there. Resist it: the flush cut's product is the panel's truth, not three pretty ends. The offcuts go to the pile, the panel goes to the stack, and the next panel takes its place. The cut exists once per panel, not once per stick, and treating it as a production step is what keeps the run of Chapter 31 at its pace.

Why this cannot be done first

The order is not arbitrary, and the reason is the whole tolerance argument in miniature.

The flush cut cannot be made before assembly, because before assembly there is nothing for the cut to be true *to*. A member cut to its final length at the split is a member cut against a tape's idea of where its neighbours will be — and its neighbours, when they arrive, will be wherever the wood actually is. The panel is the only object that knows the member's true length, and the panel only exists after the butts are located, the overlaps screwed and the clamps closed. So the cut must come last: it is the final measurement, and the fixture performs it.

It cannot be made *after* the panel leaves the bench, either — off the jig, the panel's own edges are the only reference the saw has, and a saw referenced to the thing it is cutting is a saw cutting in circles. The bench's fence is the fixed truth; the panel's edges are the variable; and the order that works is the one where the fixed truth cuts the variable.

So the sequence stands as the book has built it: split long, butt against the stop, assemble, flush against the fence. Cut long, locate, assemble, cut true. The flush cut is the last word of the sentence, and the sentence is the method — every member of every panel final-cut in place, against the only truth that could know it.

Panel Drawings

One flat drawing per distinct panel, full size

The book's last word on the panel is a drawing, not a paragraph — one flat, full-size drawing per distinct panel, exported directly from the simulator, so the drawing on the page and the panel on the bench are the same object.

The drawings are the check for everything the jig chapters described. The jig locates members against stops; the drawing is what the stops were set from. The panels come off the bench; the drawing is what they are measured against before they are stacked. And because the drawings export from the same solved model that produced the cut list and the seam schedule, the whole build has one source of truth, and it is printed on these pages.

Panel drawing, type one

The first of the two distinct panels, flat, full size, exactly as the simulator exports it — the member lengths, the overlaps, the vertex gap, the seam allowance all in their true positions.

Panel drawing, type two

The second distinct panel. Between them, the two drawings describe every one of the frame's 40 panels: the 10 of one type, the 30 of the other, the counts of Chapter 8.

Reading a panel drawing

What every line on the drawing is, so the page can be read at the bench and not only admired.

The outline is the panel's finished edge — the triangle the neighbours will meet, gasket allowance included. **The member lines inside it** are the three sticks: butts located on sides, heads running to the fence line, the pinwheel's overlaps shown as the overlaps they are. **The vertex gap** is the empty corner of Chapter 15 — drawn, not described, so nobody looks for a hub there. **The seam**

allowances are the half-gasket insets at every edge, the 0.75-inch key's share of each side, which is why the outline is slightly inside the triangle the members describe.

The drawings print at reduced scale on these pages; they are full size in the vector files, where they scale to the bench without loss. A full-size print laid on the jig's surface should match the jig's stops line for line. That is their real job — not illustration but *witness*. When a panel comes off the bench, it is laid on its drawing; when a jig is first set, the drawing is what the stops are checked against; and when the frame finally closes, every panel in it has, somewhere in its history, been held up against one of these two pages and found to be the thing the pages describe.

The reference part of the book ends here, with the drawings — because after the drawings there is nothing left to know about the panel. The next chapters leave the bench and follow the frame to the ground it stands on.

Where Error Goes

A tolerance budget for a building made of split logs

Every chapter so far has mentioned the method's tolerance — the overfit absorbing a sixteenth, the key absorbing the fold, the panel owning its edges. This chapter is the budget behind the mentions: a complete account of where error can land in a frame built from split logs, which of those places the method absorbs, and which one it does not.

A building method that cannot say where its error goes is a method that discovers the answer at raising. This book's method can say it, because it was designed around the saying: every operation in the fortnight either sends its error somewhere specific or refuses to make the cut that would. The chapter walks the four landing places — stock length, butt angle, seam gap, and the ring — marks three of them as absorbed and one as accumulating, and ends with the check that keeps the last one honest. It is the most structural chapter in the book, in both senses of the word.

Four places error can land

The complete list, because a tolerance budget that omits a landing place is a budget with a hole in it.

Stock length. The member arrives at the bench longer than it should be — the split drifted, the bucking was generous, the plan was optimistic. This is the commonest error in the build and the cheapest, because the overfit exists for exactly it: the head runs long, the assembly locates the butt, and the flush cut of Chapter 36 removes the error with the offcut.

Butt angle. The compound cut of Chapter 34 lands a degree off, and the butt no longer sits flat on its neighbour's face. The error is real but bounded: the butt's job is to present area to the side, and a butt a degree off still presents nearly all of it — the screws pull the overlap closed, and the contact settles into the wood. What the method refuses is the opposite error: a butt cut *short*, which is why the allowance stays on until the bench decides.

Seam gap. Two panels meet and the gap between them is a fraction wider or narrower than the key's share — the fold is right but the wood is itself. The seam absorbs it by design: the key compresses, the

gasket gives, and the 55 seams of the frame share the difference so no single seam carries it all. This is the landing place the whole seam system was built for, and the one the book spent its middle chapters engineering.

The ring. The base ring is where the frame meets the ground, and it is the one landing place that does not give. A sixteenth of error in the ring is a sixteenth of error in the circle the whole dome stands on, and the dome cannot compress its way out of a wrong foundation. The ring is the subject of the last section, because it is the budget's one honest failure point.

Three of them are absorbed

Why three of the four land safely, and what "absorbed" actually means in wood.

An absorbed error is one that leaves the build as something else: offcut, compression, distribution. The stock-length error leaves as the flush cut's offcut — Chapter 35 priced it, and the price is the board feet in its table. The butt-angle error leaves as contact: the overlap's screw row pulls the butt into the wood until the bearing is full, and a degree of bevel error becomes a fraction of an inch of settlement that the frame never sees again. The seam error leaves as compression: the key takes up the difference, the gasket breathes it, and the dome closes with the error spread invisibly across 55 joints.

The three share one property: each is absorbed at the *place it was made*, by the design of that place, without travelling. The stock error never leaves the bench. The butt error never leaves the joint. The seam error never leaves the seam. That locality is the whole secret — an error that dies where it is born cannot compound with its neighbours, and a frame built from rough wood needs exactly that: forty small absorpions instead of one long accumulation.

The tolerance budget

The drawing is the budget as a flow: error enters at the four places, three of them send it sideways into absorption — the offcut pile, the bearing, the gasket — and one of them, the ring, passes it on.

Read the arrows as the method's real structure, because they are. Every chapter of the build is somewhere on this drawing: the overfit is the stock-length arrow, the screw row is the butt arrow, the key is the seam arrow, and the ring is the arrow with no exit. A builder who has internalised the drawing stops fearing the ordinary errors of rough timber — they are budgeted — and starts fearing only the one the drawing marks in the colour of warning.

The one that accumulates

The ring, named and checked, because it is the budget's single honest failure point.

The base ring's error does not absorb; it *accumulates*. The ring is a closed circle of 10 panels, and an error anywhere in it is an error everywhere: the dome above it rises from whatever circle the ring actually is, and every course multiplies the ring's truth by the frame's own height. A ring set a half inch out of round is a dome that leans a half inch at the base and more at the crown — the one error in the method that grows as the building does.

So the ring is where the book's checking discipline concentrates. The check is simple and it is done three times: when the ring is first set, measure across every opposite pair of its 10 sides — the numbers must agree within the gasket's give, and any pair that does not is the ring telling you where it is wrong while the fix is still a shim. When the first course closes, measure the ring again — the courses above have already begun leaning on it, and a ring that moved has moved the whole frame's opinion of round. And before the top closes, once more, from outside: the dome's last chance to be told, cheaply, what the raising is about to make permanent.

The ring's honesty is the book's last word on tolerance. The method forgives stock, forgives butts, forgives seams — and it never forgave the ring, because the ring is the one place where forgiveness would have to come from the ground. Chapter 39 gives the ring its footing; this chapter gave it its warning. Build accordingly.

The Footing and the Ring

What the dome stands on

The Footing

The frame is up — hanging, in this book's build, between the trees — and the question of what it stands on, if it ever stands, is the one the fortnight deliberately left open: the ground work comes last, and it may never come at all. This chapter is the footing for the day it does — the part of the building under everything, the part the photographs never show, and the part where the book's own warning from Chapter 38 is most literally true: the ring's error accumulates, and the footing is the ring's floor.

A dome's weight arrives at the ground along one circle — the base ring's 10 edges, ten points where the panels meet the earth. The footing's whole job is to receive that circle and hold it: level, round, and against the wind that wants to lift the whole thing away. The chapter covers the two ways to do it, the layout that keeps the circle true, the uplift a dome invites, and the point where the book stops and the engineer starts.

Ten points, or one ring

Two approaches carry a dome's weight to the ground, and the book prices both, leaving the choice to the build that comes down from the trees.

Ten points. Each of the ring's ten base vertices sits on its own small footing — a pad block, a pier, a driven post — and the frame bridges between them. It is the cheapest material answer, it works where the ground is firm and level, and it leaves the middle of the circle empty — no concrete where nothing stands. Its cost is precision: ten separate points must land in one plane and one circle, or the ring that spans them comes out twisted, and Chapter 38 has already explained what a twisted ring does to everything above it.

One ring. A continuous footing under the whole base ring — a ring beam of concrete, or a wooden ring on blocks — receives the dome's weight as one circle and spreads it as one circle. It costs more material than ten points, and it is *forgiving* in exactly the way ten points are not: the ring beam

carries the frame's local imperfections around the circle instead of exposing them at every vertex. For a build that stays on the ground, the continuous ring beam is the forgiving answer — the priced ground option this chapter offers — because a ground build's error budget, the one the chapter just spent on, is worth more than the concrete it saves.

The honest note, kept from the dome books that skip it: a dome puts its whole weight on the *outside* of its circle. Pouring a slab across the middle is concrete nothing stands on — the centre of the floor carries no frame load at all, and the money it costs is money the ring beam could have used.

The two approaches are priced properly when the dome graduates to a permanent pad; here the rule is only: foot the circle, not the disc.

Setting out a circle accurately

The layout, because the footing's truth is the ring's truth, and the ring's truth is the whole frame's.

One stake, one string. The dome's centre goes in first: a stake driven where the middle of the floor will be, and it never moves again for the rest of the build — every measurement of the footing, the ring and the raising is taken from it. Then a string the length of the dome's radius, a loop at each end, and the circle is drawn the way circles have been drawn since before anybody wrote this down: walk the string around the stake, marking as you go.

Set the ten points from the circle. The ring's ten base vertices sit on the circle at equal spacing, found by stepping a second string — the chord between neighbouring vertices — around the circumference and adjusting until the last step lands exactly on the first. The step-and-adjust is the check: a ten-point layout that closes on itself with the last step is round; one that misses its starting mark by an inch is telling you where the circle isn't.

Check the circle before anything is poured. Measure across every opposite pair of points — the diameters — and demand they agree. Then check the level of the ten points against a water level or a laser, and shim until they share one plane. The checks are the chapter's whole discipline: a footing that is round and level on the day it is made is the only thing in this part of the build that cannot be fixed later, so it is checked the way things that cannot be fixed later are checked — twice, from different directions, before the concrete is mixed.

Anchoring against uplift

The sentence the footing chapter must say plainly, because the shape itself refuses to let it be skipped: **a dome is a wing.**

A hemisphere is a lifting body. Wind over the crown does to the dome what it does to an aircraft: pressure drops on the curved top, and the whole building feels a pull toward the sky. The frame's own weight resists — green pine is heavy, and the skin chapters add their share — but a dome that is only *sitting* on its footing is a dome the right wind can pick up by the brim, and the stories of domes that went rolling are not folklore. The footing's job is therefore two jobs: hold the weight *down*, and hold the building *down against the wind*.

The anchor is the ring's connection to the ground, and it is not optional. Every base vertex — or the ring beam at intervals — needs a mechanical tie into the footing: a cast-in anchor, a bracket, a strap over the ring member. The tie's job is uplift, not weight, and it is sized for the worst wind the site's records hold, which is exactly the calculation the last section of this chapter hands to somebody else. For a build that stays on the ground, straps at every base point are the anchor, and the straps are the one part of the footing the book will not let a reader improvise.

This is where to ask somebody

The safety page, and it is shorter than the chapter because the book's rule from the front matter simply applies here with full force.

Footings are load and soil, and neither is a matter of carpentry. The dome's weight, the wind's uplift, the ground's bearing capacity, the frost line, the drainage — every one of them is a site-specific number that belongs to someone who can read the site and the code together. This book tells you the *shape* of the footing — a circle, not a disc; anchored, not sitting; checked, not assumed. It does not tell you how deep, how wide, or how much steel, because those answers come from your ground, not from a book that has never seen it.

The division of labour is the one Chapter 10 drew for loads, applied to the earth: the geometry is the book's, the engineering is the engineer's, and the boundary between them is a page you cannot miss. Build the circle the book describes. Have the circle's ground signed off by someone who can. Then, and only then, stand the frame on it.

And remember the order the title page promised: the hanging cabin never owed this chapter anything. The foundation is the last decision of the build, made when the cabin finally comes down from the trees — and the chapters that follow this one are the reason the decision could wait.

Closing the Shell

Forty triangles of something over forty triangles of nothing

The standing frame is a skeleton, and this chapter is its skin: 40 triangles of something, over 40 triangles of nothing, each panel closed so the weather stays on the outside.

The options are few and the book ranks them the way it ranks everything — by cost and behaviour, computed, not by feel. Sheet goods dominate, because they are cheap, local and flat, and the chapter's first two sections are the craft of sheathing forty triangles from rectangular stock — and of saying plainly that no panel of this dome fits a four-by-eight sheet. The table prices the whole skin in sheets. The last section runs the alternatives — fabric, shingle, living roof — because each one exists for a reason and each one asks the frame for something different, and the builder who knows all four picks the right one instead of the first one.

Filling the opening with wedges

Wood infill belongs in the wedge method's stock plan. State the face that must close, the orientation of each wedge, and the continuous coverage of a pair. Two wedges can occupy one strip; count both pieces. A bowtie's span can include a gap, so its span cannot be substituted for face coverage.

The shared example uses a 72.00 inch nominal long chord and a 12.00 inch log split 8 ways. Its half-chord-width framing inset is an assumption, leaving 427.63 square feet to fill. Under the stated continuous-pair-pitch model, infill ranges from 1,117.4 to 2,234.9 wedge feet. The endpoints are planning bounds, not proof of seating or weather closure.

Reserve the 720.00 feet of uncut frame blanks separately from the 664.31 feet of solver cut-stock. With a uniform 60.00 foot usable trunk, framing plus that conditional infill takes 229.7 to 369.4 trunk feet. These are ideal equivalents before kerf, taper, defects and course-cut waste. The Triangle Infill calculator supplies the real tray, piece shapes and stock allocation. An opposed-point cavity experiment and the calculator's alternating-point fill are different arrangements from two points both inward.

Sheathing a triangle

The craft problem the rectangle-building world never has: every panel of a dome is a triangle, and sheet goods come in rectangles.

The answer is the panel drawing, again. Chapter 37 printed the two panel shapes flat and full size; the sheathing job is those shapes, cut in sheet, one per panel, with the seam allowance carried around the edge. The procedure that keeps the waste honest:

One template, not forty measurements. Trace each panel shape once onto scrap, cut it, and use it as the pattern for its whole class — the 10 of one shape, the 30 of the other. The template is the sheathing's jig: the shape made physical, so the saw never re-derives an angle it already cut correctly once.

Cut inside the line, then trim to fit. Sheet cut to the template's outside is always slightly big; sheet cut inside it can be trimmed with a knife in place. The panel gets its skin offered up, marked against its own frame, and trimmed to the frame's actual edges — locate first, cut second, the same maxim the jig chapters ran, applied to skin.

Seam allowance stays outside the panel. The skin's edge belongs to the *seam*, not to the panel: the gasket's share, the flashings, the overlap of the next panel's skin — all of it lives beyond the frame's own edge, and a skin cut flush to the frame is a skin that leaves the seam naked. Cut generous, trim at the seam, and the 55 seams get the overlap the weather chapter demands.

Screw to the frame, not through it. The skin fastens into the members' faces — the wide flat of the wedge — at the spacing the sheet's own spec asks. The skin is not structural; it is the panel's face, and its screws' only job is to keep it flat against the frame while the seams do the weather work.

No panel fits a four-by-eight sheet

The second craft problem, and it is pure geometry: a dome's panels are bigger than the sheet goods the rest of the building world reaches for first.

No panel of this dome fits a four-by-eight sheet, and the book says so plainly because the claim is easy to make and wrong. The equilateral panel is 72.0 inches on a side — about 62.4 inches tall — so it overhangs the four-foot width of a sheet before any seam allowance is added, and the isosceles panel is nearly as tall. Neither shape nests two-up in a sheet, either.

The honest version is that the panels are not cut from sheet goods at all. They are framed from the members themselves — the split-log sticks of Chapter 12 — and sheet goods appear only later, as the

skin. The skin triangles inherit the same limit: a skin panel the size of a frame panel does not fit a four-by-eight sheet singly, so the skin is cut from larger stock or pieced from more than one sheet, with the seam allowance carrying the joins. Where the piecing does not reach — the odd counts, the rim panels with their different edge — the waste goes to the offcut pile and the pile goes to blocking, the same honest destination as every other offcut in the book.

Sheet count

The table is the skin's bill: shell area and sheets, computed from the solved geometry, not estimated from the drawing.

The numbers are the answer to the question the chapter opened — what it costs, in sheets, to close 40 triangles. The area is the dome's own surface, the sheet count is what cutting those triangles from stock actually requires, and the difference between the two is the waste the previous section promised to keep honest. The figure is worth reading twice, because it is the first number in the book that is *bought* rather than *cut* — everything until now came out of the trees, and the skin is where the fortnight's ledger first touches a lumberyard.

The alternatives

Sheet goods are the default, not the only answer. The other three exist for other builds.

Fabric. A membrane skin — canvas, shade cloth, a soft shell pulled over the frame — closes the frame for the price of cloth and rope, sheds rain where the seams are taped, and gives up insulation and years. It is the right skin for a seasonal dome, a greenhouse's first year, or a frame that is still deciding; it is the wrong skin for a home, and the book says so rather than romancing the tent.

Shingle. Shingling a compound surface is slow, and it works — each shingle is small enough to follow the curve, the overlaps shed water the way the frame's ridges want, and the result is the most repairable roof there is: one shingle at a time, forever. The cost is the labour, which is why shingle is the upgrade for the dome that has already earned its place, not the first skin it wears.

Living roof. Soil and plants over a waterproof layer, the heaviest and coolest of the four, and the one that asks the most of the frame: the chapters on load have already said who decides whether the members carry it. It is listed because it is real and because somebody always asks; it is not recommended for the first build, and the reason is the engineer's sentence, not the gardener's.

The chapter closes where the shell does: the frame is covered, the seams are taped, and the dome is a *room* in the only sense that matters outdoors — it keeps weather on its outside. The next chapter cuts

the holes that make it usable, and the chapter after that is the one that decides whether the shell's honesty holds up to the weather it was built for.

Doors, Windows and the Rim

Cutting holes in something that works by being closed

Doors and Windows

Every building needs holes, and a dome is the building that resents them most: its strength is its closure, its panels are its structure, and a hole cut wrong does not just let the weather in — it un-triangulates the frame at the exact place the frame was leaning on itself.

So this chapter is the craft of opening a dome without breaking it. The principle, stated once and used throughout, is the one the companion book's own chapter taught: **an opening is a panel you did not close** — take a whole panel out and the frame loses nothing structural, because the panel was always its own independent unit. Cut *into* a panel, and the triangle stops being a triangle, and the load that was travelling through it now has nowhere to go. The chapter runs the openings in order of mercy: the door at the rim, the window in a panel, the risers that cheat height — and the drawing that shows all three without a single un-triangulated triangle.

A door at the rim

The easy case, and the book says so plainly: the door belongs at the rim, because the rim is the one place a hole costs the frame nothing.

The base ring's 10 edges join nothing — Chapter 8 separated them from the seams for exactly this reason. Leave one rim panel out entirely, and the gap is a doorway — but a triangular one, not a rectangular one. It is the missing panel's own shape: a triangle with its point at the top, roughly 52 to 62 inches tall at that point, so there is no flat header across the top edge and no pair of jambs. The two sloped sides are the neighbouring panels' edges, and the top is the point where they meet. The head height is what it is — about five feet at the apex, falling away to the rim at the sides — and a person steps through the widest part of the triangle, not under a lintel.

Nothing has to be framed, because the opening is a whole panel removed, not a hole cut in one. The panels on either side are complete, their seams close as they always did, and the triangle above the

door is still a triangle, doing the lintel's work the way every other panel does — by staying a triangle. The dome loses a panel that was never structural in the way the crown panels are, and the ring's continuity is carried by the two neighbouring rim panels and the course above them, exactly as it always was. In a taller opening — a door raised onto a riser, the chapter's last section — the engineer's sentence from Chapter 10 reappears, and rightly. The rule for the chapter: take the whole panel, and the gap is a door with nothing to frame.

A window in a panel

The harder case, and the craft is keeping the triangle a triangle.

A window wants light where the wall leans — mid-panel, high on the dome — and a hole in the middle of a triangle is exactly what the frame's logic forbids: the triangle's load runs around its edges, and a hole in its middle interrupts the members that were carrying it. So the window is not cut into a panel; it *replaces* part of a panel, and the part it replaces is re-triangulated around the hole.

The procedure is the panel's own logic, applied smaller: frame the window opening inside the panel with members that run from the window's corners to the panel's edges — a smaller triangle above the window, two beside it, one below — so every region of the panel is still triangulated, and the panel's loads still travel along members, now detouring around the glass. The window unit itself is fixed into its own sub-frame, flashed to the panel's skin, and the panel remains one independently-built, independently-lifted thing — the property Chapter 18 paid for, spent here on daylight.

The honest cost, said aloud: a windowed panel is a panel built twice — once as the triangle, again as the triangle-with-a-hole — and its member count rises accordingly. That is why the windows go high and few, where the glazing earns its keep in a leaning wall and the view, rather than everywhere the floor plan's habit would put them.

Three openings that work

The drawing collects the chapter: the rim door, the windowed panel, and the risers option of the next section — three ways to open the dome, each one keeping the frame's triangulation intact.

Read the drawing against the rule it illustrates. The door takes a whole panel and frames the gap; the window sub-frames inside a panel and re-triangulates around the glass; the risers lift the whole shell and put the openings in a *straight* wall, where doors and windows stop being dome problems at all. Three openings, three strategies, one invariant — no hole ever interrupts a load path without a smaller triangle being drawn around it.

Risers: cheating height at the rim

The third strategy is not an opening but an avoidance: raise the whole dome onto a short vertical wall — a riser, a knee wall, a stem wall — and the openings stop fighting the curve.

A riser is a straight cylinder of wall under the hemisphere's brim: knee-high for a shed door, waist-high for furniture to lean against, full-height where the building wants ordinary windows and an ordinary door. The dome above it is unchanged — the same frame, the same panels, the same seam schedule — and the wall below it is ordinary stud framing, which is the entire point: the expensive part of the building keeps its shape, and the parts that want rectangles get rectangles.

What the riser costs is the ring beam's reappearance: the dome's base ring now sits on a wall instead of on the ground, the wall needs a top plate to receive it, and the wall's own top must be as round as the ring it carries — the same layout discipline as Chapter 39, moved up one storey. What it buys, Chapter 53 prices in full: usable floor at the edge, headroom where the dome's wall leans hardest, and the cheapest square footage in the building, bought by the foot of wall height.

The book's order of operations closes the chapter: frame, shell, then openings — and if the openings want a straight wall, the riser is the answer, because it is cheaper to lift a whole dome than to argue with forty curved triangles one window at a time.

Weather and Water

Where it leaks, and where the water goes

The honest chapter about domes and rain, because the dome's reputation was made by this subject and the subject deserves the book's full discipline.

The title page inverted the usual order on purpose: the frame is the part you are allowed to get wrong, and the skin is not — a frame out of line stands for fifty years, and a skin that leaks one winter takes the frame with it. This chapter is the skin's side of that bargain, and it is the reason the book treats the seams here with more care than anywhere else.

Here is the fact the frame chapters have been circling: a dome's roof is one surface made of 40 flat panels meeting at 55 interior seams, and every seam is a place water can get in. A rectangle concentrates its roof's risk in two valleys and a ridge; the dome spreads the same risk across 55 lines. The answer is not fifty-five seals; it is the two principles this chapter teaches — shed, don't seal; and where water must be managed, manage it deliberately, at the rim. The chapter closes with the inside face, because the dome that survives the rain outside can still be ruined by the water its own breathing makes inside, and the book says that before the reader discovers it in month three.

Every seam is a potential leak

Say the number, because the number is the strategy: 55 seams. Fifty-five lines on the outside of the building where two panels meet, fifty-five keys, fifty-five weather details — and the honest way to think about them is not "one roof" but "55 small roofs, all meeting".

The frame chapters already did the structural half of each seam: the key locates the geometry, the gasket takes the tolerance. The weather half is a different question, and it is the one that decides whether the building lasts. A seam that fits is not automatically a seam that sheds — water moves downhill, and a seam's fit matters only where the water actually runs. The dome's seams run in two families: the shallow ones near the crown, where the shell is nearly flat and water moves slowly, lingers in the key's shadow, and waits for a gap; and the steep ones near the rim, where the shell stands near-vertical and water runs off fast. The strategy follows the slope: the crown seams need

weathering — a drip edge, an overlap, a path that takes the water off the seam before it can sit on it — and the rim seams, steep as they are, need only alignment, because their own slope carries the water away.

The count is the discipline. A builder who treats 55 seams as one repeated detail does the same detail wrong 55 times; a builder who separates them into the two families the geometry gives — the steep and the shallow — has two details to perfect and a schedule to repeat them by. Chapter 16 is the schedule. This chapter is the weather it has to survive.

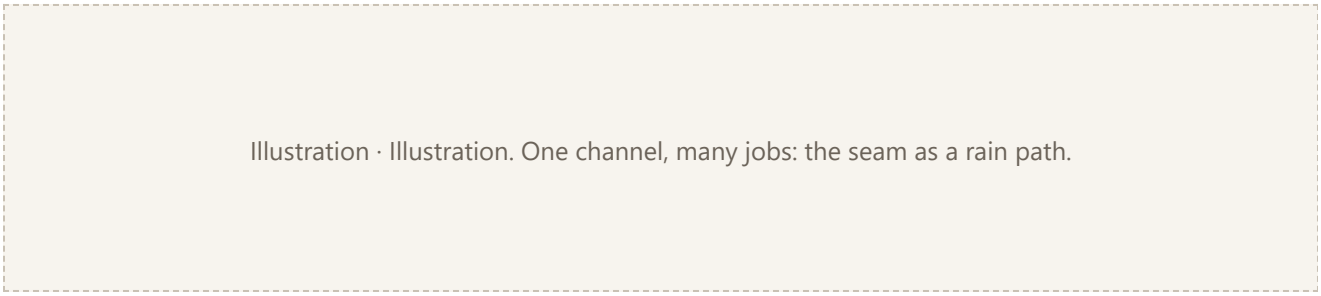


Illustration. One channel, many jobs: the seam as a rain path.

Shedding, not sealing

The principle that actually works on a compound surface, and the reason “waterproof” is the wrong word for what this book builds.

A sealed surface tries to stop water at the membrane — one barrier, perfect, everywhere. Sealing fails on a dome because a dome moves: the frame dries, the panels settle, the seams open and close with the seasons, and a seal that was perfect in July is a cracked line by January. The answer the roof-building world already knows — and shinglers have known for centuries — is to stop trying to hold the water back and instead *hand it downhill*: every surface overlaps the one below it, every seam wears a shingle, and the water is always on top of the roof, never inside it.

So the dome is weathered as 55 little roofs, each seam flashed like a valley: the upper panel’s skin laps over the lower’s, the key is protected rather than depended on, and the seam’s own slope is the drainage. The skin chapters supplied the panels; this chapter supplies the lapping order; and the result is the honest claim the book makes about weather: not “the dome never leaks” — no roof can promise that — but “where water reaches the seam, the seam hands it down, and the lowest seam hands it off the rim”. Shedding is a path, not a promise, and paths can be walked, checked and repaired — which seals, once breached, cannot.

Harvesting it instead

The one place the shape helps with water, and it is not small: a dome is a very good rain collector.

Every drop that lands on the shell runs down the panels to the brim, and the brim is a circle — one continuous gutter, already built, waiting only for the collection hardware the gutter chapter provides. Chapter 54 computes the annual catch: the footprint, the local rainfall, a declared efficiency, and the answer comes out in thousands of gallons a year off a building this size. The dome does not merely *tolerate* rain; it gathers it, and the gathering is free — the roof was already a gutter, the shape was already a funnel, and the only purchase is the tank.

The honest note, stated plainly: the catch is not drinking water without treatment. The filter makes it clear; the lamp makes it safe; neither does the other's job — and the book's water system, when Chapter 45 reaches it, is a first-flush diverter and a UV lamp downstream of the tank, not a rain barrel and optimism. For everything else — irrigation, flushing, washing, the buffer against a dry well — the catch is as good as the roof, and the roof is the dome's.

Condensation and the inside face

The last section is the problem people discover in month three, and it is worth a whole section because it is not rain.

A dome is warm inside and cold outside, and the meeting of the two happens *inside the skin*: moist room air reaches a cold surface, the water condenses, and the builder who sealed every outside seam perfectly wakes up to water running down the inside of the shell. The seams were not the leak. The air was.

The answer is the seam channel, and it is the reason the later chapters of this book engineer it deliberately rather than sealing it away: the cavity between the frame's wedges is the one continuous void that connects the whole shell, and it is exactly where condensation belongs — collected, drained at the rim, vented at the crown. The full climate system — the dew points, the levels, the drawer, the plate — is spelled out in the Complete Edition, and it is the book's real answer to the inside face. Here the principle is enough: **never seal the inside face against the air; give the water a path, and drain it at the rim.** The dome that leaks from rain failed its seams. The dome that leaks from condensation was built by someone who thought the inside face was not part of the weather.

The chapter ends where every weather chapter should: outside, shed; inside, drain; at the rim, catch. The dome is not a box, and its water is not a nuisance to be sealed away — it is a flow to be directed, and the direction is downhill, through 55 honest seams, off the brim, into the tank.

Finishing the Wood

What oil does to split timber, what it does not do, and the afternoon it costs

An oiled member is not a protected member. It is a member that has been finished — and the difference between those two sentences is the whole chapter.

The weather chapter said where the water goes and what keeps it out: a skin that sheds, flashing at the openings, a ring that drains, and a member allowed to dry. None of that changes here. What changes is the bark face on the outside of every member — the curved back that carries the bark — which is charred in the yakisugi pass and then oiled, so the weather side darkens and sheds while the sawn faces stay bare. The sawn faces are the building's inside look: pale, honest, slightly furry from the saw, and left alone on purpose, because they are also the joint faces the seam key and its gaskets have to bond to.

Finishing, then, is not how you decide what the inside of the building looks like; the inside look is the bare sawn wood. Finishing is the weather-side decision, with a structural edge: an oiled bark face is easier to wipe, a sealed surface can trap water where an oiled one lets it go, and a member with a coat on it can still be inspected through it. This chapter is the whole practice, from reading the can to the rag station, with the quantities worked out for the frame in this book: 664 square feet of sawn face left bare, and 8.0 hours of active finishing on the bark.

The source material for the finishing practice is a video claiming 20 things about linseed oil. This chapter is what survives when those claims are checked against product labels and against the arithmetic of the building — including the ones that do not survive.

Read the can before the rag

Linseed oil is three products wearing one name, and the differences matter before you buy anything.

Raw linseed oil is what comes out of the seed press. It cures slowly — it takes oxygen out of the air and slowly sets, which is why it is safe and quiet and why a coat in a cold damp workshop may still be tacky next week.

Boiled linseed oil is not boiled in the sense of a pot on a fire in any modern product. It is raw oil with metallic driers added so that it sets in hours rather than weeks. That is a real convenience with a real cost: the driers are the reason the label carries warnings, and they are the reason the drying behaviour of a can is not something you can assume from the word “linseed”.

Heat-polymerised oil — the third thing you will meet — is oil that has been thickened with heat and no driers, and it is what several traditional woodworking products actually are. A name like *natural*, *boiled* or *Danish* does not describe a formulation. The label does.

So the rule is the one the label gives you, not the one the video gives you. Two real cans make the point better than any warning could:

the can says	wait before wiping	minimum cure
Danish oil	5 minutes	8 hours
oil and wax	60 minutes	24 hours

The same furniture finish, the same manufacturer, and two different schedules by a factor of twelve. If you pick one number and apply it to every can on the shelf, you will either be rubbing off a coat that has already started to set, or waiting a day for nothing.

And the same video’s cleanest rule is too broad: it says raw oil only for anything that touches food. That is a sensible instinct and it is not what the labels say — some polymerised finishes are sold specifically for kitchen wood, and the question a buyer should ask is not “raw or boiled” but “is this product rated for this use”, which is printed on the can.

The rag station comes first

This is the dangerous part of the chapter and it comes before the pleasant part on purpose.

Linseed oil cures by taking oxygen out of the air, and curing gives off heat. A single rag on a bench sheds that heat into the room. A *heap* of rags traps it in the middle of the pile, and a pile of rags with no spark anywhere near it can reach ignition temperature and catch fire — often hours after you have left the shop. It is the classic workshop fire, and it is the one part of finishing that can burn a building down.

The prevention is three objects and one habit, and all four are cheaper than a fire:

- **A lidded metal tin.** Not plastic, not a bucket, not a bag. A tin with a lid that closes.

- **Water in the bottom of it**, enough to cover what goes in.
- **An oil-breaking detergent** in the water, so the rags stop being a pile of oil and become a wet heap of cloth.
- **The habit**: rags go in the tin in the same session they were used, the tin stays outside and away from the wall, and it is emptied into whatever your local rules say is right. A rag that dries out in a heap becomes hazardous again — which is why the habit matters more than the container.

Set the station up before the first coat is poured. A rag station improvised after the fact is a rag station that gets improvised at the end of the afternoon, when you are tired and the shop is about to be shut with forty oily cloths in a bucket by the door.

How much oil, and how long it takes

The frame in this book is 120 members and 673 feet of stock. Each member carries one bark face — the curved back the char and oil go on — and two sawn faces 5.00 inches deep, the flat radial faces the saw left. The sawn faces total 664 square feet, and under this chapter's rule they stay bare: the bark gets the finish, the sawn faces get none. The bare figure is printed so the reader sees the scale of what is deliberately left unfinished, not so it can be oiled.

The allowance for the bark faces:

- **Coverage** is planned at 350 square feet per gallon per coat. That is an estimate for rough-sawn timber, which drinks more than a planed board.
- **2 coats**, which is what a working finish means in practice on a softwood face.
- **15 per cent handling**, for the spillage, the brush loss and the end grain, which takes up oil like a wick.

The bark is a single curved face per member — a smaller surface than the two sawn faces put together — which is the point of the estimate being printed rather than guessed at: the finish is a small, contained job, not a whole-building interior one.

Now the number that argues against the whole business, and it belongs here rather than in a footnote. The can's label claims 1,000 square feet per gallon, which is 2.9 times the planning figure — and that claim is a manufacturer's coverage on a prepared surface, not a measurement on a rough-sawn wedge. If you plan to a label you will run out halfway through the pile, which is the sort of thing that turns a pleasant afternoon into a second trip.

And the labour: 2 minutes per member per coat comes to 8.0 hours of active work for the frame, and that figure excludes every part of the job that is not rubbing oil into wood — preparation, getting to the members, inspection, and the cure time between coats. A cheap can of oil can still make an expensive job. That is not an argument against finishing; it is an argument for knowing that you have bought eight hours of work when you buy the can.

What oil cannot do

Four things the video implies and the building refuses:

It cannot restore strength. A member that has split along its pith, or that has decay in it, is a repair — and if the decay is in a member doing structural work, it is a replacement. Oil on a compromised member is paint on a cracked beam: it hides the problem from the next person who inspects the building. The grading passes in the small-units chapter are where that decision is made, and this chapter does not get a vote.

It cannot stop decay or insects. Nothing in a linseed oil can carries a preservative rating. The ground-contact ring at the base of the building, the footings, and any timber that stays wet want a product rated for that job, not a food-grade oil.

It is not an air seal and it is not a rain seal. This is the one that matters most, because the seal is the thing that keeps the water out and a finish is the thing that makes the wood look right. An oiled face will still pass air and still pass water vapour — which is what you want on the inside faces, and exactly the wrong thing to rely on for a weather surface.

It cannot be used where something else has to bond. Keep oil off the lands that a sealant, tape or gasket has to stick to, and keep it off the printed key and its gaskets. Marine epoxy systems say it plainly: bonding surfaces must be free of oil and wax. A coat of oil in the wrong place turns a sealed joint into a joint that leaks in year two — which is precisely the failure the seam chapter spent its whole length preventing.

So the honest summary of what a finish buys is this: a face that wipes clean, that stays pale rather than greying, that lets the wood dry, and that can be renewed in an hour. That is a real thing to want. It is just not protection, and a builder who confuses the two will leave a weather surface unfinished because it “has oil on it”.

Inspection, and the second coat

The finish record is the last piece of kit and it costs nothing. Write down the product, the batch, what you did to prepare the wood, the date, and where the coat went. Keep a sample board with the same treatment on it, because grey wood darkens under oil and it is easy to convince yourself that a weathered face has been protected when it has only been darkened.

Then the renewal question, which is where most finishing advice goes wrong by being too tidy. An oiled bark face on a dry frame does not need a schedule; it needs an inspection. Once a year, look at the faces that see water — around the openings, under the floor edge, at the base ring — and treat what is actually dry and worn rather than the whole building on a timetable. And when you do renew, clean first: a second coat over dust and grease sets the dirt in rather than sealing it under.

The one place to be genuinely suspicious is the base of the building, because that is where water sits and where a member is closest to the ground. If a member there looks dark and soft rather than merely grey, the answer is a repair and the answer is this year, not a coat of oil and a note in the record.

Which is the shape of the whole chapter: the finish is what you do to a sound building to keep it looking like one, and the inspection is how you find out whether it still is one.

Inside a Round Room

Living with curves

This is the chapter about living in the thing, written the way the rest of the book is written: the numbers first, the compromises named, and the romance earned rather than assumed.

A round room is not a rectangular room with curved wallpaper. It changes where you can stand, how furniture fits, how sound behaves, and where the light falls — and the changes are measurable. This chapter measures them: the honest floor, the standing-height map, the furniture problem and its real solutions, and the acoustic quirk a hemisphere brings for free. It is the chapter to read *before* building, because the fixes it describes are layout decisions that cost nothing on paper and a renovation once the walls are up.

Where the usable floor really is

The honest number, because the floor a dome advertises and the floor a person can use are different things.

The dome's plan area — 365 square feet for this build — counts the whole circle. But the shell meets the floor at a tangent: the wall leans inward from the rim, and the last ring of floor around the edge has almost no headroom over it. The usable floor is the smaller circle where the ceiling is over your head, and the difference between the two numbers is the dome's real estate tax — floor you paid for, in frame and skin, that only a dog can use.

The map on the next page draws the honest version: the full circle, the standing-height contour inside it, and the dead ring between. The dead ring is not waste in the moral sense — it is where low storage, benches and beds belong, the furniture that does not need to stand up — but it is real, and the chapter states it because the alternative is the surprise every dome builder watches a visitor have: they step in, they look up, and they say the roof is lower than the picture suggested. The picture suggested the plan area. The roof was always the geometry.

The fix the book already spent a chapter on is the riser of Chapter 53: lift the shell a few feet and the tangent moves outward, the dead ring shrinks, and the floor becomes genuinely usable to the wall. The round-room chapter recommends it for exactly this reason — not for looks, but because the standing-height map improves more per dollar than anything else in the whole fit-out.

Standing height, mapped

The map is the chapter's central object: a plan of the 365-square-foot floor, coloured by headroom, with the standing-height contour drawn where the ceiling passes head height.

Read the map as a furniture plan before it is a chart. The central disc is the room: full height, 10.8 feet at the crown, the space where a person stands, works, cooks. The ring between the contour and the rim is the compromise zone: overhead clearance shrinking smoothly to nothing at the wall, exactly the profile a bed, a bench, a desk-with-a-view or a row of shelves wants. And the rim itself is the dead zone the previous section named — reachable, useful, not standable.

The map also settles the oldest argument in dome living: where the bed goes. The answer is where the map says — in the ring, head toward the wall, because the only dimension a sleeping person needs at the wall is the one the dome has plenty of: length. The furniture section follows the same logic for everything else.

Furniture, storage and the wall that curves away

The practical answers, including the ones that are honestly just compromises, because a chapter that only offers custom joinery is writing for a budget this book does not assume.

The curve costs a wedge of dead air. Any straight-backed piece against a curved wall leaves a tapering gap behind it — the round-room tax of Chapter 11, paid here in inches. The cheap answers: pull furniture off the wall entirely and let the gap be a cable run and a shadow line; angle the piece so one edge touches and the gap opens somewhere unreachable; or accept the gap and fill it with the things that are soft. The honest answer for the pieces that matter is built-in storage on the curve — the one place custom work pays for itself, because it converts the dead ring into drawers.

Low is the ring's whole vocabulary. Beds, benches, window seats, shoe racks, the wood stove's hearth — everything that does not need headroom lives in the ring, and the centre of the room stays open, which is how a round room ends up feeling larger than its number: the walls are furnished, the middle is air.

The middle wants to be the table. A round room's centre is its tallest, brightest, most reachable point, and the instinct to put the table there is the geometry's instinct, not a style choice. Everything radiates from it; every path through the room crosses it; and the light from the crown falls exactly on it. Furnish the ring, leave the centre, put the table in the middle — the round room's layout writes itself once the map is accepted.

Acoustics, and the whispering problem

The last section is the quirk nobody expects and everybody notices in the first week: a hemisphere focuses sound.

A dome is half of a whispering gallery. Sound from one side of the room runs up the curved wall and comes back down at the centre — the geometry that lets a whisper cross a cathedral — and in a house-sized dome the effect is not magical, it is *tiring*: conversation pools at the middle of the room, a kitchen tap at the rim is audible everywhere, and the building has no quiet corner, because corners are where quiet lives and the dome has none.

The fixes are the same ones theatres use, scaled down. Soft surfaces break the focus: the quilted layers of Chapter 55 do double duty as acoustic treatment, rugs and fabric kill the floor's reflection, and the furniture in the ring breaks up the wall's curve the way baffles do. The one thing not to do is nothing — an empty dome is an echo chamber, and the echo arrives before the furniture does.

The chapter closes with the round room's honest summary: a floor that is smaller than its number, a wall that leans, a centre that rings — and a room that, furnished the way the map dictates, is the calmest, most enveloping space most people will ever sleep in. The curves are not the problem. The curves are the room, and this chapter is the manual for living inside them.

Heat, Power and the Small Systems

Making a shell into somewhere you can be in February

The Small Systems

A shell keeps weather out. Somewhere you can be in February needs the small systems: insulation in the walls, power in the dark, heat in the cold, and a decision about the grid. This chapter is those systems, at the scale the book's method implies — small, deferrable, and run through the same voids the frame was built around.

The dome's gift to its systems is the wedge frame itself: the cavity between the members, the channel of the seams, the column of the centre. The systems ride in those voids instead of requiring a stud wall's worth of drilling, and the chapter's sections follow the voids: insulation between the wedges, services in the seam channel, heat in the round volume, and the grid decision stated without romance. The chapter closes the home-making half of the book; after it, the dome is a house.

Insulating between wedges

The frame's wedges are 5.00 inches deep, and the depth is a cavity: between the skin outside and the inside face lies the whole thickness of the member, interrupted by nothing — no studs, no plates, no bays that were not drawn in from the start. Insulation goes in that cavity, and the shape does the rest.

The options are the ladder of Chapter 55, rung by rung: nothing first, then the quilted layers, each one a size up, each winter warmer than the last. The book's own preference is the quilt — layers of recycled clothing, sewn by hand — because it is the one insulation that matches the frame's own economy: it fits the cavity loosely, it comes off in order, and it costs what a sewing machine costs. Rigid board fits too, cut to the panels' triangles and pressed into the cavity; it insulates better per inch and gives up the removability the quilt was designed for. Either way the principle is the cavity's: **insulate the space between the members, never in front of them** — the members are the frame's structure and its look, and burying them under a blanket of foam throws away the only thing the building's walls ever had to say.

The one rule the chapter will not soften is the moisture rule from Chapter 42: insulation without a path for vapour is a rot factory. The cavity must breathe toward the seam channel, and the seam channel drains at the rim — the vapour path the channel was built to carry. Insulate the dome as a living shell, not as a sealed box, and it gets warmer every winter instead of wetter.



Illustration. It gets warmer every winter: the quilt between the wedges.

Running services in a frame with no studs

A conventional wall hides its wires and pipes in the stud bays. A wedge frame has no stud bays — and it does not need them, because the seam channel is the chase, and it runs everywhere.

The seams are the building's service corridor: a continuous void between the panels that follows the frame from rim to crown, wide enough for cable, small pipe and the sensor wiring the later chapters assume. The rules from the channel chapter bind here: water and wire fit, a round duct does not, no pressure fitting inside a seam, and the cable-in-chase question is the inspector's call, asked before the skin closes over it. Power runs from the utility column — the centre, where the core lives — outward along the seams to the points of use, and the dome's own geometry makes the routing obvious: centre to rim, like the spokes it is.

The discipline the chapter adds is the *deferral* discipline, borrowed from the stem-cell chapters and stated once here: run the services where the building will never be cut open again — the channel, the column, the chase — and leave the end connections surface-mounted, where a change of mind is a screwdriver, not a renovation. Threaded inserts in the wedge faces, set at panel time, are the cheap version of the same promise: the services bolt on, they unbolt, and the frame never learns their names. The building that can change its mind is the building whose services were run as if change were coming, because it always is.

Heating a round volume

Why a dome is easier to heat than a rectangle, and the one way it is not, stated together because both are true.

The easy half is geometry: the dome has less skin per floor than a box — Chapter 52 computed the margin — and heat loss is a skin phenomenon. Less skin, less loss, less heat to buy; and the round volume mixes itself, so a single small source warms the whole room instead of leaving the far corner cold while the thermostat argues with the window.

The hard half is the one the book's own audit keeps repeating: a dome is not *automatically* cheaper to heat. The skin is cheaper to insulate well, and the shell ladder exists because an uninsulated dome loses heat faster than an uninsulated box — less skin, but skin with a seam every few feet, and air moving through a seam carries heat with it. The heating claim the book will make is narrow and honest: **the same build quality, spent on a dome, buys more warmth per dollar than on a box** — the margin is geometry's, the build quality is yours, and the metered version of the claim is one this book has not measured — it is marked as such here rather than asserted.

The practical system follows the shape: a small stove or a single heat source at the centre, the round volume mixing the air, the quilted layers doing the retention. February in the dome is a solved problem, not a promised one.

Off grid, or not

The last decision, treated without romance, because the dome's audience includes both the grid-connected and the gridless, and the book refuses to pretend the choice is free.

Off grid is a battery question before it is a panel question. The panels are cheap and getting cheaper; the storage is the real purchase, and the honest version of a solar budget — the demand, the array, the bank, the price, and the days of autonomy the bank actually buys — is printed later in this chapter. Off grid means the lights go out when the bank is empty, the fridge becomes a weather gamble, and the system's owner becomes its operator — none of which is a romance, all of which is survivable for the person who chose it.

On grid, the dome is simply a small building with a small bill, and the chapter's recommendation for the first build is the unglamorous one: connect, if the line is there, and let the off-grid system be an upgrade bought with the money the dome saved. The deferred-order rule applies: run the conduit for the future array while the walls are open, buy the panels later, and never cut the shell twice for a decision that could have waited.

The small systems close the home chapter the way the dome closes its shell: one layer at a time, each one bought when the last one proved itself. Insulate first — the cheapest warmth. Then power — the difference between a room and a home. Then heat, then the grid question last, when the building's own behaviour has answered it. The dome does not need its systems all at once, and the builder who remembers that spends the winter warm instead of broke.

The set, priced

The project's own material quotes a small off-grid set for a building like this one, and the model's own components reproduce it line by line: an 800-watt array at 0.95 dollars a watt self-installed, a 10 kilowatt-hour bank at 410 dollars a kilowatt hour, and a 3,000-watt inverter. That is $4,100 of storage, $630 of inverter, and the array on top: **5,490 all in** — printed here because the project's own material prints it, and because a reader deserves to see that the *battery* is three quarters of the set.

Three things about that set are worth knowing before buying it.

The battery is the purchase. Power electronics have got cheap and cells have not, which is why the storage line dominates. Every kilowatt hour added to the bank costs 410 dollars and nothing else in the system cares — so a reader sizing a system should decide the days of autonomy they want and buy *that*, rather than buying panels until the lights stay on.

A derated array is the honest array. The model applies a 78 per cent derate to a nameplate rating, because wires, heat, angle and dust are real. Sizing on nameplate watts is how a system comes up short in its first February — and February is the month it was bought for.

And the surplus is the real argument. On this building's roof the array makes 40 panels' worth of power across 292 square feet, and 54 per cent of what it makes is surplus the building cannot use at the moment it is made. That surplus is worth 0.14 dollars a kilowatt hour if it displaces bought power and 0.045 if it is exported — and the gap between those two numbers, not the price of the panels, is what decides whether an off-grid system pays. It is also why a fixed array loses at grid prices and wins off grid: the same numbers, read at two different rates.

For a reader who is genuinely off grid, the surplus argues for storage rather than for more array — the same conclusion the battery-dominates-the-set note reached from the other direction, and the reason this book's off-grid advice is "bigger bank, smaller array" rather than the reverse.

Build one utility core

Everything in this chapter so far has been about the building's skin and its services as *decisions*. This section is the one place where the services are a *build*, because the utility core is the one part of the building an owner cannot make, and it is worth seeing exactly why.

The core is the centre column: the chase, the drain stack, the water manifold, the electrical panel, the riser and the seal cap. Everything the building needs passes through it, and it is assembled and tested as one object before it is stood up on the pad's service port.

It is 14 steps across 5 stages, 7.2 hours once you have done it once and 17.2 hours the first time, for 1,405 of parts. The stage that decides the build is power, at 110 minutes. And the build uses 9 tools, of which the two people skip are worth naming, because skipping them is the exact failure this chapter exists to prevent.

The chase first, on the bench, upside down. The whole core is built on a bench and stood up at the end, which is the difference between a four-hour job and a day of kneeling. Every penetration is drilled and grommeted while the sections are apart and at waist height, because the same hole later is a hole saw inside a finished core. The chase is also the one part where an error at the bottom is an inch of error at the apex, where the seal cap has to seat: plumb is checked over the whole run, dry, before anything is fastened.

The drain next, because it cannot be re-routed. A two-inch stack running the full height with the trap at its foot, dry-fitted and marked across every socket first — solvent cement gives you about five seconds and no second chance at alignment. Work bottom to top so gravity is not pulling a fresh joint apart, and then do not touch it for fifteen minutes.

Then water, and prove it before anything closes. Manifold at chest height, isolating valve at knee height, both opposite the panel, and four capped tails so a building with no plumbing still ships with them. Every crimp gets the go/no-go gauge — this is the first tool people skip, and an uncrimped ring will pass a pressure test and fail in a year. The proof is a gauge on the system for half an hour, and it happens *before* the panel goes in, because after that a leak costs the panel.

Then power, into a core known dry. Every lug torqued to the figure printed inside the panel door rather than by feel — the torque screwdriver is the second tool people skip, and a terminal that feels tight is not tight. Four breakers: outlet ring, lighting, the window unit, and one spare, so the first snap-on module never needs the door opened again. Label them in writing now, not from memory in a year.

Then close it and prove the cap. Sleeve on the chase, cap on the sleeve, six catches set so the gasket compresses evenly, and then open and close it five times on the bench. The cap is the one thing between the building and the weather.

Two things follow from doing it in that order, and both are the chapter's argument in miniature. The first is that **the sequence is the specification**: drain before water, water before power, power before closing, closing before standing up. Get the order wrong and you are cutting into a finished assembly, which is exactly what the chase's grommets exist to prevent. The second is that **the core moves**: it unbolts from four joints and a lift and goes into the next dome, so those 7.2 hours are spent once for a lifetime of buildings rather than once per building. Which is why, in any built version of this building, the money goes into this part and almost nothing into the frame.

A reader who has built the frame in this book can build the core. A reader who has built neither should practise on the core first: it is small, it is on a bench, its mistakes are recoverable, and it teaches the same lesson the whole building teaches — that the order you do things in is part of the design.

The plates, and what they are honestly worth

One piece of hardware in this chapter's list deserves its own page, because the project's own material spends two chapters on it and this book had only mentioned it in passing: the thermoelectric plate that lives in the seam channel.

It is a heat pump, not a cooler. A Peltier plate moves heat from one face to the other, and reversing the current reverses the direction. That single fact decides how it is used: throw heat to the side you want warmer — into the room in winter, outside in summer. The project's first published material showed the plate heating the room and called it right, which was right in winter and wrong in the other half of the year, and the correction is worth knowing before anybody wires one up.

What it makes, and what that is worth. A dome's worth of plates is 651 watts, and run six hours a day it condenses 0.785 litres a day — 76 gallons in a year. That sounds like a water supply until it is put next to the roof: one inch of rain on this shell is 227 gallons, so a year of plate output is the rain of a single 0.33-inch shower. And it costs 4.98 kilowatt hours per litre, against the roughly one a compressor-driven dehumidifier uses — which is already a poor way to make water.

So the honest statement, and it is this project's own: **the plates are a dehumidifier that happens to make water, and the roof is the water supply.** Buy them to keep the wood dry where nothing else can — in the seam channel, in the wettest weeks of the year — and let the gutter do

the water. A reader who installs plates to be independent of a well has bought the wrong machine, and will find out in August.

The wall that could be a filter

The last idea in this chapter is the one the project has computed and never built, so it belongs here with its arithmetic and its caveats side by side.

The idea: run air in at the base ring, let it cross the whole shell, and the building's wall becomes a distributed filter — warmer and drier on the way in. Reverse it and the sky is the supply, drawn inward through the structure: dynamic insulation, in which the wall is warmed by the air being drawn through it. Both directions are one machine — a fan, a ring main, and a shell that is porous on purpose.

The airflow side is arithmetic a reader can check. The shell encloses 2,622 cubic feet behind 730 square feet of surface, and the fresh air a dome needs is about 17.6 cubic feet a minute — so air would cross the wall at 0.02 feet a minute, 0.3 inches a minute, which is 0.4 air changes an hour. That is a crawl no occupant can feel, and feeling it is what would make the idea unacceptable. So far so good — and that is where the arithmetic stops, and where this book stops too.

Four things decide whether the idea is *good* rather than merely quiet, and none of them is computable from the airflow:

The wall is untested as a plenum. Nobody has built a strut lattice at building scale and measured how air actually spreads through it. A model that treats the shell as an even filter is an assumption, and the first place it will be wrong is at the joints.

Direction is the risk. Air moving outward through a warm, damp structure carries moisture into the wood; the same flow in the other season carries it out. A dynamic-insulation wall works because it is engineered to keep the outer face warm and dry — a designed assembly with a moisture profile, not a gap between wedges.

A sealed skin cannot breathe. The weather membrane that keeps rain out also kills this idea, so the wall-as-filter needs a deliberately permeable shell, which is the opposite of what the weather chapter spent its length achieving.

An unreachable filter is never cleaned. If the shell is the filter, the filter is the building. Nobody takes a dome apart to clean it, and a filter nobody cleans is a blockage with a fan behind it.

The position this book takes is the one the project's own material took: the airflow arithmetic stands, the idea is not refuted, and the project has not built one. A reader who wants to try it should treat the first winter as the experiment — instrument the members, vent the cavity rather than sealing it, keep the filter somewhere reachable — and should not build a house on the assumption that it works. The chapter's rule about honesty applies to an idea as much as to a number: say what is computed, say what is not, and leave the reader the decision.

CHAPTER 46 · REFERENCE

The Money, Both Ways

What was spent, and what was avoided

> ***This chapter corrects something.*** *The $50/hour figure this project once publicised, which doubled a volume ratio for safety rather than computing the section.*

The Money

The accounting chapter, and it begins by saying how it counts, because the two numbers a build produces — what was spent and what was avoided — are different species, and most dome books breed them into a hybrid that flatters both parents.

This chapter counts cash out, itemised, with the dashes left where the original build did not log a line. It counts the value of the frame three ways, with the disagreement between the ways printed rather than averaged away. And it corrects its own project's most-quoted hourly figure, on the page, the way this book corrects everything. The number that does not help the argument is here, several of them are, because the book's reputation was staked on printing them, and a chapter that only argued one way would un-stake it.

What was actually spent

The table is cash out: the saw, the fuel, the chain, the fasteners, the gaskets, the sheeting — every line the fortnight actually paid, with the source of each number and the honest dashes where the original build did not keep a line.

The table's headline row is the one the whole method exists to produce: timber, zero dollars. The structural wood cost the fuel to cut it, and the fuel is one of the small lines above. Everything else in the table is the *conversion* — the saw, the chain, the screws, the keys — which is the entire point of Chapters 2 and 13 made into a receipt: the building material was free, and the receipt is for the machine that made it building material.

The dashes are not a gap to be filled. They are lines the original build did not log, and this book will not fill them with plausible numbers — the declared-constants discipline from the front matter applies to money exactly as it applies to geometry. A cost that was not recorded is a cost that is not claimed.

Substitution value is not income

The most important paragraph in the chapter, and it gets its own section because every honest number after it depends on the distinction.

What the frame *avoids* is not money the builder *has*. The frame bought at the national rate would cost 9,798 dollars; the fortnight spends 492; the difference — 9,306 dollars — is what the build *keeps*. (That cash figure is the project's own published ledger, which counted two trees; this book's build fells 3, and the third tree's stumpage — 23.23 dollars a ton at the researched rate, on a tree the size of a car — barely moves the line. The book says so rather than silently rounding.) It is a real number, and it is not income. It buys groceries only if the builder was going to buy the frame anyway, and the builder who was never going to buy the frame has not earned 9,306 dollars; they have simply not spent it. The same distinction runs through the whole book: the strut's value and the hour's rate — both of them are substitution values, stated as such, and the one sentence that converts a substitution value into a lie is the sentence that drops the word "substitution".

The useful, honest use of the number is as a *rate on time*: 9,306 dollars kept, over 37.5 hours of hands-on work — the wage the fortnight paid the builder in avoided cost, and a rate this chapter will not dress up further. Against a carpenter's median 29.12 dollars an hour, and a typical worker's take-home 19.69, buying that frame would have cost 498 hours of ordinary pay. Those comparisons are the chapter's real product: the fortnight's work, priced against the hours a wage would have needed to buy the same frame. The cash never existed. The hours did.

The strut, valued three ways

The table is the same arithmetic run at the strut's scale, and it is printed with the disagreement left in, because the disagreement is the lesson.

One strut, valued against a nominal two-by-four, against a dressed one, and by the board foot — three methods, three answers, and the reasons they differ are printed beside them. The nominal comparison is flattered by the fact that a nominal board does not exist; the dressed comparison is the honest one, and it comes out *higher*, because a dressed board is smaller than its name and a strut replaces more of them — exactly the direction a sceptic should check first, and the table invites it. The

board-foot row refuses to give a single number, because the market does not. The last row is the estimate this project started from, kept in the table so the errata below has something to correct against.

The fifty dollars an hour claim

The errata, named and corrected, because this project once said fifty dollars an hour and the arithmetic says otherwise.

That fifty was reached by doubling a volume ratio for safety rather than by computing the section — a shortcut that made the number up, generously, and the project's own published material repeated it. The computed rates are 30.65 dollars an hour against nominal two-by-fours and 46.70 against dressed ones — smaller numbers, arrived at by measuring the actual cross-section and pricing the actual replacement. The honest note, stated the way the strut table stated it: the dressed number is the true comparison, and it is the *higher* of the two computed rates, because the dressed board is the smaller board. That fifty overshot even that.

The correction is the model for the chapter and the book: the claim, the error, the arithmetic that produces the smaller number, all on the page, and the corrections live in the chapters where the claims were made. The money chapter's own verdict closes the circle: the method's economics are strong enough to be measured properly, and measuring properly is the only version this book will print.

Where the difference actually comes from

The fortnight's own receipts say what was spent. The comparison asks a different question — what would the same floor have cost *bought* — and the answer is a stack rather than a number, which is worth printing because a reader can check every line of it.

the same floor, 300 square feet — the reference floor this comparison runs on, a little smaller than this book's own 365	dollars	whose it is
Bought whole	65,100	the price of the building
The builder's and seller's share	16,600	work you did not buy
Your own labour	17,300	the hours the book counted
Your own trees	3,200	the wood you already had
The shape itself	4,300	less skin, less framing, shorter runs
Left to pay	**23,700**	materials, services, finishes

The four middle lines add to the difference exactly, and that is the point of printing them as a stack rather than as a total: **37 per cent of what a house costs does not care what shape it is** — the fees, the windows, the cabinets, the appliances, the driveway — and the four lines above are where the rest goes. A reader who has been told "domes are cheap" has been told nothing. A reader who can see that 16,600 of the saving is the builder's own share, 17,300 is their own hours and only 4,300 belongs to the geometry has been told the whole truth about this method's economics, including the part that is about them rather than about the dome.

And it reconciles against the cheapest thing a family can buy instead:

	dollars
The same floor at a manufactured home's average price	25,986
The dome way, before any of your hours	16,439

That gap is what the project calls the small one, and it is right to call it small: the manufactured home is genuinely hard to beat on price, and the honest comparison is not price but what the money buys — a building you can take apart, re-skin and move, made from wood that was standing on the land. The money chapter prints both numbers and refuses to pretend the gap is bigger than it is.

The Hours

Where fourteen days actually went

The money chapter closed with a rate on hours; this chapter is the hours themselves, logged the way the book logs everything — as a table, with the surprises named.

A fortnight is fourteen days, and the days did not go where the plan said. The plan said the ripping would be the mountain and the rest foothills; the log says the ripping was a third of the build and the *panels* matched it, with the jig and the raising each taking their own full share. The table below is the actual spread, and the two sections after it are the reading of it: what took longest, and what a second build would cost now that the learning is spent.

The forty hours

The title's claim, counted before the calendar is, because the title is an arithmetic: 37.5 hours of hands-on work, inside the 40-hour week the book is named after.

The clock, in full: the frame's 120 members split at the measured 30 wedges a session is 16 hours at the log, and the whole frame, split to standing, runs 30 straight hours at 15 minutes a member. The 25 per cent error allowance — a first build's tuition, paid in the stick that splits off the radius and the panel that goes up twice — brings the whole frame to 37.5, with 2.5 hours spare inside the week. The splitting is inside that clock, not ahead of it; the forty hours is the whole frame, from split wedges to standing. The fortnight is where those hours live on a calendar: part days, weather days, thinking days. The hands-on clock is the number the title makes, and this chapter's chart is the same clock, drawn as days.

The fortnight, hour by hour

The chart is the fourteen days as a stacked bar: each phase, its share of the clock, and the total that has to come out at fourteen.

The shape of it is the chapter's whole argument. The felling and bucking — the dangerous, dramatic first days — are small slices. The ripping is the longest single block, as the plan promised. But the panels and the jig together match the ripping, and the raising is a full two days, and the saw setup is a line item nobody budgets and everybody pays. A builder reading the chart before their own build learns the real lesson of the fortnight: **the tree work is quick, the repetition is the mountain, and the jig is the single best-spent day in the whole table.**

What took longest and why

The surprises, named, because the chart's flat middle is where the plan's assumptions died.

The ripping took its 16 hours — the 7.5-struts-an-hour rate of Chapter 29, sustained across four afternoons. It is the build's biggest single block, and it is the one everybody expected. The surprise is that it is *only* the biggest block — the panel run that followed matched it nearly hour for hour, because forty panels of eight motions is not fast, it is merely *constant*, and constant work still sums to days. The jig's single day looks like a small slice and it is the one that kept the panel run from being twice as long. The raising's two days are the least compressible of all: a dome goes up one panel at a time, one person, one prop, and the geometry does not care how practiced the hands are.

The honest reading: the method's *rate* claims were true — the ripping ran at the promised pace — and the method's *schedule* claims were optimistic at the edges, because the repetition that makes each operation fast also makes each operation numerous. The fortnight is the size it is; the chart is the size's explanation.

What a second build would take

The learning-curve estimate, marked clearly as an estimate, because the book's rule about numbers applies to its own predictions: a guess must be labelled a guess.

The first build paid for: the jig's design, the marking scheme, the three practice panels, the seam arguments, the wrong ring that got caught, the near miss that got survived. A second build pays for none of them — the jig exists, the scheme is written, the arguments are settled on the page — and the honest estimate is that the repetition days shrink most: the ripping stays at its physical pace, but the panels and the raising lose their learning time, and the total lands near the rate this project always claimed and the first build only approached.

The estimate is not a promise and the book will not dress it as one. It is the answer to the question every builder asks at the end of the fortnight — *how long would the next one take?* — and the answer's form matters more than its number: the second build is the first build minus its tuition, and

the tuition is what this chapter's chart was actually recording. The hours were never wasted. They were the price of the next dome, and the next dome inherits them.

Bigger, Smaller, Other Frequencies

Changing the dome without changing the method

Other Frequencies

The whole book so far has built one dome: the 2V hemisphere of Chapter 8, subdivided once from the icosahedron. This chapter opens the question the outline's own logic raises: what happens to the method when the frequency changes — and what does not change at all.

The frequency of a geodesic dome is how many times the icosahedron's edges are subdivided before projection. 2V splits each edge once; 3V splits it twice; 4V three times. The method this book teaches — split logs, pinwheel panels, one jig — does not care which frequency it frames, and that is the chapter's first finding: the method is the same, and only the numbers change. The rest of the chapter states the trade the numbers make, and the place where the method stops working altogether.

More panels, shorter sticks

The trade, stated as arithmetic, because "3V is smoother" is an adjective and this book trades in counts.

Higher frequencies subdivide harder: more triangles per face, more panels in the hemisphere, and — this is the part that matters — *shorter* members, because every edge is a smaller fraction of the sphere. The panels multiply, the members shorten, and the dome approaches a true sphere, which is the entire point of the higher frequencies for anyone whose skin or aesthetic wants roundness. The cost is the pile: more panels means more seams, more keys, more screws, more motions of the jig — and fewer of the frame's parts repeat in the way 2V's two-lengths-two-shapes economy allows. The table below runs the counts, and the honest reading of it is that 2V is not the *best* dome; it is the *cheapest* dome at the scale the method was designed for, and the higher frequencies buy smoothness with repetition.

What does not change is the more important list. The split, the sector, the pinwheel, the jig, the overfit, the seam key — none of them cares whether the frame is 2V or 3V. A 3V dome built this way is

still one log, seven cuts, one bench, forty panels and a fortnight; it just has more panels in the fortnight. The method transfers, which is the sentence the chapter exists to prove: the wedge is not married to the frequency, it is married to the triangle, and every frequency is made of triangles.

Frequency comparison

The table is the trade in full: 2V, 3V, 4V, each with its panel count, its member classes and its longest member, all computed from the same geometry so the rows differ only by frequency.

The row that matters is the member-length row, because it is where the frequencies meet the method's own constraint. 2V's longest member is 72.00 inches — the size the tree chapter bucked for. The higher frequencies' members are shorter, which would let the same trees build a *bigger* dome at the same handling limit, applied through the frequency knob. That is the real decision the table supports: not "which dome is rounder" but "which frequency lets these trees and these arms build the floor I want", and the answer is read off the table the same way every other sizing decision in this book is read — by looking at the row, not the adjectives.

Where the method stops working

The honest limit, stated so the chapter does not over-sell its own transferability.

The method stops working at the *small* end, and the reason is the sector's own geometry. A dome too small for its split count has panels whose edges are shorter than the members are *wide* — the sector's bark face is 3.83 inches across, and a panel whose sides approach that width can no longer accommodate the pinwheel's overlap, the inset, and the key. The joints crowd the corner, the overlap swallows the triangle, and the method's arithmetic quietly runs out of room. The fix is not a smaller split — a dome that small needs a different member, not a thinner wedge.

At the large end the method does not stop; it *delegates*. Big domes want higher frequencies for the member-length reason above, and beyond the handling band they want crews, which is Chapter 50. The method itself keeps working — the split, the jig, the seam — all the way up; what changes is the schedule, not the procedure. The chapter's closing sentence is the one it opened with: the method is the same at every frequency. Only the numbers change, and the numbers are in the table.

Where the two lengths actually come from

Every frequency above is built by the same five moves, and it is worth walking them once with numbers, because the project's own material spends a whole chapter on it and this book had only ever

printed the *results*.

Start with a solid. The icosahedron has 12 corners, 30 edges and 20 faces, and V − E + F = 2 however many times you subdivide it. That identity is the cheapest bug detector in geometry, and the book uses it as one.

Put it on a sphere of radius one. Divide every corner by its own distance from the centre and the parent edge measures 1.051462 of the radius — the only number in this section that is not a length a reader can hold.

Halve every edge. The midpoint of a straight edge is the average of its two ends, and that average falls *inside* the sphere: the midpoint sits 0.850651 of the radius from the centre, a shortfall of 0.149349 — 14.93 per cent of the radius. On a small model that is nothing; on a building it is the difference between a faceted ball and a dome, and it is the arithmetic answer to the reader who assumes a sphere must be rounder. A sphere drawn with straight sticks is not round at all, and the input to that error is a straight line drawn where the surface curves.

Push each midpoint back out to the surface — divide it by its own length. That one operation is the entire difference between an icosahedron and a geodesic dome, and the project calls it the geodesic step. It leaves the edge line kinked, and the kink is where the second length comes from.

Measure the two. The chord from a corner to a projected midpoint is 0.546533 of the radius; the chord between two projected midpoints is 0.618034. Two lengths and no third: that is why the cut list has two classes rather than forty, and it is the arithmetic behind every table in this chapter.

Two corrections for things a reader may have heard, both checked rather than asserted.

Phi is not the strut ratio. The golden ratio is 1.618034, and it has a real home in this geometry: the twelve corners of the icosahedron are three golden rectangles at right angles to each other. But the ratio *between the two struts* is 1.130826 — about 30 per cent below phi, near enough to look like a coincidence on a drawing and nowhere near enough to build on. A two-frequency dome is not a golden-ratio dome; it is a dome whose parent solid happens to be built from golden rectangles.

The boards are not the factors. Cut 72 inches and 63.67 and you are cutting the solved lengths. Measure two boards on a bench and you get 1.133858 — 0.27 per cent above the factor ratio, because a tape, a kerf and a real log are not a unit sphere. The worked-build chapter fits both boards by least squares rather than trusting either, and that 0.27 per cent is exactly what it is fitting around: the arithmetic is exact and the tree is not, which is the method’s whole position in one line.

Other Species, Other Sections

Hardwood, hemlock, and a trunk that is not round

Other Species

The book's tree has been pine throughout — the windfall of Chapter 1, straight-grained, soft, forgiving. This chapter is what changes when the tree is something else, because the method was never pine's; it was the *sector's*, and sectors grow in other forests.

The chapter covers the three differences that matter: how other species split, dry and hold; how to split a trunk that is not round; and what building with green wood actually does over the first year. The honest rule of the chapter, stated once and applied throughout, is **use what grows where you are** — the method works with whatever straight-grained stock the local woodlot offers, and the differences are adjustments, not prohibitions.

Density, splitting and the drying schedule

Species by species, the practical differences, because a book that says "any species" owes the reader the list of what "any" asks.

What splits well. Straight-grained softwoods — pine, spruce, fir, cedar — split cleanly along the radius, which is what the seven-cut procedure assumes. The grain runs the length of the log, the pith is honest, and the sector comes off whole.

What splits poorly. Interlocked or spiral grain — elm, some eucalyptus — fights the rip: the grain runs around the log instead of along it, the split wanders, and the sector arrives ragged. The tree's bark usually tells the story before the saw does — spiral bark means spiral grain — and the rule is to read the bark and let the tree opt out.

What lasts. Rot resistance is a species property, not a build quality. Cedar, larch, black locust and osage orange sit in the ground and shrug; pine and spruce do not. The rot-prone species belong above

the damp line, on dry footings, with the bark charred or the face oiled the way the finish section describes — and the rot-resistant ones are the free upgrade the local woodlot sometimes hands over.

Density and the drying schedule. Hard, dense wood dries slowly and splits harder; soft wood dries fast and forgives. The schedule follows: dense species get longer in the stack, thinner stickers, more patience — and the key-cut-last rule below becomes more important the denser the tree, because a dense sector moves more in its first year than a pine one does.

A trunk that is oval

The tree is not always round, and the split procedure's first assumption — a circle divided evenly into 8 — meets reality in the oval trunk.

An oval section still splits into eighths; it just produces *unequal* eighths if the cuts are spaced by angle the way a round log's would be. The fix is to space by *area*, not by angle: mark the end grain into 8 sectors of equal wood, not equal degrees, so every wedge carries the same section even though the trunk was an ellipse. The marks are made by eye against the rings — the pith is the centre, the rings are the map, and the sector boundaries are drawn where the wood divides evenly — and the ripping passes follow the marks instead of the habit.

The consequence travels, and the book says so: unequal spacing means the sectors' *angles* vary slightly, which means the seam keys vary slightly, which is exactly the tolerance the gasket chapters exist to absorb. The oval trunk costs a little more attention at the split and nothing downstream, because the downstream was designed for "nearly" — the same word the round trunk's own imperfections already earned.

Green versus dry

The last difference is not a species; it is a state, and every build in this book happens in it: the wood is green.

Green wood moves as it dries — shrinks across the grain, barely along it — and a wedge moves *least* of all the shapes a log could become, because a split follows the grain instead of cutting across it. The sector's angle closes slightly as the stick dries; a sawn board cups and twists; a boxed heart tears itself open with a check. The method's geometry was chosen for this behaviour: the frame is full of seams that can take the movement, the keys are the last things cut, and the whole structure is allowed to settle for a year before the skin is final.

The one rule that matters most, stated for the bench: **key and screw as you go, then let the frame dry before you cut the replacements.** The raise fits its scheduled keys as the panels go up — that part cannot wait for the wood to dry — but a key cut to a green seam fits a dry seam the way a green pair of shoes fits next year. Let the frame stand its first months, let the sectors close their angles, then cut replacement keys to the seams that actually exist. The rule costs patience and saves the dome; it is the whole lesson of drying, in one sentence.

And the finish, because the wood's bare faces are a joint, not a surface: char the outside bark face — the yakisugi pass — and oil it with linseed, and leave the sawn faces bare, because a finish on the joint faces traps the moisture the seam channel is trying to move. The one warning the finish carries is the safety one: oily rags self-ignite, and the can lives outside the building. The species change, the sections change, the state changes — and the method, all the way through, is still the one this book has been teaching: read the tree, split the sector, let the wood be wood.

Building With Other People

What changes when there are three of you

Other People

The method was designed for one person, and every number in the book so far assumed that constraint — the six-foot member, the one bench, the props a single pair of hands can set. This chapter is what changes when there are three of you, because crews happen: a partner, a neighbour, a build somebody else asked for.

The changes are fewer than the fear suggests, and they cluster in three places: the benches multiply, the teaching becomes a step in the build, and the liability conversation stops being optional. The method itself — the split, the jig, the seam — does not change at all, which is the chapter's first and most useful sentence: **the method was built from processes, not from a person, and processes hand over.**

Parallel jigs

Two benches, and the sorting problem that follows — because a crew does not make the panels faster by crowding one jig; it makes them faster by running two.

The second bench is a copy of the first, set from the same panel drawings of Chapter 37, checked against each other before production starts — two jigs that disagree produce two populations of panels, and the raising is where they meet. The check is the same one the circle layout used: measure both benches against the drawing, not against each other, and demand they agree within the gasket's give.

The sorting problem is the real new work. Two benches pull from the pile twice as fast, and the pile stops being a thing one person holds in their head; it becomes a shared inventory that two people mark, move and count — which is where the marking scheme of Chapter 31 earns its keep. The rule for the crew: one person owns the pile, one person owns each bench, and the pile-owner's count is the

number the day ends on. Parallel work without one shared count is how a dome ends up with thirty-nine panels and an argument.

Teaching the eight motions

How to hand the procedure to somebody in an afternoon, because the method's own claim — eight motions, no decisions — is exactly what makes it teachable.

The teaching order is the jig's order, and it starts with the *why* in one sentence each: the bench replaces the tape, the overfit is the crumple zone, the flush cut is the only true cut. Then the eight motions of Chapter 31, demonstrated once, watched twice, and the watcher's first panel is the tuition panel — the fourth-panel rule applies per person, not per build, and a crew that budgets three practice panels per new hand is a crew that plans for the method instead of against it.

The one thing not to teach is the decisions, because there are none: the stops decide, the schedule decides, the marking scheme decides. The teaching is handing over *habits*, and a habit handed over correctly survives the hand that taught it. An afternoon, then: the method was designed to be this teachable, and the teachability is part of the design — a method that needed its inventor present at every joint would be the frankendome again, wearing a crew.

A build that is not yours

The third change, and the chapter's serious close: helping somebody else build, and the conversation that must happen first.

The liability conversation is not legal advice — the book is not a lawyer, and it says so — it is the plain speech version of the safety and engineering pages, delivered before the first cut: this method is a prototype, not an engineered standard; the loads are the engineer's job; the felling is the most dangerous part and the rules of Chapter 27 are not optional; and the person whose tree, whose site and whose back are on the line is the person who owns the decisions. Helping is not owning, and the difference must be said out loud, because silence is how a favour becomes a lawsuit.

Then the build itself is the easiest of the three changes: the guest's dome is the same cut list, the same jig, the same fortnight — and the guest learns the method by building it, which is the only way anybody ever really learns it. The chapter's last sentence is the book's own: **we make the method; you build on it.** And when the build is somebody else's, the gift is the procedure — the eight motions, the honest numbers, the warnings — and the rest is their fortnight.

Who the method is actually for

The method fits three kinds of people, and the list is worth printing because a reader wondering whether any of this applies to them will recognise themselves in one of the three or in none.

The person with trees and no money. The method's original audience and the one it fits best: land, standing timber, a small saw, and more time than cash. A dome is a way of turning the first three into a building without turning them into a loan first, and the money chapter is the arithmetic of that trade. This reader needs no permission and no contractor to start; they need the cut list and a fortnight.

The person with money and no trees. The reader who wants the building rather than the process, and for whom this book's honest advice is the product chapters: buy the standard article, buy the core, buy the modules, and skip every chapter about splitting. The method still serves them — it is what the thing they are buying is made of — but the labour arithmetic is irrelevant, and the Complete Edition's two kit rungs exist precisely to price that difference.

The person who is neither, and wants the method anyway. The quilter with a sewing machine and an evening, the host with a serviced pad and no dome, the builder who wants to make them for other people. This is the audience the method reaches last, and it is the one the project has the least evidence about: quilts need quilters, pads need hosts, and both need somebody to go first.

Two levers connect those three, and both are about *who does the assembly*.

The customer-assembly lever. Almost all of the cost of a delivered building is the labour of putting it together, and all of that labour is work a determined owner can do — which is why the Complete Edition prices the same kit thousands apart depending only on whether the timber comes with it. The lever is not a discount. It is the price of the work, offered as an option.

And the little-guy lever. One owner who wants a dome is a customer; fifty owners who want domes are a market for pads, quilts, cores and kits, and the pad is the unit that can be shared. Nobody needs this book to build one dome. Somebody might need it to start the thing that builds the fiftieth.

CHAPTER 51 · STORY

What I Would Do Differently

The honest closing chapter

The story strand closes where it opened: one person, one saw, one tree that was already down. Between the first page and this one the tree became a frame, the frame became a method, and the method grew a book around it — and the honest way to close a story like that is the same way the book has handled everything: name what would change, keep what was right, and say what the next one is.

This chapter is not a confession and not a victory lap. It is the builder's review of the build, written while the review is still worth something — the five decisions that would change, the parts that stay exactly, and the second dome the method has already earned.

The five decisions I would change

Specific, named, with what each would have cost — because "learn from your mistakes" is advice until the mistakes have names.

The marking scheme came too late. A panel built backwards — the mirror of the one the frame wanted — was the tuition for a scheme that should have existed on day one: mark the sawn faces, one stripe per class, bark up. It cost an afternoon to discover and it would have cost ten minutes to do first. The next build starts with the crayon.

The keys were cut too early. The green-frame rule of Chapter 49 — the raise keys go in as you go, and only the replacements wait until the frame has dried — was learned the expensive way, with a full set of keys cut up front that fit a frame that no longer existed by month three. The rule is now in the book; it would have been in the build, if the build had had the book.

The ring was trusted once too often. The tolerance chapter names the ring as the one place error accumulates, and the build proved it: a ring checked once, at setting, moved a fraction under the first course, and the correction cost a morning of shims. The three checks of Chapter 38 would have caught it in the first one, for free.

The skin waited for the wrong perfection. The shell was delayed while the "right" skin was decided, and the frame stood in the weather longer than it should have. The shell ladder of Chapter 55 now says what the build learned: the first skin's only job is to keep the rain off, and the perfect skin is the one you buy after the first one has taught you what you actually need.

The fuel was not logged. The dashes in the money chapter's table are this decision: the build kept its hours and not its fuel, and the calories the work cost had to be modelled where it should have been measured. The next build logs every tank, because the book's rule about numbers applies to the builder's own: what is not recorded is not claimed.

What I would keep exactly

The parts that were right the first time, listed with the same specificity as the changes, because a review that only lists failures is a review that learned nothing about what worked.

The split count. Eight, for the reasons Chapter 12 printed — the key fits, the stick carries, the cuts stay "through the middle". Not one afternoon of the build ever regretted it.

The overfit. The 6 inches of Chapter 35 are the cheapest insurance in the method, and the build spent them exactly as designed: every error that had somewhere to go, went there.

The jig, as built. One flat bench, stops instead of measurements, the day that paid for the other thirteen. The build's single best decision, and the one the second build will not have to make again.

The order of the fortnight. Fell, buck, split, jig, panels, raise — the sequence the book now teaches — survived contact with the weather, the wood and the calendar without a single step being moved. The order was right before the method existed, and the method is mostly the order, written down.

The honesty habit. The corrections printed where the claims were made, the dashes in the money table, the near miss reported — that discipline was the build's own habit before it was a book's policy, and it is the one thing in this chapter that costs nothing to keep and everything to lose.

The next one

What the second dome is, and why — the question every builder asks at the end of the fortnight, answered the way the build answered everything: with the numbers, and without the romance.

The second dome is the same method at the size the first build's own leftovers suggest: the spare struts and the offcut pile already hint at a smaller companion — a greenhouse, a workshop, a guest

pod — and the second frame's parts list is the first frame's parts list, with the problems already solved. The second build's real change is the one Chapter 47 estimated: the tuition is paid, the jig exists, the scheme is written — and the fortnight that took fourteen days the first time is the same fourteen days, spent where they were always supposed to go, on the repetition instead of on the learning.

The next one is also the book's promise to its reader, stated plainly: the method was never for one dome. It was for the *second* dome — the one built without the discoverer, from these pages, by somebody who has never watched it done. Every number in this book was written for that person, and the chapter's last line is the book's whole thesis in one sentence:

A windfall taught the method; three trees, felled on purpose, built the frame. The saw cost 199 dollars. The rest is in here.

The dome, a year on

The closing plate is the one image in the book that must be a photograph rather than a computation: the dome, a year on, standing where the tree stood. What the frame chapters only promised shows there — the wood greyed and settled, the keys seated, the skin doing its one job, and the building looking less like a fortnight's work and more like the land's own idea of a room.

Until that photograph is taken, the story strand closes on the frame at day fourteen — the standing plate of Chapter 32 — because that is what a year looks like at its beginning, and this book is the beginning's record, kept honest all the way to the end.

PART 2

Why It Scales

“The parts list does not grow with the house. Everything downstream of that one fact is this part.”

Four chapters on the shell itself: why a dome skins cheaper than a box at the same floor, what a pony wall buys at the edge, how the brim throws water clear and catches it in one move, and how the shell gets warmer one affordable layer at a time. Everything else this project makes -- the network, the pads, the manufactured systems -- lives in the Complete Edition on the site.

Less Skin for the Same Floor

A dome wraps a given floor in noticeably less outside surface than a box does, and the wall you never build never costs anything to build, heat or repair

A wall gets paid for four separate times.

It is paid for the day the sheathing is bought. Again the day it is insulated. Again every time it wants paint, flashing, membrane or a walk round with a caulk gun. And again every winter for as long as the building stands, in heat that leaves through it. Four bills, one surface, and only the first of them ever appears on a materials list.

Which makes the cheapest square foot of wall in any building the one that is not in the drawing.

That is the whole argument for building round, stated without enthusiasm. A sphere is the shape that wraps a given volume in the least surface. Cut one in half, stand it on the ground, and most of that property survives the cut. The same floor, enclosed the round way, takes less skin around it.

It is true. It is also smaller than the people who repeat it believe, a good part of it turns out not to be efficiency at all, and it is a property of *small* buildings rather than a property of domes. There is a size above which the box wins outright, and this chapter finds it.

None of that is an argument against building round. It is an argument against quoting the good number on its own, which is what I did for about a year, in public, with a straight face.

The same floor, two envelopes

So here are two buildings with exactly the same floor.

One is this book's dome: 365 square feet of floor, 10.8 feet to the crown — which for a hemisphere is also its radius, a coincidence that stops being charming later in this chapter. The other is a square house of identical floor area, 19.1 feet on a side, with 8-foot walls and a gable roof reaching 12.8 feet at the ridge.

The box is drawn fairly. It is square rather than oblong, which is the least perimeter a rectangle can have for its floor, so it is the best case for its own shape rather than a straw man. Its corners are not penalised, even though corners are where boxes leak heat and where boxes rot. And the dome is measured as what it actually is — flat triangles — rather than as the ideal sphere it approximates, which costs it a little surface it would not have if geometry were doing the building.

The dome's shell comes to 677 square feet. The box's comes to 1,110. The difference is 433 square feet of building that simply does not exist in the round version: **39.0 per cent less.**

That is the number. Now the part that gets left off the poster.

Read the volumes in the drawing. The dome holds 2,622 cubic feet. The box holds 3,790. At this size the dome is not only wrapping the same floor in less skin — it is wrapping 30.8 per cent less building. Some of that saving is shape. Some of it is that you bought less house.

Ask the question the other way round, then. Not skin per square foot of floor, but skin per cubic foot enclosed. The dome still wins, and it wins by 11.8 per cent rather than 39.0. That is the shape-only figure at this size, and it is roughly a third of the headline.

The two do not stay apart. Past 1,015 square feet of floor the dome begins enclosing *more* air than the box rather than less, because a hemisphere has to gain height to gain width and a stud wall does not. Above that line the volume deduction disappears and the surface margin is real all the way through. Below it — which is where this book's dome sits, and where almost every dome anybody actually builds sits — part of the win is a smaller building wearing the same floor plan.

Both of those are worth knowing. Only one of them ever gets said.

What the saved skin is worth four times over

Take the 433 square feet at face value for a moment and walk them through the four bills.

Bought once. The dome's shell is 21 sheets of four-by-eight against the box's 35, so it starts 14 sheets lighter before anything else has happened. Each of those sheets is also a delivery, a lift, a set of fasteners and a stretch of work that is paid for in labour as well as in money.

Insulated once. The same area over again, in whatever is chosen to fill it, priced per square foot like everything else.

Sealed repeatedly. This is the bill nobody budgets and everybody pays, and it is the only one of the four that arrives forever.

Heated every winter. At a wall assembly of 0.060 BTU per hour per square foot per degree — an ordinary insulated wall, handed to both buildings identically — the dome loses 40.6 BTU an hour per degree against the box's 66.6. Across a season of 4,200 heating degree days that is 4.1 million BTU against 6.7.

And this is where I have to take something back.

39.0 per cent less heat is not a second finding. It is the first one wearing warmer units. Heat through an envelope is area multiplied by a U-value, both buildings were given the same U-value, so the heat margin *cannot* come out any different from the surface margin. It is arithmetically forbidden. Saying "less skin, and less heat" sounds like two wins and is one win said twice, and I have said it that way in public.

The wind figure is genuinely independent. 0.42 against 1.05 — 60.0 per cent less drag — comes from published coefficients for a hemisphere and a cube and has nothing to do with area at all. That one may be quoted beside the surface number without double counting.

Then one deduction larger than everything above it.

Equal floor area is not equal usable area. A hemisphere meets the ground at a tangent, so its ceiling arrives at its floor around the whole perimeter. Call standing room 7 feet — generous; habitable ceilings are held higher than that — and only 225 square feet of this dome's 365 qualify. That is 61.7 per cent of the floor, and 140 square feet of it that you can reach on your knees. Chapter 44 draws the whole curve; the shaded rim in the drawing above is the same fact in elevation. Every square foot of the box clears the line.

So grow the dome until it stands up over as much ground as the box does. It needs 505 square feet of floor to get there, and 937 square feet of skin to wrap it, and the margin falls to **15.6 per cent.**

Still a win. Roughly half the one on the poster, and the only version of it I would put in front of somebody who builds for a living.

Where the box wins

The margin is not a property of domes. It is a property of small domes, and the chart says so more bluntly than I would like.

11 sizes, from 80 square feet to 5,000, both buildings holding the same floor at every point. The saving starts at 62.5 per cent for something hut-sized. It is 39.0 at this book's dome. At 707 square

feet — past the one-person band — it is 27.8. By 1,015 it is 21.7, and by then the dome has stopped being the smaller building too.

And at 4,288 square feet it is gone. Zero. Above that floor area **the box takes less skin than the dome**, and every sentence in this chapter runs backwards.

The reason is in the drawing and not in the arithmetic. A stud wall is 8 feet tall whether the house is small or enormous. A hemisphere cannot do that: to cover more ground it must stand taller, and at the crossover it is 37 feet to the crown and buying skin to wrap a column of air that nobody will ever use, heat or reach.

That is not the only place the box wins, and the others do not need a chart.

Boxes stack. A second storey doubles the floor for the price of a deck, adding no roof and no foundation, and a dome cannot do it at all — a dome's second floor is a mezzanine inside the same shell, competing for the same volume. Boxes share walls: two terraced houses each delete an entire elevation, and a row of them beats any dome on skin per floor so decisively that there is no argument to have. Boxes take sheet goods whole, where a triangle takes a rectangle and hands you the offcut. And boxes fit furniture, appliances, cabinets, baths and doors, every one of which is manufactured for a flat vertical wall that a dome does not have.

The honest summary is narrow and worth having anyway. For a single, small, detached, one-storey building, on its own ground, the round shape wraps the same floor in less skin, and that saving lands four times on every bill the building will ever send you.

Outside those conditions, build a box.

Inside them, the dome still has the rim problem this chapter kept pointing at — 140 square feet of floor with no room to stand up in. The fix for that is not a bigger sphere. It is the cheapest wall in the whole build, and it is 53.

The sheet that does not fit

One practical consequence of the shape belongs in this chapter, because it is the difference between a saving on paper and a saving at the bench.

No panel of this dome fits a four-by-eight sheet. The triangles are larger than the sheet once a margin and a kerf are allowed for, which means the skin cannot be cut by laying a template on a plywood sheet and running a saw around it. Every panel is a cut somebody has to get right, and on this building that cut is wanted forty times.

Which is why the panel drawings in this book matter more than a reader expects of a drawing, and why buying the cut instead costs real money on a router with a four-by-eight bed. The honest options for a self-builder are three: cut the panels oversize and trim them to the frame, which wastes material and requires the frame to be right first; choose a size whose panels *do* fit a sheet, which is exactly what a smaller dome does; or buy the cut, in which case the sheet's size stops being your problem and becomes somebody else's line item.

What this chapter will not do is pretend the saving is free. Less skin is less material, less money and less to heat — and it is also a harder cutting job than siding a box, which is one of the places where the box genuinely wins.

The whole case in ten lines

The case this book has made, in one place, because a reader who has come this far deserves the summary rather than the tour.

1. **Less exterior for the same floor.** 39.0 per cent less surface, and at equal insulation the heat follows the surface exactly.
2. **The parts list does not grow with the building.** The frame is 40 panels and 120 members at any diameter, and the hardware count rides with the seams — 55 of them — not with the floor.
3. **8 operations, repeated.** Not a hundred — eight, and the list is genuinely closed.
4. **It is rigid by shape rather than by bracing.** A triangle cannot change shape without a side changing length, so the frame needs no shear panels, no diagonal bracing and no racking check.
5. **Little for the wind to push on.** 60.0 per cent less drag than a box of the same floor, before any hold-down is counted.
6. **It costs what the parts cost.** The stack in Chapter 46 shows the builder's own labour and trees as the two biggest lines, with nothing in the number for a shape premium.
7. **The cheapest floor in the building is a short wall.** 791 buys 97 square feet of standable floor — the arithmetic Chapter 53 worked through line by line.
8. **Less skin to heat and cool.** 39.0 per cent off the load, which is the same number as the surface saving and arrives for the same reason.
9. **A roof that can refuse the sun.** Pale, and optionally radiative: a passive option priced at the value of the air conditioner a reader does not buy.
10. **A brim that collects rain.** 7,259 gallons a year off an overhang the building already wears, into a 1,000-gallon tank, through a gutter that is not an accessory.

Ten claims, all computed, and four of them carry a caveat this book has already printed in full: the skin margin dies above the envelope crossover, the dead ring at the rim is real until a wall is built, the paint is comfort rather than payback, and the whole case assumes a small single-storey building on its own ground. A summary that dropped the caveats would be the sales document this chapter has spent its length not being.

The Cheapest Square Footage in the Building

Standing the shell on a short straight wall buys floor, headroom and usable edge for the least money per square foot of anything in the build

A dome has one weakness, and it is honest about where it is: the edge. A hemisphere comes down to the floor at a tangent, so the ring of floor around the rim has a ceiling sloping straight at it. You can stand in the middle of the room. You cannot stand anywhere near the wall.

On the reference floor — 314 square feet, a little smaller than this book's own 365 — only 174 of those feet are under enough ceiling to stand up under. That is 55 per cent of the floor you paid for, and the rest is a dead ring.

The fix is the cheapest wall there is. A short, straight stem wall stood under the rim of the shell, lifting the whole dome a few feet off the ground. It carries no thrust, because the ring at the base of the frame already takes that. It is just a wall of height. And a wall of height is exactly what converts a dead ring into the cheapest square footage in the building.

The pony wall ladder

The ladder is four rungs, all computed from the same geometry, at the declared cost of 4.20 dollars a square foot of wall:

wall	usable floor	share usable	the wall costs	floor gained	$/sq ft gained
none	174 sq ft	55%	nothing	—	—
2 ft	245 sq ft	78%	$528	+71	$7.4
3 ft	**272 sq ft**	**87%**	**$791**	**+97**	**$8.1**
4 ft	292 sq ft	93%	$1,055	+117	$9.0

Read the bold row. The 3-foot wall is 188 square feet of ordinary wall, 791 dollars. It buys back 97 square feet of floor you already paid for and could not use — a 56 per cent increase in usable room — at 8.1 dollars a square foot.

Nothing else in this build sells floor at 8.1 dollars a foot. Not the frame, not the skin, not the glass. The wall is the one part of the building where the floor is already there and only wants lifting.

The other rungs are worth a look, because each has a different character. The 2-foot rung is the bargain: 71 feet for 528 dollars, the cheapest dollars on the ladder. It leaves a fifth of the floor unusable, but for a workshop or a greenhouse — rooms where you bend over the work and do not care about the walls — it is the right rung. The 4-foot rung is the wrong one at the far end: it spends 1,055 dollars to buy 117 feet, the last of them at 9.0 dollars a foot, and the next section explains what else that height starts costing. The 3-foot rung in the middle is the one for a room people live in, and the reason it is the bold row is the third section of this chapter.

There is one more thing the wall buys that the table does not show, because it is not a square foot: a band of straight wall around the room. On a bare hemisphere the ceiling curves down to the floor everywhere, so a door has to be cut out of a curved surface and a cupboard has to stand away from the wall. On a 3-foot stem the first three feet of every wall is a vertical, rectangular surface — doors are rectangles in it, windows are rectangles in it, shelves and counters stand against it. The stem wall does not just buy floor. It buys *furniture wall*, and that is the difference between a room in a dome and a tent with a dome over it.

The same band is where the heavy and rectangular things live. A door is a rectangle; cut one in a bare shell and you are building a curved door, or boxing a square out of a surface that slopes two ways at once. On the stem the door is just a door, and the windows are stock units instead of specials. The band also swallows the clutter a dome has nowhere else to put — the water tank, the batteries, the shelves, the coat hooks — which would otherwise stand off the curved wall and eat the floor the wall was bought to save.

Why the edge is the expensive part of a dome

It is worth understanding *why* the edge is dead, because the number comes out of one simple fact: a circle meets a flat floor at a tangent.

Headroom in this comparison means a 6.7 foot ceiling. On a dome 10 feet in radius, you can stand only where the shell is at least that high. Walk from the centre toward the wall and at some radius the ceiling drops through 6.7 feet and keeps going, all the way to zero at the rim. Everything past that radius is floor with a sloping lid over it — geometrically fine, humanly useless.

That dead band is not a strip. It is the whole outside of the circle, and it is where the *most* floor lives, because a circle's area lives at its outside. It is why the dome's cheapest room is bought by moving the shell up, not by anything you do to the frame.

The same geometry explains why the wall is cheap compared to every other way of buying that floor back. Want the floor by making the dome wider instead? Then the frame grows with the diameter and the skin grows with the square of it — every extra foot of room is bought with longer sticks *and* more skin, which is the most expensive way there is to buy a foot. Want it by the wall? The wall grows as a line — one more foot of wall all the way round is a fixed, small number of square feet of plain wall at 4.20 dollars each. The shape of the dome makes the rim the expensive place to stand; the shape of the wall makes the rim the cheapest place to fix.

A taller shell would do the same job, but a taller shell means a bigger frame everywhere — every member, every panel, the whole fortnight. The stem wall buys the height in exactly one place: at the rim, where the room is missing. That is the whole trick, and the ladder table is what the trick costs.

Where the ladder stops paying

The ladder is not free forever, and its own columns say where it stops.

Look at the right-hand column. The rate climbs: 7.4 dollars a foot, then 8.1, then 9.0. Each added foot of wall buys less floor than the one before it — 71 feet, then 97, then 117 — because the ring of new standing room gets thinner as the ceiling profile flattens out. The wall is still cheap. It is just not as cheap.

And past a certain height the wall stops being a free ride in a second way: it starts acting like a real wall. A 3-foot stem is a curb. A 6-foot stem is a cylinder the wind pushes on, one that needs its own ties into the ring, its own sheathing, its own foundation edge. Somewhere past that you have built a round box and set a hat on it, and you are paying for a conventional wall *and* a dome frame on the same site.

There is also a water detail the wall introduces that a bare dome does not have, and it belongs to the chapter on the brim (Chapter 54): where the shell meets the top of the wall there is now a horizontal seam between two materials, and it is exactly the kind of joint water finds. The brim throws the rain past it; without the brim, the wall is a gutter fed by its own roof. The two chapters are one decision: the wall buys the floor, the brim keeps the wall.

The top of the wall wants a cap, and it is the one piece of metal the rest of the dome never asks for: a strip of flashing lapped down over the wall face and up under the shell edge, so the end-grain of the

wall's top plate is never the thing the weather meets. It is a cheap part, and it is the part first builds leave off, because it hides under the hat and nobody is checking.

The honest stopping point is where the next rung costs more per gained foot than the next size of dome would. This book's own dome — the one 3 trees fill — is 365 square feet for the same fortnight of work as the 314 square foot reference floor. If you want more than the 3-foot rung buys, the better answer is usually not a taller wall. It is the next dome up, where the same hundred and twenty members stand further out and the floor comes along for free.

The wall buys back the edge of the floor you already have, for 791 dollars, and nothing else moves. The bigger dome — the 21.6 foot dome 3 trees surround — buys that edge by buying the room over again, spending the one thing a dome exists to save, the subject of Chapter 52. The wall is the answer when you are short of edge; the bigger dome when you are short of room.

So the ladder has a last rung, and on this floor it is the 3-foot one: 272 usable feet for 791 dollars, at 8.1 dollars a foot. That is the cheapest room in the building, and this chapter is the receipt.

The Roof Is Already a Gutter

An overhanging brim throws water clear of every joint and collects it in the same move, and the annual catch off a dome this size is not a trivial number

The brim around a dome — the overhanging ring of the hat — looks like decoration. It is not. It is the detail that keeps water out of the building, and it does two jobs at once.

The first job is the boring one that matters most. Rain runs down a dome shell, and without a brim it runs straight into the ring of joints at the base: the seams, the brackets, the screws, the point where timber meets footing. Every one of those is a place water can sit, wick and work on the frame for years. The brim throws all of it clear of the wall before it ever gets there.

The second job is the one this chapter is about. The same overhang is a catchment. It catches the rain it throws, and the annual number off the reference dome is 7,259 gallons — a figure that scales with the floor, so a bigger dome catches proportionally more. Not a trivial number, and it costs no extra roof to collect it, because the roof was already a gutter.

Where the water goes, with and without a brim

Follow one raindrop through both buildings.

Without the brim. The drop lands on the upper shell, runs down a panel, across a seam, down onto the lower panels and off the rim — straight onto the base joints and the splash line at the bottom of the wall. Some of it wicks into the seam keys. Some of it follows the brackets down the inside face. Over a wet season the bottom ring of the frame is damp more often than it is dry, and the footing is in a permanent puddle of its own making. The building is not leaking into the room. It is leaking into its own skeleton.

With the brim. The same drop lands on the shell, runs down to the hat, and hits the overhang. The hat's edge is further out than the wall — 10.4 feet from the centre, against 8.9 feet for the rim of the hat itself — so the drop leaves the building at a point where there is nothing below it but ground. It drips clear. The base joints stay dry because the rain never arrives.

The brim does not stop at dripping clear. Put a gutter lip on the hat's edge — the same metal the hat is already made of — and the drip becomes a stream, and the stream runs into a tank.

The lip is also the only gutter this building will ever have, and it wants the same care as any gutter. Leaves, needles and dust wash down the shell and arrive at the lip, and a gutter is a gutter — it clogs where the downpipe narrows. This one is a ring at the hat's edge, so the autumn job is a walk round the building with a ladder and a broom. In snow country the brim is a flat shelf that holds ice, and an ice dam on the hat can weigh more than the metal was priced for.

One of those two buildings has a water system. The other has a slow leak into its own frame. The difference is the width of the hat, 18 inches.

Wider is not automatically better, and this is the place to say why, because it is the same trade everywhere on the shell. A wider brim catches more — every extra inch of radius is real catchment — and it also catches more *wind*, and it shades more wall, which is a virtue in summer and a cost in winter. The 18 inches this book prices is the width that clears the wall, shades the windows, and stays on the right side of the wind math. The number is a choice, declared once and used everywhere, not a discovery.

The annual catch

The catch is worth working out properly, because the number sounds like a mistake the first time you see it.

The inputs, all declared before they are used: 40 inches of rain a year, the regional average; a runoff coefficient of 85 per cent, which is what a metal roof sheds — the rest evaporates, splashes and never reaches the lip; and the 314 square foot reference dome, whose hat stops at a plan radius of 8.9 feet.

The catchment is not the sloped area of the hat. Rain falls straight down, so what catches it is the *plan* area — the circle the brim's edge describes over the ground. At 1.5 feet of overhang that circle is 343 square feet.

Now the arithmetic. 40 inches times 0.623 gallons per square foot per inch of rain, times 343 square feet of catchment, times the 85 per cent that actually sheds:

7,259 gallons a year. 19.9 gallons a day, on average.

It is worth saying what that number is not. It is not metered — nobody has stood under this dome with a bucket for a year. The rainfall is a regional average, not a guarantee, and a drought year is

exactly the year the catch goes dry. But the arithmetic is the honest plan-view arithmetic, with every efficiency declared on the line rather than buried.

Size the tank against use rather than rain. Two people, using 50 gallons a day between them, empty a 1,000 gallon tank in 20 days — so a full tank is a three-week buffer against a dry spell, and in an ordinary year it refills 19.9 gallons a day.

At utility prices the whole year's water is worth about 44 dollars. Say it plainly: collected rain is nearly free water either way, and the money argument is a weak one. The real argument is the 20 days of water that keep flowing when the well does not. That is what the number buys.

The same 7,259 gallons appear in the ten-points list in Chapter 52, computed by this same function. That is not a coincidence and it is not a copy: the two chapters read the one water model, so they cannot drift apart. Wherever you meet this number, it is the same dome, the same rain, the same arithmetic.

What the catch is actually good for

The honest uses first, because the dishonest one gets said first everywhere else.

Not drinking, without treatment. Rain off a roof has been through bird droppings, dust and whatever the shell sheds, and the tank is warm and dark, which is a polite description of a biology experiment. Filter and treat it or keep it out of your mouth.

Everything else. Irrigation is the obvious one, and the dome's own garden sits in its shadow. Flushing and washing take the 19.9 gallons a day without a second thought. And the quiet use that matters most on a rural site: a buffer against a dry well. A well that fails in August is a crisis with an empty tank and an inconvenience with a full one.

The tank belongs under the pad, not beside it. Three reasons, all boring and all true. The pad is already excavated, so the hole is half dug. The ground under a pad is shaded and frost-protected by the pad itself, so the tank neither cooks in summer nor splits in winter. And the weight sits on compacted ground that is already carrying a building, instead of in a second hole you had to dig and backfill for no other reason.

Two details make the difference between a tank and a pond, and both are cheap compared to the misery they prevent. A first-flush diverter — a simple standpipe that captures the first few gallons of every storm, the ones carrying the dust and the droppings, before the clean water is allowed into the tank. And an overflow, sized like it is going to be used, because 40 inches of rain will overflow any

tank ever built and the water has to go somewhere that is not the pad. Skip either and the tank teaches you the lesson within a year.

The plumbing between the lip and the tank is simple, and it has two jobs nobody thinks about until the tank overflows. A leaf screen at the lip keeps the big stuff out of the pipe, because the first-flush diverter catches only the first of the dirt, not the last. One downpipe carries the whole ring's 19.9 gallons a day down the wall and under the pad to the tank, and it wants an air gap so a full tank cannot back up into the pipe and out the lip. That is the whole of the plumbing: a screen, a pipe, a gap.

One more job the brim does that is worth a line, because it is free with the same metal: shade. The overhang keeps the summer sun off the upper wall in the middle of the day, and it keeps the rain off the base ring all year. Every one of those jobs is done by a piece of the building that was already there, doing the first job — keeping the frame dry.

That is the whole point of this chapter. The gutter is not an accessory a dome can have. It is the brim the dome already wears, and the only question is whether you let the water run off it or into a tank.

The top half is the half worth glazing

One geometric fact makes the skin cheap if you use it, and it falls out of the subdivision rather than out of a design decision.

Sort the shell's panels by height and the bottom half of them come to almost exactly half the surface. That is a property of the way a sphere is divided rather than a coincidence: an even subdivision puts its widest ring at the equator, so the panels below the waist cover the same area as the panels above it and the sorting comes out even. Which means the question of what to do with the shell splits cleanly in two:

The upper half is the roof — seen from outside by nobody, lit from above, and the place where glass does the most work. Light comes in over the furniture rather than across the floor, and on a dome the upper ring is already the roof, which is why a skylight here costs nothing but a panel.

The lower half is the wall — where a reader wants insulation, siding, a hatch and the one solid door. Nobody sees the bottom of a dome from the air, and glass down there is glass paid for and then curtained over.

So the honest rule for a first build: glaze the top, side the bottom, and spend the difference on the brim. A reader who does that buys the *lighting* of a much larger building for the price of the half of

the shell that was worth glazing, and leaves the other half doing the job a wall is supposed to do. The same rule explains why the panel catalogue in this book lists sixteen surfaces for one geometry: the shape is decided, and the shell is a set of forty separate decisions that this one fact sorts in half.

A House That Gets Warmer Every Winter

Build the skeleton once and add layers to it for as long as you own it: the shell ladder turns a shelter into a house in steps you can afford one at a time

Most houses have two states and nothing between them: finished, or unfinished. Unfinished means tarps and regret. Finished means the whole invoice at once.

A dome shell can be neither, permanently and on purpose.

The skeleton of a dome is the frame; the frame carries the skin; and nothing about the skin needs to be decided on the day the frame goes up. You can stand a bare shell this autumn, quilt a first layer under it this winter, add a second next year, and keep going for as long as you own the place. Each step is affordable on its own, each step makes the building better, and none of them ever requires undoing the one before.

The ladder also becomes a habit, and that is the part the table does not show. One layer a winter is a small, repeatable job — a weekend under the shell, the same as splitting firewood or cleaning a chimney — and it lands in the season that needs it. By the time the house is fully quilted you have stopped noticing the cold and started noticing the ritual: the first hard frost is the cue, the layer goes up, and the house is a little warmer than the winter before. You buy a winter at a time, in the order the cold arrives.

That is the shell ladder, and this chapter is the ladder with its prices. The numbers are for the Split-Log Homestead — a 39.4 foot dome with 1,217 square feet of floor — because that is the dome the park model computes its ladder for, and a heating number means nothing without the building it came off.

This book's own dome is smaller than the flagship the table prices, and the ladder scales to it the same way: the per-square-foot price holds, the total is smaller because the shell is smaller, and the order of the rungs does not change.

The shell ladder, rung by rung

Each rung is one quilted layer of recycled fabric under the shell, at a declared 1.6 R-value and 1.1 dollars a square foot. The ladder runs to 7 layers:

layers	R-value	heating, $/yr	cost so far
0	1.5	8,260	nothing
1	3.1	3,955	$2,486
2	4.7	2,600	$4,973
3	6.3	1,937	$7,459
…	…	…	…
7	12.7	958	$17,404

Read the first and last rows. The bare shell is R-1.5 and it wants 8,260 dollars a year to stay warm. The same shell with all 7 layers is R-12.7 and wants 958 dollars a year. The layers removed 88 per cent of the heating bill, and they cost 17,404 dollars spread over however many winters it took you to add them.

The point of the ladder is not the top row. It is that there *is* a ladder. Every rung exists on its own: one layer is a real improvement, not a down payment on seven. You buy the rung you can afford, in the winter you can afford it, and the house is warmer than it was the winter before.

What the next layer costs, and what it returns

The table above prices the ladder from the bottom. This one prices it from the *rung you are standing on*, which is the question a reader actually asks in year three: what does one more layer do for me now?

layers	R	heating a year	the layer that got you here	saved a year	pays back in
0	1.5	8,260	—	—	—
1	3.1	3,955	2,486	4,305	7 months
2	4.7	2,600	4,973 cumulative	1,355	—
3	6.3	1,937	7,459 cumulative	663	—
7	12.7	958	17,404 cumulative	139	17.9 months

Read the last two columns together and the ladder stops being a table of prices and becomes a decision rule. **The first layer saves 4,305 dollars a year and pays for itself in 7 months** — inside the first winter, by a wide margin, which is why the first layer is not a question. **The seventh layer saves 139 dollars a year and takes 17.9 months** to pay for itself, and it is being asked to justify itself against a shell already holding R-6.3 at three layers. Both of those are true, both are in the same table, and the honest reading is in between: the ladder is worth climbing while the next rung pays back inside the horizon you intend to keep the building, and it stops being worth climbing when it does not.

Which is the reason the quilt is shaped the way it is. A layer bought in a cold year for 2,486 is a different proposition from a fourth layer bought for a house that is already warm, and the quilt's whole economic argument — 50 dollars a layer for 132 shirts' worth of somebody else's evening — is that the *first* rung of that ladder is cheap enough to be worth doing for a building somebody else lives in. The ladder is what makes insulation a sequence of small decisions rather than one large one, and the payback column is where a reader decides how far up to go.

It is also worth noticing what the ladder is made of, because it matters that it works at all. These are not batts stuffed between studs. A quilted layer is a continuous sheet hanging under the whole shell, and continuity is the difference: a stud wall insulates between the studs and conducts through every one of them, so its *assembly* R-value is well below its cavity number, while the quilt has no studs in it to bridge. The shell frame is the same story — the struts touch the skin at the seams, but the quilt covers them too, on the inside, in one unbroken layer. That is the quiet reason a ladder of cheap fabric layers does what the table says it does.

The quilt is worth one more sentence for what it is made of, because it changes the price. It is recycled fabric — clothing and bedding that left the waste stream and came back as insulation — so the 1.1 dollars a square foot is not the cost of a manufactured foam product. It is paying somebody to shred,

clean and stitch what a city was throwing away. The ladder adds a thickness of second-hand fabric to a frame that was already second-hand timber.

Why the bones are the thing to get right

Every layer in that ladder attaches to the same thing: the frame. The quilt hangs from the struts, the seams key into the panel edges, the inner lining screws to the members. There is no layer that does not touch the skeleton.

That is why the frame is the one part of the building worth being fanatical about, and it is the cheapest time to be fanatical about it. If the frame is straight, every later rung has something straight to land on, and nothing is ever blocked. If the frame is wrong, the error does not stay in the frame. It walks up the ladder with you: the quilt puckers over a twisted strut, the lining opens a gap at a seam that never quite closed, and the tenth winter you are still paying to heat the mistake you made on day five.

There is a second reason the bones matter, and it is the one the marketing never says. The ladder's arithmetic prices *conduction* — heat moving through the shell material. It does not price airtightness. Airtightness is not bought with layers; it is built into the frame, in a continuous shell with no corners to leak and seams you made yourself. That is the shell's job, not the quilt's. A leaky dome with seven layers is a quilted leak. This book has not metered a winter in one, and it says so plainly rather than making an energy claim it has not measured.

The rung that pays back fastest

Now rank the rungs, because they are not equal and the order surprises people.

The first layer is the best buy in the building. It costs 2,486 dollars and it saves 4,305 dollars in the first year alone — it pays for itself in 7 months, and everything after that is free. No other rung does that.

The second layer costs the same 2,486 dollars and saves 1,355 dollars a year. Still good, but the layer that took 7 months to repay now takes years, because each layer of insulation removes a smaller share of what is left.

The third saves 663 dollars a year. The fourth, fifth and sixth continue the same slide, and the seventh saves 139 dollars a year on its 2,486 dollar cost — a payback of 17.9 years, which is most of a mortgage.

That slide is not a failure of the ladder. It is the physics of insulation, and it is the same in every building: the first inch does the most work, and every inch after it works on the smaller leak that is left. What the ladder buys you is the *option* to stop exactly where your winters say to stop. A mild climate might own the first two layers and never miss the rest. A hard one buys all seven and the top rungs still make sense there, because a cold climate's heating bill is big enough that even 139 dollars a year compounds over the thirty years of a maintained dome.

There is one more column the table cannot show, and it is the reason the ladder beats a one-time insulation purchase in this particular building. Every layer attaches *after* the shell is up and *after* you have lived in it, so every rung is bought with information the first rung did not have — which walls are cold, which windows sweat, which winter hurt. A conventional house makes its one insulation decision before the walls are closed, on guesswork. The ladder makes seven smaller decisions, each one with a year of living in the house behind it. That is what "warmer every winter" actually means: not a promise about the weather, but a building that lets you keep paying for warmth in the order your own experience says to.

A one-time foam purchase is the other way, and it is the way a conventional house has to take. Foam is bought once, sprayed once, and sealed behind a wall the same day; if it was the wrong thickness, or it shrank, or a seam opened, you will not know until the bills are in, and you cannot reach it to fix it. The quilt is the opposite: it hangs where you can see it, it comes down in the same layer it went up, and a bad layer is a morning's work to replace. The ladder's rival was never the foam itself. It was the single invoice.

So the honest ranking is: the first layer pays back fastest, by an enormous margin, and it is rarely the one people do first. People paint the inside. They buy a wood stove, which is a lovely thing and does nothing to the envelope. They build a porch. The layer that returns 4,305 dollars a year on 2,486 dollars waits, because a quilt hanging under a shell is not something anybody has ever been proud of at a party.

That is the argument for the ladder, stated plainly: the house gets warmer every winter, one rung at a time, in the order your money allows — and the first rung is the best one, so even a broke first winter is a good winter to start.

The Master Tables

Four tables carry the whole build, and they are the ones to come back to at the saw. Every number in them is generated at build time by the same solver that drew the 3-D model, so a table and a drawing

can never disagree. Print these pages and take them to the bench.

Full cut list

The whole frame in one table: every member, by class, length and count. The frame has 120 members in 4 distinct lengths — two classes, short and long, each cut once per panel. The table is sorted the way the work is sorted: by class first, then by length, so a member on the pile can be found from the table in one look. Mark each class with its own colour at the saw, and keep the classes in separate stacks; the raise is where a mixed pile turns into a slow afternoon.

The master cut list

The list the saw is set to, as this build finished it. 3 lengths, and every stick in the frame is one of them:

length	what it is	where it goes
72.0 in	the long member	all three members of an equilateral panel; the **base** of an isosceles panel
63.5 in	the middle member	the **left** side of an isosceles panel
61.5 in	the short member	the **right** side of an isosceles panel

These are resultant lengths: cut a member to one of them and the panel is the size the dome wants, with nothing left to trim.

The rule, stated once and followed at the bench: **shortest on the right, middle on the left, longest at the base**, laid with the base nearest you. The left member is the longer of the two sides, so it reaches over to the top point, which sits off the middle of the base; the right member meets it there, and the base butts up to that short right side. The two sides of an isosceles panel differ by two inches over five feet, so nothing about the pair is obvious by eye — pick them from their own stacks, in order, and the panel is right. Equilateral panels have no rule to remember: three long members.

The fully seated schedule the pinwheel model prints is in the table above. Where the two disagree the master list is the report of the build and the model is the model of the joint.

Full seam schedule

Every seam in the frame, with its fold angle and its key. The frame's 55 seams fold at only two angles — 18.03 degrees, 25 of them, and 22.46 degrees, 30 of them — so the brake is set twice and the schedule is how you know which setting a given seam wants. The key column is the gasket: 0.75 inches of compressible strip in every seam, which is what lets a frame of green timber close and stay closed while the wood moves.

Butt cut setups

Every distinct saw setup in the build, in the order the work meets them. The butt cut is the one compound angle in the frame, and this table is the one place its setups live: stock left long by 6 inches at the head end, allowance of 8 inches past the cut, and the angle itself. The rule from the cutting chapters applies here in table form: the angle lives in the tool, not in the operator. Set the jig once per row, run every member of that row, and never measure an angle at the saw.

Declared constants

The inputs the whole book is built on, repeated here for reference. These are the numbers that are *declared* rather than derived — the gasket thickness, the handling limit, the prices and rates that could not be computed from geometry because they are facts about the world. Every one of them is stated with its reason where it is used, and this table collects them so a reader can change one and see what moves. That is the promise the front of this book makes and the arithmetic of this book keeps: no number is typed into a sentence. Change a constant here and every figure that depends on it — in the chapters, in the tables, in the 3-D model — changes with it.

Glossary

The words this book uses in the senses it uses them, for the terms a reader may not have met.

2V. Two-frequency. A geodesic hemisphere whose icosahedron edges are divided in two, so the sphere is carried by two lengths of chord. This book's frame is 2V: 120 members in two edge classes.

Board foot. A volume of wood: one foot by one foot by one inch. The unit mills sell by.

Brim. The overhanging lip of the shell, the hat's edge, which throws rain clear of the base joints and can catch it in the same move. Chapter 54.

Bucking. Cutting a felled trunk into sections of working length, cross grain.

Butt cut. The one compound-angled cut in the frame, made on the butt end of a member so it seats flat on the side of its neighbour. Once per member, off the jig, at a handful of fixed settings.

Cap. The rainproof outer layer of the shell, pulled over the quilt and the frame like a shower cap and strapped at the base.

Channel. The groove a seam's key and gasket sit in, and the path water takes down the seam toward the brim.

Chord. The straight line between two points on the circle; the edge a member spans. A pinwheel member is always shorter than its chord, because it is inset from the vertices it spans.

Declared constant. A number the book states as an input rather than derives: a price, a rate, a handling limit. Every declared constant is named with its reason where it is used, and collected in the master tables.

Dihedral angle. The angle between two faces meeting at an edge; in this frame, the fold between two panels at a seam.

Fold angle. The angle two panels close through at a seam, which the key is cut to fill. This dome needs only two of them: 18.03 and 22.46 degrees.

Frankendome. The author's first dome: mismatched salvaged boards joined with bent sheet-metal brackets, forty triangles and no two members alike. It stood, and it was awful, and it is why the wedge exists.

Frequency. How many times a geodesic edge is subdivided: 1V, 2V, 3V. Higher frequency means more, shorter members for the same dome. The flat parts list holds across diameter, not across frequency.

Gasket. The compressible strip in every seam — 0.75 inches – that lets a green-timber frame close and stay closed as the wood moves.

Hat. The upper part of the shell, the twenty upper triangles, whose edge is the brim. The hat's overhang throws water clear and collects it.

Hub. A junction where several members meet through one piece of hardware. The wedge frame is hubless: members lap past the vertices instead of butting into a hub.

Jig. A purpose-built guide that holds a tool or a workpiece at one setting, so a cut or a fold repeats exactly. The rule of the whole build: the angle lives in the tool, not in the operator.

Kerf. The width of the cut a saw removes.

Key. The strip of timber seated in a seam between two panels, cut to the seam's fold angle. The key locates the geometry; the gasket beside it seals the gap.

Limbing. Cutting the branches off a felled tree.

Member. One stick of the frame, cut and dressed for its place. The frame has 120 of them.

Overfit. The deliberate extra length left on a member's head end so error leaves as offcut when the flush cut is made against the jig.

Pinwheel. The frame's corner pattern: members lap past the vertices rather than butting into them, so no two members fight for the same end point. The inset the lapping requires is why a member is shorter than its chord.

Pith. The soft core of a trunk at the tree's centre. Splitting runs with the pith, and a wedge's point is the pith side.

Pony wall. A short straight wall stood under the rim of a dome, lifting the shell and buying back the low-ceiling ring of floor. Chapter 53.

Rip. To cut along the grain, the length of the member.

Runoff coefficient. The share of rain that sheds off a roof rather than evaporating or splashing: 85 per cent for a metal roof.

Saddle. The strap that spreads a cable's load over a tree's bark, so the trees carry the hanging frame the way they carried their own crowns. Saddles, never holes.

Seam. The joint between two adjacent panels; a fold angle plus a key.

Sector. One wedge split out of a round log section, pith to bark.

Setback. The inset of a pinwheel member from the vertex it laps past.

Shell. The enclosed skin of the dome, frame and covering together.

Shell ladder. The sequence of insulation layers hung under the shell, one affordable rung at a time, until the cabin is as warm as its owner wants. Chapter 55.

Strand. The reading track a chapter belongs to: story, howto, explain or reference. The contents page promises each track reads on its own.

Strut. A frame member loaded along its length — the load a triangulated frame puts on every stick in it.

Token. A figure in double braces in the manuscript that resolves, at build time, to a number computed by the software. The reason no number in this book was ever typed into a sentence.

Vertex. A corner point of the frame where members converge.

Wedge. A member split from a round log so its cross-section tapers, bark face wide, pith edge narrow. The wedge is the method: a structural section no mill sells, cut to be loaded along its length.

Wedge method. The whole system this book documents: split logs into wedges, lap them into a pinwheel frame with no hubs, and let the triangles do the structural work.

Worked example. A page that carries one number all the way through a calculation, showing every step rather than asserting the answer.

Yakisugi. The char-and-oil finish for the bark face only, leaving the sawn faces bare so the wood can breathe.

Where the Tools Live

Every number in this book came out of software: the raw-wedge solver, the tree model, the pinwheel arithmetic and the costing tables. That software is not hiding, and it is not gone. It lives in one place, and this page is the signpost.

The codebase. The geometry, the solver and the tables are published, open source, at github.com/dkzeanah/DomeSimulator. Anyone who wants to check a number in this book — or change the tree and rebuild every table around their own — can run the same software that printed these pages.

The Complete Edition. The site sells *The 40 Hour Cabin: Complete Edition*: the same method carried further — every finish, every material and panel priced, the systems this paperback leaves out

— together with the codebase that computed all of it, and every future edition of that book. It is the other half of the project this paperback introduces.

The design session. For one dome, designed with the author using the project's own tools, the site also sells a one-on-one design session.

Nothing in this book was invented for the page. The numbers were computed, the corrections were found by the same arithmetic, and the tools that did it are one address away.

Index and Colophon

Index

Chapter numbers, because a reader who has the page open can follow a number as easily as a page. The contents list carries the pages.

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Colophon

This book was not typeset from a manuscript that could drift from the things it describes. It was built, in one step, from the same software that draws the models and solves the geometry, and that is worth one page of explanation because it is the book's one unusual promise.

Every figure in these pages is computed at the moment the book is built. When a chapter says the dome is 21.6 feet across, the number came out of the solver a second before it reached the page; when a table prints an angle or a price, each one is resolved against the geometry, and a misspelt figure stops the build instead of printing as a blank. The arithmetic is a test suite, and it runs before any copy of this book is allowed to exist.

The book itself is the same material in three forms: the HTML you can read in a browser and print from, the PDF typeset from that same HTML, and the Markdown source. All three are produced by one function, so the version you read and the version you send cannot be different books.

The tools that made it are open source at github.com/dkzeanah/DomeSimulator, and the Complete Edition — the same method carried further, sold with the codebase on the site — continues where this paperback stops. The declared constants can be changed — a different tree, a different handling limit, a different rainfall — and the book rebuilt around them. That is what makes this a manual rather than a memoir of one build: the same geometry that produced this dome will produce yours.

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